

UM JCShell

JCShell Standalone

5.32.0.X — 13/01/2020

User Manual,
License Terms and Conditions

[Nwww./terDocument information](#)

Info	Content
Keywords	JCOP, JCShell, Standalone
Abstract	Help for JCShell usage



1. Contents

1.	Contents	2			
2.	Introduction.....	5			
3.	General	5			
3.1	Supported OS.....	5	5.27	/list-readers	53
3.2	Installation on Windows.....	5	5.28	/list-vars.....	53
3.3	Installation on Mac OS X/Linux	5	5.29	/manage-channel	54
3.4	Configuration.....	5	5.30	/mode	55
3.5	Command prompt.....	5	5.31	/printf.....	55
3.6	Script files.....	5	5.32	/r-echo	56
3.7	Abbreviation of commands.....	6	5.33	/register	56
3.8	Script parsing	6	5.34	/reset	57
3.9	Command line options.....	7	5.35	/reset-transaction-time	58
4.	Features	8	5.36	/select.....	58
4.1	Managing Command Line	8	5.37	/send	59
4.1.1	Commands supported	8	5.38	/set-channel	61
4.2	Variables	9	5.39	/set-var	62
4.2.1	Variable modifiers.....	9	5.40	/set-random-seed	65
4.2.2	Special Variables.....	11	5.41	/sleep	65
4.2.3	Variable Scope	13	5.42	/start-timer	66
4.2.4	Arrays	14	5.43	/stop-timer	66
4.3	Line parsing.....	14	5.44	/start-scr-timer	67
4.4	Control Flow Statements	14	5.45	/stop-scr-timer	67
4.5	Expressions.....	16	5.46	/start-mp500-timer	67
4.6	Special Operators.....	17	5.47	/stop-mp500-timer	67
4.6.1	The #() Operator.....	17	5.48	/start-mp300-timer	67
4.6.2	The %() Operator	17	5.49	/stop-mp300-timer	68
5.	List of commands.....	18	5.50	/start-mpreader-timer	68
5.1	/applet	18	5.51	/stop-mpreader-timer	68
5.2	/atr	18	5.52	/terminal	68
5.2.1	Behavior of /reset, /atr and /card	19	5.53	?	83
5.2.2	Old behavior of /atr and /card	20	5.54	>	83
5.3	/cap-cipher	21	5.55	auth	83
5.4	/cap-clear	22	5.56	card-info	84
5.5	/cap-del-auth	23	5.57	delete	86
5.6	/cap-extrdt-auth	25	5.58	delete-key	87
5.7	/cap-info	26	5.59	ext-auth	87
5.8	/cap-inst-auth.....	28	5.60	flush	88
5.9	/cap-load-auth	38	5.61	get-cplc	89
5.10	/cap-reg-update	40	5.62	get-data.....	89
5.11	/cap-sign.....	42	5.63	help	90
5.12	/cap-patch	44	5.64	init-update	90
5.13	/card	45	5.65	install.....	91
5.14	/channel.....	47	5.66	ls	99
5.15	/clear-vars	47	5.67	print-key	101
5.16	/close.....	48	5.68	put-key	101
5.17	/change-protocol.....	48	5.69	put-keyset	102
5.18	/echo	49	5.70	put-pub-key	103
5.19	/error.....	49	5.71	quit	104
5.20	/exchange-warmreset.....	49	5.72	registry-update	104
5.21	/exchange-poweroff.....	50	5.73	select.....	106
5.22	/exec.....	51	5.74	send	106
5.23	/expr	52	5.75	session-info.....	107
5.24	/get-transaction-time.....	52	5.76	set-applet	109
5.25	/glob	52	5.77	set-key	109
5.26	/identify.....	53	5.78	set-scp	110
			5.79	set-sd-state	111
			5.80	set-security.....	111
			5.81	set-state	111
			5.82	store-aid	112

5.83	store-data	113	5.141	/viso-get-protocol-version	144
5.84	store-keyset	114	5.142	/viso-eSeReset	144
5.85	upload	114	5.143	/set-ifsc-ifsd-size	145
5.86	upload-install-make-selectable	120	5.144	/switch-logical-interface	145
5.87	manage-upgrade-elf-start	123	5.145	/configure-se-gpio	145
5.88	manage-upgrade-elf-resume	124	5.146	/set-se-gpio	145
5.89	manage-upgrade-elf-abort	124	5.147	/get-se-gpio	146
5.90	manage-upgrade-elf-status	125	5.148	/end-of-apdu	146
5.91	version	125	5.149	/set-nfcc-mode	146
5.92	get-crs-state	125	5.150	/simu-get-device-state	147
5.93	set-crs-state	126	5.151	/swp-control-recv	147
5.94	put-key-ecc	127	5.152	/src01-set-ifsc	147
5.95	/get-screxec-time	128	5.153	/scr01-set-protocol-parameter	147
5.96	/get-scr-serialnum	128	6. Crypto Plugin Commands	148	
5.97	/switch-swp-gate	128	6.1	hash	149
5.98	/get-clf-param	128	6.2	sign	149
5.99	/select-scr01	128	6.3	keygen	154
5.100	/set-scr01-icctrigger	129	6.4	mac	156
5.101	/set-scr01-vtptrigger	129	6.5	pad	158
5.102	/set-scr01-trigger1	129	6.6	unpad	160
5.103	/set-scr01-trigger2	131	6.7	encrypt	161
5.104	/set-scr01-vcc	131	6.8	decrypt	165
5.105	personalize	131	6.9	gen-cert	168
5.106	get-sd-cert	131	6.10	crcgen	170
5.107	create-ecka-certificate	131	7. Provider Security Domain Plugin Commands		
5.108	perform-security-operation	133	(Delegate Management)	171	
5.109	mutual-authenticate	133	7.1	delete	172
5.110	internal-authenticate	134	7.2	install	172
5.111	notifications	135	7.3	upload	172
5.112	/set-swp-activation-time	135	7.4	registry-update	175
5.113	/set-swp-actsync-time	135	7.5	extradite	175
5.114	/get-swp-activation-time	135	8. Simulator Plugin Commands	175	
5.115	/get-swp-actsync-time	135	8.1	simu-reset	176
5.116	/set-scr01-baudrate	135	8.2	simu-restart	176
5.117	/send-hci-event	136	8.3	simu-terminate	176
5.118	/get-hci-event	137	8.4	simu-change-protocol	176
5.119	/clear-hci-event	138	8.5	simu-dump-memory	176
5.120	/get-last-event	139	8.6	simu-get-dump-path	177
5.121	/connect-device	139	8.7	start-tearing	177
5.122	/disconnect-device	139	8.8	stop-tearing	177
5.123	/felica-polling	139	8.9	simu-load-memory-dump	177
5.124	/set-scr01-rf-error-indicator	140	9. eUICC Plugin commands	178	
5.125	/prepare-tearing	140	9.1	authenticate	178
5.126	/set-io-pin	140	9.2	get-euicc-info	179
5.127	/set-default-nad	141	9.3	get-euicc-challenge	180
5.128	/send-jrcp-frame	141	9.4	authenticate-server	180
5.129	/modify-scr01-iso7816-timeout	141	9.5	bound-profile-package	182
5.130	/modify-scr01-protocol-params	141	9.6	prepare-download	185
5.131	/scr01-update-clpp	142	9.7	SCP80-send	186
5.132	/set-scr01-clockmode	142	9.8	tr-proactive	191
5.133	store-certificate	142	9.9	fetch-proactive-data	192
5.134	store-rsa-pub-key	143	9.10	scp81-tls	193
5.135	/Update-clpp	143	9.11	envelope	195
5.136	/viso-switch-interface	144	9.12	scp81-http-response	196
5.137	/viso-reset	144	9.13	scp81-alert	197
5.138	/viso-end-of-apdu-session	144	10. Generation of .pem file	197	
5.139	/viso-get-atr	144	11. Generation of PKCS#12 file in openssl	198	
5.140	/viso-get-allowed-power-state	144			

12.	Generation of PKCS#8 file using OpenSSL ..	198	19.	Changelog.....	203
13.	Storage of DAP blocks and DM tokens in capfile	199	20.	Legal Information	212
14.	Ciphered Cap File	200	20.1	NXP Semiconductors Netherlands B.V.	212
15.	Read JCShell Script Contents	201	20.1.1	END USER LICENSE AGREEMENT	212
16.	Additional Resources.....	202	20.1.2	APPENDIX A	216
17.	Known Limitations	202	20.1.3	APPENDIX B	216
18.	Restrictions.....	203	20.1.4	APPENDIX C	216

2. Introduction

The JCShell is a powerful tool to interact with your JavaCard. You can type commands interactively or you can run scripts against a JavaCard. The JCShell allows you to exercise functions of a JavaCard on APDU level.

This document delivers some general help and in addition to every command some help and an example of use.

3. General

3.1 Supported OS

- Windows XP SP2 and Windows7
- For Linux and Mac OS X only connection to the simulator and PCSC card reader is supported. Interaction with the real cards is limited to PCSC card readers only on Linux/Mac OS X
- No other operating system is supported

Note: On Linux platform, to detect PC/SC readers set the environment variable 'JAVA_OPTS' with libpcsc-lite.so library along with library folder path.

3.2 Installation on Windows

To install the software on Windows, please double click on the "jcshell.bat"

Note: Please Install Visual C++ Redistributable package for Visual Studio 2012.

3.3 Installation on Mac OS X/Linux

To install the software on Mac OS X/Linux, give the executable permission for the file "jcshell.sh", and execute the same "./jcshell.sh".

3.4 Configuration

The optional file `jcshell.rc` contains the configuration and is executed once per start of the JCShell. It is located in the same directory as the JCShell itself. The `jcshell.rc` file holds a sequence of shell commands, for example you can always turn on trace mode by adding `/mode trace=on` to the `jcshell.rc`.

3.5 Command prompt

When the JCShell starts, it will display a simple prompt, designated by a dash (-), that indicates that no smart card reader and card is yet attached. Upon the successful connection to a terminal the prompt changes to a caret (>) indicating connection to a smart card reader, but not to a card. If you are connected to a card, the prompt changes to (cm>) indicating successful activation of the CardManager plug-in. By default 1024M bytes of heap size had been allocated for the JVM.

3.6 Script files

The JCShell supports shell scripts. A script file typically contains a sequence of JCShell commands and must have the extension `.jcsh`. The JCShell maintains an internal special variable (path) which defines the path it searches for script files. The variable can be set via the `/set-var` shell command (e.g. `/set-var path c:/JCShellScripts`). The JCShell always searches the current directory for script files. To execute a script just type the name of

the file (without extension) in the JCShell command line. User can also call a script by specifying the full path to a script. For example `./scripts/test.jcsh` will do the same as typing `test` if the path variable is set to `./scripts`.

Examples:

The JCShell script files are stored in the folder `C:/JCShellScripts`

Set the path variable as below.

```
/set-var path C:/JCShellScripts
```

Run the script say `test.jcsh` which is stored in the folder `C:/JCShellScripts` from JCShell command prompt.

```
test
```

The above command runs the `test.jcsh` script from `C:/JCShellScripts` folder.

If user wants to run any JCShell script at the start of the JCShell, user has to call the JCShell with the `-f` option and provide the complete path of the JCShell script.

JCShell `-f <jcshell script>`

Examples:

Start JCShell with `-f` option followed by JCShell script `test.jcsh` stored in `C:/JCShellScripts` folder as below.

```
jcshell.bat -f C:/JCShellScripts/test.jcsh
```

JCShell shall report the error and abort the script execution for any failing command in the JCShell script. Preceding hyphen (-) in front of the command in the JCShell script shall ignore the error and allow the script to continue the script execution.

Examples:

Using a delete command like `"delete -r <AID>"` aborts the script in case the package/module AID is not found (not present) or could not be deleted.

By adding a hyphen in front like `"-delete -r <AID>"` the command error will be ignored and the script will continue even though the package/module was not present.

3.7 Abbreviation of commands

All commands may be abbreviated except the name of scripts. But abbreviations have to be unique. For example `/term` behaves like `/terminal` but `/cap-i` will produce an error as it could mean `/cap-info` or `/cap-inst-auth`.

Each syllable (separated by dashes) of a command name can be abbreviated separately. Legal abbreviations of `/list-vars` are: `/list-v`, `/l-vars`, or `/l-v`.

3.8 Script parsing

The input lines read from a script are processed in the following way by the shell:

- Lines ending with a backslash are continued in the next line. These are combined into a single logical line.
- Comments are identified and ignored. A comment starts with a `#` character.
- Leading labels are identified and are removed. A line may start with a label. The first word in a line is recognized as a label if it ends with a colon.

3.9 Command line options

There exist several optional arguments which are very helpful if you want to use the JCShell in a batch-script.

Following are optional arguments listed when JCShell batch file is executed with option “-h” from command prompt.

Example: C:\JCShell>jcshell.bat -h

```
usage:  jcshell  [-h|--help][--gui][-v|--verbose][-n|--norc][-x|--echo][-g|--go][-e|--edit][-t|--
terminal term][-D|--define var[=value]][-R|--rc file][-d|--debug][-s|--rng-seed seed][-c|--
commDetails][-f|--script file][--iacceptthelicenseterms][-p|--Dpath path] [-o|--compatible][-m|--
heapmem memSize]arg..
```

-h | --help

Prints a short or long description of all options and arguments.

-- gui

Start a graphical user interface for entering options and arguments

-v | --verbose

Turns on the *verbose* mode. Same as `/mode verbose=on`

-n | --norc

Disables the execution of the default `jcshell.rc` startup script.

-x | --echo

Turns on the *echo* mode. Same as `/mode echo=on`

-g | --go

Tells the JCShell to keep on going. The JCShell is not going to exit in errors.

-e | --edit

Option to enables command line editing

-t | --terminal term

The terminal specified by the mandatory *terminal* argument is opened on startup.

For example: `-terminal pcsc:4`

-D | --define *variable_name*[=*value*]

Defines the specified variable. Can be added several times to define more variables.

-R | --rc *file*

Set the `.rc`-file given by the mandatory argument *file* to be executed. This script is processed before any other script or command is executed.

But other commands from the command line are executed before this script. It is not possible to overwrite variables from a `.rc`-file for example by adding `-D var=val` to the command line options. The `.rc`-file is executed after the variable definition. So the `.rc`-file may overwrite the variable not the other way round.

-d | --debug

Turns on the *debug* mode. Same as `/mode debug=on`

`-c| --comDetails`

Option defines the communication details mode whether on/off.

`-f | --script file`

The specified script file is going to be executed.

`-s | --rng-seed seed`

Initializes the internally used random number generators to the given seed.

`--iacceptthelicenseterms`

Accepting the license terms and condition.

`-p| --Dpath`

Option to set the directory path where the script files to be executed are present.

`-m| --heapmem`

Option to allocate the heap memory for JVM

Note: Please note that the argument “-m/--heapmem” is optional and if provided shall be first argument to JCShell.

`arg...`

Option to define the script file parameters.

4. Features

4.1 Managing Command Line

JCShell supports managing of command line. JCShell command line handling supports full history of previous commands including searching, TAB completion of command depending on context. This managing command line is supported in all platforms Windows, Linux and MAC OS X.

4.1.1 Commands supported

Following commands are supported for managing command line.

Command	Description
Ctrl + A	Go to the beginning of the line you are currently typing on
Ctrl + E	Go to the end of the line you are currently typing on
Ctrl + L	Clears the Screen
Ctrl + U	Clears the line before the cursor position. If you are at the end of the line, clears the entire line.
Ctrl + H	Same as backspace
Ctrl + R	Lets you search through previously used commands
Ctrl + C	Kill whatever you are running
Ctrl + D	Exit the current JCShell

Ctrl + W	Delete the word before the cursor
Ctrl + K	Clear the line after the cursor
Ctrl + T	Swap the last two characters before the cursor
Alt + F	Move cursor forward one word on the current line
Alt + B	Move cursor backward one word on the current line
Tab	Auto-complete commands

4.2 Variables

Variable references have the following form:

```
${name}
```

A variable reference starts with a '\$' character and the variable body is enclosed in curly braces. The variable body inside the curly braces is subject to variable expansion.

Examples:

```
/set-var x1 Hello
/set-var x2 1
/echo ${x1}           prints Hello
/echo ${x${x2}}       prints Hello
```

4.2.1 Variable modifiers

Modifiers can be used to change the substitution results of variables. A variable and its modifier are separated by a ';' character. Avoid spaces between modifier and separator.

The following modifiers are defined:

Modifier	Description	Example
Dirname	path of a canonical filename (content of the variable must be a filename)	<pre>> myscr=\${script.name;qq} > /echo Script: \${myscr} Script: D:/eclipse/test/test.jcsh > /echo Dir=\${myscr;dirname} Dir=D:/eclipse/test</pre>
Filename	filename of a canonical filename (content of the variable must be a filename)	<pre>> myscr=\${script.name;qq} > /echo Script: \${myscr} Script: D:/eclipse/test/test.jcsh > /echo Fn=\${myscr;filename} Fn=test.jcsh</pre>
Basename	basename of a canonical filename (content of the variable must be a filename)	<pre>> myscr=\${script.name;qq} > /echo Script: \${myscr} Script: D:/eclipse/test/test.jcsh > /echo</pre>

Modifier	Description	Example
		<pre>Base=\${myscr;basename} Base=test</pre>
extension	extension (including dot) of a canonical filename (content of the variable must be a filename)	<pre>> myscr=\${script.name;qq} > /echo Ext=\${myscr;extension} Ext=.jcsch</pre>
L	Length of the variable content or number of elements of a variable array	<pre>> name=12345678abc > /echo \${name;l} 11</pre>
Q	quote variable content with "	<pre>> /set-var name "this is my test" > /list-var Global:_____ txt1=this is my test > /set-var txt2 \${txt1} jcsch: Bad expression: is > /set-var txt2 \${txt1;q}quoted > /list-var Global:_____ txt1=this is my test txt2=this is my testquoted</pre>
Qq	quote variable content with '	<pre>> /echo The script name is: \${script.name;qq} The script name is: D:\eclipse\test\test.jcsch</pre>
Lc	Change variable content to lowercase	<pre>> name=john PAUL Ringo > /list-var Global:_____ name=john PAUL Ringo > /echo \${name;lc} john paul ringo</pre>
Uc	Change variable content to uppercase	<pre>> name=john PAUL Ringo > /list-var Global:_____ name=john PAUL Ringo > /echo \${name;uc} JOHN PAUL RINGO</pre>

Modifier	Description	Example
h[x]	Change the variable content to hex-characters. The output length (x) specifies the number of characters in the output. The output will be prefixed with '0' characters if the length is less or equal to 8.	<pre>> ins=21 > /echo FF\${ins;h2}FF FF15FF > /echo FF\${ins;h4}FF FF0015FF</pre>
sOFF[,LEN]	Return the substring from position <OFF> in length <LEN>. The first character is offset 0. If you omit length, the rest of the string is returned. If LEN is greater than the length of the string, only the length of the string is returned.	<pre>> text=this is my test > /echo \${text;s5,7} is my t</pre>
eval	Evaluate given hex string to HEX data	<pre>>/s-v data "%(AA#(DDEE ascii))" >/echo \${data;eval} 09AA07DDEE6173636969</pre>

The special use of the modifier ":" allows access to JCShell-Plugin variables. The form is <plugin_nickname>:<varname>

4.2.2 Special Variables

The JCShell will maintain a set of special variables. The following variables are always declared.

Variable	Description
version	Returns the JCShell version
Channel	If a card is connected returns the result of "card.getChannel()" otherwise " "
JCOPDIR	Path to the JCOP directory (only when JCShell was started directly from console).
last.error	Returns the content of the error message of the last command. If the command has been successful this variable evaluates to an empty string. (<lastErrorMsg>)
last.errorlog	Returns the content of the last script error <lastErrorLog>
last.executiontime	Time taken in execution of last command. Units of time will be in millisecond by default.
last.executiontime.unit	Unit of time used in last.executiontime variable. Default value is millisecond. Valid values for last.executiontime.unit are usec,msec,ms,sec,s,min,m
last.apdu.executiontime	Time taken in executing last C-APDU and receiving R-APDU. Units of time will be in millisecond by default
last.apdu.executiontime.unit	Unit of time used in last.apdu.executiontime variable. Default value is millisecond. Valid values for last.apdu.executiontime.unit are usec, msec, sec, min
last.return	Returns the content of the last return <returnValue>

Variable	Description
last.response.data	Returns the data field of last APDU response
last.response.status	Returns the SW of last APDU response
timer.system.time	Stores total time recorded is the system time between the /start-timer and /stop-timer commands.
timer.device.time	Stores total processing time is the processing time of all the APDUs between the /start-timer and /stop-timer commands.
mp300timer.exec.time	Stores time measured for APDU execution for MP300, when /start-mp300-timer and /stop-mp300-timer commands are used.
mp500timer.exec.time	Stores time measured for APDU execution for MP500, when /start-mp500-timer and /stop-mp500-timer commands are used.
mptimer.exec.time	Stores time measured for APDU execution for MP300/MP500, when /start-mpreader-timer and /stop-mpreader-timer commands are used.
mp300timer.exec.time.unit	Unit for time measured in mp300timer.exec.time Default value is millisecond.
mp500timer.exec.time.unit	Unit for time measured in mp500timer.exec.time Default value is microsecond.
mptimer.exec.time.unit	Unit for time measured in mptimer.exec.time, Default value is millisecond.
Path	A semicolon separated list of directories. The JCShell checks these directories for script files.
Plugin	The nickname of the current plugin or an empty string if no plugin active.
Random	Returns a 16 byte random value in HEX [max=nn len=mm] are optional modifiers \${random;len=12} returns a 12 byte random value \${random;max=4378266} returns a numeric random value lower then the given maximum
Response	Returns the last APDU-response
script.line	If a script is executing returns the line-number + 1 otherwise 0
script.name	Name of the executing script file
script.dir	Path of the executing script file
Argv	Arguments given to the script execution
Terminal	If a terminal connected returns the result of "openTerminal.toString()" otherwise "" Example - /term Remote --Opening terminal > /echo \${terminal} Remote(Socket[addr=localhost/127.0.0.1,port=8050,localport=2116]/JCOP (built: Jul 12 2006 17:51:03))

Variable	Description
Time	Returns the current system timestamp (default fmt="y-m-d H:M:S") Format modifiers are possible. They must start with "fmt=" and may contain the key-values {y, d, m, H, M, S} Example > /echo \${time;fmt=d.m.y H:M} 09.05.2006 15:04
*name	Returns the value of the system environment variable <name> (result of "System.getenv(name)" or "System.getProperty(name, "")")
0	Returns the name of the script which is currently executing
1, ..., N	If the variable name is a number then it refers to the positional as passed to the script or shell process.
jrcp.version	Returns the JRCP protocol version supported. If JRCP version 2+ is active then jrcp.version = 2+ otherwise it is 1.
jcop.v242	Indicates JCOP version supported. By default, to support JCOP3 or newer version "jcop.v242" variable is set to false. Effectively JCShell uses SHA-256 as default hashing algorithm. To support JCOP 242 or older version set this variable to true.
cap.package.aid	Returns package aid for cap file used in /cap-info command. This variable is available only after successful execution of /cap-info command
cap.applet.aids	Returns list of applet aid(s) for cap file used in /cap-info command. This variable is available only after successful execution of /cap-info command
pseudo.random	Returns a 16 byte pseudo random value in HEX [max=nn len=mm] are optional modifiers \${pseudo.random;len=12} returns a 12 byte random value \${pseudo.random;max=4378266} returns a numeric random value lower then the given maximum
exec.errorlog	Contains the stderr log message while using the /exec command
exec.output	Contains the stdout log message while using the /exec command

4.2.3 Variable Scope

- Any local variable is inherited to called functions (or script files)
- If in the called function (or script file) a variable with the same name is defined, a new variable is created with the scope limited to that function (and called ones). On return, the previous value is restored.
- If the variable is unset, the variable in the current scope is unset. If another global variable of the same name existed from the scope "above", that variable is still present, and its value will be visible.

Subsequently unsetting variables of the same name unsets the variables in the innermost scope moving outwards

4.2.4 Arrays

JCShell supports creation of array variables and assign values to them.

Arrays are declared using [/set-var](#) command.

- a) **Operator *** is used to create new array variables. Following is an example to create an array variable.

```
Ex: /s-v array1[*] 1 2
```

Above example creates an array variable array1 and initialize array1 with values 1 and 2.

Note: If an array is created within a function then its scope is limited to the function and cannot be accessed outside it.

- b) **Operator #** in array is used to append values to an existing array.

If the array is not existing, then it shall create a new array and assign values to it.

```
Ex: /s-v array2[*] 8 8
    /s-v array2[#] 5 6 7 8
    /echo "array2: ${array2[*]}"
    array2: 8 8 5 6 7 8
```

Note: when a # operator is used within a function (in the context of arrays), then it would check if the variable is existing inside the function, if yes then it would append the values to it, else it would check for the variable which are declared outside the function and append the values accordingly.

4.3 Line parsing

White space splits a line into words. Strings enclosed in "\"" and "'" become part of words. Characters with special meaning can be quoted using a backslash in strings. The backslash is removed only if applied to special characters. In other cases the character sequence remains unchanged.

Labels and control flow keywords are not subject to variable evaluation. In strings enclosed in single quotes (') variables are not evaluated. All other words may contain variable references.

Examples:

```
/set-var x1 Hello
/set-var x2 ${abc}           x2 has value "Hello"
/set-var x2 "${abc} ${abc}"  x2 has value "Hello Hello"
/set-var x2 '${abc}'"${abc}" x2 has value "${abc}Hello"
/echo '\${}'\'' '\${}'\''    prints the following '\${}'\'' '\${}'\''
```

4.4 Control Flow Statements

```
if expression
...
```

The elseif and else part is optional. An elseif part may occur several times. The

<pre>elseif expression ... else ... end Example: if \${3} /echo "Called with at least 3 arguments" elseif \${2} /echo "Called with two arguments" elseif \${1} /echo "Called with one argument" else /echo "Called without arguments" End</pre>	expressions are evaluated to a Boolean value.
<pre>while expression ... end Example: /set-var x 3 while \${x} /echo \${x} /set-var x \$(/expr \${x} - 1) end</pre>	The expression is evaluated to a Boolean value.
<pre>for varname words ... end Example: for num 1 2 4 8 16 /echo \${num} End</pre>	The for-loop is executed once for every value of the <code>words</code> list where <code>varname</code> holds the actual value.
<pre>break [label] continue [label] Example:</pre>	The label is optional and if present must refer to label in front of a <code>while</code> or <code>for</code> statement. If the label is omitted then the statement refers to the innermost <code>while</code> or

<pre> myFor: for Num 1 2 4 8 16 /echo \${Num} break myFor end </pre>	for loop.
<pre> goto label </pre> Example: <pre> goto myLabel /echo skipped myLavel: /echo jumped </pre>	Continue execution at the specified label. The destination should be in the same block and the same block nesting level. You should not leave or enter an if, while, for, or try block.
<pre> try ... catch expression ... catch expression ... end </pre> Example: <pre> try /error test catch \${last.error} == test2 /echo nothing catch \${last.error} == test /echo caught end </pre>	If a command between the try and the first catch terminates abnormally then execution is resumed at the first catch whose expression evaluates to true.

4.5 Expressions

Shell expressions consist of a set of words. Each operand and each operator must be a separate word. Sub expressions can be enclosed in parenthesis which also must appear as separate words.

The following table lists the supported operators. If nothing is said the listed operators have the same semantics as in the C/Java language. The higher the row the stronger is the operator binding. Operators usually work on integer values. If an operator works also on strings its meaning is explained.

Unary operators	
- ! ~	purely numeric

Binary operators	
* / %	purely numeric
+ -	purely numeric
<< >> >>>	purely numeric
< <= > >=	If both operands are numbers then a numeric comparison is performed. Otherwise an alphanumeric comparison is done.
== != =~	The operator =~ checks if the left operand (string) matches regular expression on the right side (string). This operator is only defined on strings. If both operands to == or != are numbers then a numeric comparison is performed. Otherwise an alphanumeric comparison is done.
&	purely numeric
^	purely numeric
	purely numeric

4.6 Special Operators

JCShell supports some of special operators in the commands. The following sections lists the supported special operators and their behavior.

4.6.1 The #() Operator

usage: #(DATA)

The #() operator is used to calculate the length of the DATA specified inside the parentheses "()". JCShell will calculate the length of the DATA and place single byte indicating the length before DATA. The characters "#()" of this special operator characters are removed.

Example:

```
#(AABBCC)
This shall be interpreted by the Shell as '03AABBCC'
```

Note:

JCShell performs the specified operation before the command is executed. For example `/send #(AABBCC)` will be substituted by `/send 03AABBCC` before the `/send` command is executed.

This operator is used to calculate the length of DATA up to 255. Hence the length byte which will be placed before DATA will always be 1 byte.

4.6.2 The %() Operator

usage: %(DATA)

The %() operator is used to calculate the length of the DATA specified inside the parentheses '()' in BER encoded. JCShell will calculate the length of the DATA and place 1 to 3 bytes indicating the length before DATA. The characters “%()” of this special operator characters are removed.

Example:

```
% (00 to 7F bytes DATA)
This shall be interpreted by the Shell as 'XXDATA', where XX being the length
ranging 0 to 127 of DATA
% (00 to FF bytes DATA)
This shall be interpreted by the Shell as '81XXDATA', where XX being the length
ranging 0 to 255 of DATA
% (0000 to FFFF bytes DATA)
This shall be interpreted by the Shell as '82XXXXDATA', where XXXX being the
length ranging 0 to 65535 of DATA
```

Note:

JCShell performs the specified operation before the command is executed. For example '/send %(AABBCC)' will be substituted by '/send 03AABBCC' before the /send command is executed.

The length bytes which will be placed before DATA ranging from 1 to 3 bytes.

5. List of commands

5.1 /applet

usage: /applet [[-p|--plugin-restore][[-r|--reuse]][nickname]*

-p|--plugin-restore

To restore plugin state.

-r|--reuse reuse plugin.

Switch to specified applet plugin.

By default a new instance of the desired plugin is created.

If a plugin instance (with the same nickname) currently active on another logical channel is to be reused, then the reuse option must be set. Reusing might make sense when communicating with a multi-selectable applet.

If no nickname is provided, all registered plugins are listed.

Example:

```
1 /applet
2 Nickname AID Plugin
3 -----
4 cm A000000003000000 com.nxp.id.jc.tools.CardManagerPlugin
5
```

5.2 /atr

usage: /atr [[-n|--nad <node address>]]

-n|--nad <node address>

Indicates the node address.

NAD is applicable only when Remote Terminal is opened.

<node address>

node address

Returns the ATR Information of the last (protocol) activation applied.

NOTE: When using type F (FeliCa) communication, no ATR is available.

Therefore, a dummy ATR is provided by /atr.

Command will perform a reset if and only if it wasn't prior activated (no /reset command called).

The terminal decides whether this will be a warm or cold reset.

Note:

Command behavior has been changed since version 4.0.0.0.

Example:

```
> /atr
ATR: 3B8F80014A434F5033205037332056362020203F
ATR:
      T = 0
      T = 1
```

5.2.1 Behavior of /reset, /atr and /card

Following table depicts the current behavior of /reset, /atr and /card.

Sequence of commands	New Implementation	
	Description	Example
/terminal <terminal type> /atr	Since no /reset command is called prior to /atr command, it does protocol activation (cold reset)	> /atr Warning: Usage of /atr with no preceding of /reset is deprecated. /reset is invoked. ATR: 3B8F80014A434F50332E332052312E30302E3144 ATR: T = 0 T = 1
/terminal <terminal type> /reset /atr	/reset is called without any options. Protocol (re-)activation (cold reset) is the default behavior. /atr returns the response of preceding /reset.	> /reset > /atr ATR: 3B8F80014A434F50332E332052312E30302E3144 ATR: T = 0 T = 1
/terminal <terminal type> /reset -w /atr	/reset is called with warm reset option. /atr returns the response of preceding /reset -w	
/atr -t	/atr command options are deprecated since version	> /atr -t jcshell: /atr: Excess arguments: -t ... > /atr -w

	4.0.0.0. Please use /reset command instead.	jcshell: /atr: Excess arguments: -w ... > /atr -c jcshell: /atr: Excess arguments: -c ...
/terminal <terminal type> /reset -c /card	Since previous reset is called with -c option. /card internally calls /atr, /app cm then select	
/terminal <terminal type> /card	/card internally calls /atr, /app cm then select. As /reset command not invoked before /atr, /reset command is invoked i.e., cold reset would be performed and then /atr, /app cm and select would be called.	

5.2.2 Old behavior of /atr and /card

Following table depicts the old behavior of /atr and /card. This is provided to give the reference to the behavior of old JCShell.

Note:

Please note that /reset was not supported for the old releases of JCShell. Only /atr was supported to reset card and get ATR.

Sequence of commands	New Implementation	
	Explanation	Example
/terminal <terminal type> /atr	/atr performs cold reset.	> /atr --Waiting for card... ATR=3B 8F 80 01 4A 43 4F 50 33 20 50 37 33 20 56 36 ;...JCOP3 P73 V6 20 20 20 3F ? IOCTL(). ATR: 0 T = T = 1
/terminal <terminal type> /atr -w	/atr -w performs warm reset and returns the ATR response	
/atr -t	/atr waits for specified time until card is inserted.	
/terminal <terminal type> /atr -r /card	/atr -r does power on reset (cold ATR). /card	

	internally calls again /atr and then /select	
/terminal <terminal type> /card	/card internally sequentially calls /atr -r and /select commands.	

5.3 /cap-cipher

usage: /cap-cipher *[[--i|--icv][--s|--clsd-aid clsd-aid][--t|--target <Ciphred CAP>][--p|--package package-name][--c|--component component-name][--h|--search search-string][--r|--replace replace-string][CAP-file [[--k|--cl-key clkey]]*]*

--i|--icv To store initialization vector data to output CAP file. Optional parameter.

--s|--clsd-aid clsd-aid

AID of CLSD to be used during install for load. Optional parameter.

--t|--target <Ciphred CAP>

If the cap file to be ciphered shall not be modified, in such case

this option can be used to specify the location and name of the resultant (ciphered) CAP-file.

Optional parameter.

<Ciphred CAP>

File name of the resultant ciphered CAP file.

--p|--package package-name

Specify the Java package name to search for in the CAP-file. If this option isn't set, the first package found will be patched.

package-name

Java package name.

--c|--component component-name

Specify the CAP component name to search for in the CAP-file. If this option isn't set, the StaticField.cap component will be patched.

component-name

CAP component name.

--h|--search search-string

Specify the search string to search for in the CAP-file.

search-string

Search string.

--r|--replace replace-string

Specify the replace string to patch in the CAP-file.

replace-string

Replace patch string.

CAP-file Plain cap file to be ciphered or encrypted. Mandatory parameter.

-k|--cl-key clkey

Ciphered Loading key. Mandatory parameter.

Example 1:

```
/cap-cipher      -i      -t      "T:\cipherwithicv.cap"      "T:\plain.cap"      -k  
0000000000000000000000000000000000000000000000000
```

Example 2:

```
/cap-cipher -i -n AABBBBBBBBAA -r 010203040506 "S:\test.cap" -k
0102030405060708090A0B0C0D0E0F10
```

Example 3:

```
/cap-cipher -i -h AABBBBBBBBAA -r 010203040506 -t "S:\testModified.cap" "S:\test.cap"
-k 0102030405060708090A0B0C0D0E0F10
```

Example 4:

```
/cap-cipher -i -c "Method.cap" -h AABBBBBBBBAA -r 010203040506 -t
"S:\testModified.cap" "S:\test.cap" -k 0102030405060708090A0B0C0D0E0F10
```

Note: Only single string shall be replaced with “/cap-cipher” command. If multiple string must be replaced, then please use “/cap-patch” command.

5.4 /cap-clear

usage: /cap-clear [-p|--package package-name] CAP-file

`-p|--package package-name`

Specify the Java package name to search for in the CAP-file. If this option isn't set, the information for the first package found will be removed.

package-name

Java package name.

CAP-file CAP file

Remove DAP and Delegated Management information from a CAP-file.

Example:

```

6      /cap-info test_signed.cap
7      ----non relevant parts deleted----
8      DAP block(s)          :
9      DAP block              :      1
10     AID                    :      1234456789
11     Signature               :      55E23EBAF6FCE3BE
12     Algorithm               :      GP 2.1.1
13     Load Token(s)          :      none
14     Install Token(s)        :      none
15     /cap-clear test_signed.cap
16     /cap-info test_signed.cap
17     ----non relevant parts deleted----
18     DAP block(s)           :      none
19     Load Token(s)          :      none

```

20 Install Token(s) : none

5.5 /cap-del-auth

usage: /cap-del-auth [-e|--delegation][--l|--cm-lock][--t|--terminate][--d|--default][--c|--pin-delete][--b|--sd-dap][--n|--tokenpin PIN][--p|--package package-name][--s|--sdid Identification Number][--m|--sdimg Image Number][--i|--tokenid Token Identifier][--f|--target CAP-file] Applet-AID CAP-file tokenspec HashAlgorithm

-e|--delegation

Security Domain with delegated management.

--l|--cm-lock

Card Manager lock permission.

--t|--terminate

Card terminate permission.

--d|--default

Implicit delete (default) applet.

--c|--pin-delete

PIN delete permission.

--b|--sd-dap

Security Domain with DAP verification.

--n|--tokenpin PIN

Specify the PIN to open the token. Only used if token is PIN protected.

PIN PIN to open the token referenced by tokenspec.

--p|--package package-name

Specify the Java package name to search for in the CAP-file. If this option isn't set, the first package found will be authorized for delegated deletion.

package-name

Java package name.

--s|--sdid Identification Number

Specify the desired Identification Number of the Security Domain with the Token Verification privilege.

Identification Number

Identification Number of the Security Domain with the Token Verification privilege.

--m|--sdimg Image Number

Specify the desired Image number of the Security Domain with the Token Verification privilege.

Image Number

Image number of the Security Domain with the Token Verification privilege.

--i|--tokenid Token Identifier

Specify the desired Token identifier/number (digital signature counter).

Token Identifier

Token identifier/number (digital signature counter).

-f|--target CAP-file

If the CAP-file to be authorized shall not be modified this option can be used to specify the location and name of the target (signed) CAP-file.

CAP-file File name of the authorized CAP-file.

Applet-AID AID of the applet in the package to be deleted.

CAP-file CAP file

tokenspec

Specify the token holding the private key to be used for signature generation.

Possible token definitions are: "<PKCS#12-file>" | "<PKCS#8-file>" | "<hex-signature>". You can also pass the signature directly.

For ECDSA algorithm, EC private key has to be provided in a .pem file containing these headers:

```
-----BEGIN EC PRIVATE KEY-----
-----END EC PRIVATE KEY-----
```

Delete token generation:

For initializing the EC curve parameters for ECDSA, Initialize the following global variables with hex strings using /set-var command, else 'P-256' curve will be used.

ECC_CURVEPARAM_P - The prime p

ECC_CURVEPARAM_A - EC curve coefficient a

ECC_CURVEPARAM_B - EC curve coefficient b

ECC_CURVEPARAM_G - The base point G(Gx,Gy)

ECC_CURVEPARAM_N - The order of G

HashAlgorithm

It specifies whether to use SHA1/ SHA-256 digest for signature generation.

Authorize a package in a CAP-file for delegated deletion of an applet of this package. The resulting Delete Token information is stored in the CAP-file.

Example:

```
/cap-del-auth |myTest "S:\test.cap" "S:\Privatekey.pem" SHA-256
Privatekey.pem is the file containing EC private key.
|myTest is the Applet-AID.
"S:\test.cap" is the cap file where the delete token generated will be
stored.
```

Note: Please refer to [Section 8](#) for generation of .pem files.

Refer [Section 11](#) to know about storage of tokens in Capfile.

5.6 /cap-extrdt-auth

usage: /cap-extrdt-auth [-s|--sdaid AID][-n|--tokenpin PIN][-p|--package package-name][-i|--sdid Identification Number][-m|--sdimg Image Number][-t|--tokenid Token Identifier][-f|--target CAP-file] application-AID CAP-file tokenspec HashAlgorithm

-s|--sdaid AID

Specify the AID of the Security Domain to be associated with the package and its applications (Card Manager is default).

AID AID of Security Domain.

-n|--tokenpin PIN

Specify the PIN to open the token. Only used if token is PIN protected.

PIN PIN to open the token referenced by tokenspec.

-p|--package package-name

Specify the Java package name to search for in the CAP-file. If this option isn't set, the first package found will be authorized for delegated extradition.

package-name

Java package name.

-i|--sdid Identification Number

Specify the Identification Number of the Security Domain with the Token Verification privilege.

Identification Number

Identification Number of the Security Domain with the Token Verification privilege.

-m|--sdimg Image Number

Specify the Image number of the Security Domain with the Token Verification privilege.

Image Number

Image number of the Security Domain with the Token Verification privilege.

-t|--tokenid Token Identifier

Specify the Token identifier/number (digital signature counter).

Token Identifier

Token identifier/number (digital signature counter).

-f|--target CAP-file

If the CAP-file to be authorized shall not be modified this option can be used to specify the location and name of the target (signed) CAP-file.

CAP-file File name of the authorized CAP-file.

application-AID

AID of the application or executable load file to be extradited.

CAP-file CAP file

tokenspec

Specify the token holding the private key to be used for signature generation.

Possible token definitions are: "<PKCS#12-file>" | "<PKCS#8-file>" | "<hex-signature>". You can also pass the signature directly.

For ECDSA algorithm, EC private key has to be provided in a .pem file containing these headers:

```
-----BEGIN EC PRIVATE KEY-----
-----END EC PRIVATE KEY-----
```

Extradite token generation:

For initializing the EC curve parameters for ECDSA, Initialize the following global variables with hex strings using /set-var command, else 'P-256' curve will be used.

```
ECC_CURVEPARAM_P - The prime p
ECC_CURVEPARAM_A - EC curve coefficient a
ECC_CURVEPARAM_B - EC curve coefficient b
ECC_CURVEPARAM_G - The base point G(Gx,Gy)
ECC_CURVEPARAM_N - The order of G.
```

HashAlgorithm

It specifies whether to use SHA1/ SHA-256 digest for signature generation.

Authorize a package in a CAP-file for delegated extradition. The resulting Extradition Token information is stored in the CAP-file.

Example:

```
/cap-extrdt-auth -s A000000151434169643064 |myTest "S:\test.cap"
"S:\WritePrivatekey.pem" SHA-256
```

```
|myTest : is the executable load file to be extradited to Security domain
with AID A000000151434169643064.
Extradite token is generated and store in "S:\test.cap".
Privatekey.pem is the file containing EC private key.
```

Note: Please refer to [Section 8](#) for generation of .pem files.

Refer [Section 11](#) to know about storage of tokens in Capfile.

5.7 /cap-info

usage: /cap-info [-p|--package package-name] CAP-file

-p|--package package-name

Specify the Java package name to search for in the CAP-file. If this option isn't set, the first package found will be examined.

package-name

Java package name.

CAP-file CAP file

Displays information of particular CAP file.

Displays only header component and LFDBH details when ciphered cap file is given.

Example:

```
- /cap-info "S:\test.cap"

CAP file name      :      S:\test.cap
CAP file version   :      X.X
Java package name  :      test
Internal pkg. name :      test
CAP file components :
    Header.cap (24 Bytes)
    Directory.cap (36 Bytes)
    Import.cap (24 Bytes)
    Applet.cap (16 Bytes)
    Class.cap (50 Bytes)
    Method.cap (147 Bytes)
    StaticField.cap (13 Bytes)
    ConstantPool.cap (69 Bytes)
    RefLocation.cap (25 Bytes)
    Descriptor.cap (206 Bytes)
    Debug.cap (892 Bytes)
Integer support    :      no
Package version    :      1.0
Package AID        :      "myTest"
Import AIDs        :
    A0000000620001 (java.lang) version 1.0
    A0000000620101 (javacard.framework) version 1.3
Applet AIDs        :
    "myTestApp"
Code size to load  :      404  bytes
Code size on card  :
    pkgAID         :      6
    applet AIDs    :      16
    classes        :      50
    methods        :      147
    statics        :      2
    exports        :      0
-----
overall           :      221  bytes
SHA-256           (Load      File      data):
097DED3D0CAAEC4A130495D9F0A093CAD796527C32D1C3F4BC5615D6257931F3
SHA-256           (Load      File      dbg)      :
792C691101A6052DCD94F656C9BB1B4117900A1962B12158CD2241FE892D864D
SHA-256           (Load      File)              :
E902D248225601C6BAEB1134D427711FE049A97B041E99F699187272C5572954
SHA-1 (Load File data):      EB1D28C34C79ACE148D14D9F799AD3F5CF5EB155
SHA-1 (Load File dbg) :      FD0F2E94BFA6DE27AFED36901EC688126B48EB12
SHA-1 (Load File)      :      42E197F0BFC4CEC3C4DD72D3938028174F768EFF
DAP block(s)         :
```

```

DAP block      :      1
  AID          :      112233445577
  Signature    :      55E970146500426F2F51975CD90F0B67
                                   CD13197BBED2A8C59EC97F397BD08F54
                                   93D5AE2A1AD4ABEA2702CD5C550922A7
                                   B680AAE812B8D57A344090B5B3F76064
                                   6C86BD480E9B461A0FB62E254C942E88
                                   CEAC4C2766679958CF38019E17C297A9
                                   DE3AAE07195A4836981C7CACDEA6D268
                                   BDE3C1509FB4C880ED6B4826FFB6C647
  Algorithm    :      RSA_GP_SE
Load Token(s)  :      none
Install Token(s):      none
Delete Token(s):      none

```

5.8 /cap-inst-auth

usage: /cap-inst-auth [-n|--tokenpin PIN][-e|--delegation][-l|--cm-lock][-t|--terminate][-d|--default][-p|--pin-change][-s|--security-domain][-b|--sd-dap][-m|--mandated-dap][-r|--trusted-path][-a|--auth-manage][-h|--token-manage][-j|--global-delete][-u|--global-lock][-w|--global-registry][-f|--final-application][-z|--global-service][-v|--receipt-gen][-x|--ciphered][-c|--contactless][-g|--contactlessself][-q|--install-param params][-i|--instance-aid AID][-o|--install-only][-ip params][-is params][-cpp params][-ra params][-cp params][-cia params][-pa params][-pb params][-pf params][-dt params][-ha params][-ga params][-gr params][-ca params][-cr params][-pr params][-ad params][-af params][-dr params][-cm params][-cn params][-k|--target CAP-file] applet-AID CAP-file tokenspec Hash Algorithm

-n|--tokenpin PIN

Specify the PIN to open the token. Only used if token is PIN protected.

PIN PIN to open the token referenced by tokenspec.

-e|--delegation

Security Domain with delegated management.

-l|--cm-lock

Card Manager lock permission.

-t|--terminate

Card terminate permission.

-d|--default

Implicit selectable (default) applet.

-p|--pin-change

PIN change permission.

-s|--security-domain

Security Domain privilege.

-b|--sd-dap

Security Domain with DAP verification.

- m|--mandated-dap
Security Domain with mandated DAP verification.
- r|--trusted-path
Trusted Path permission.
- a|--auth-manage
Authorized Management permission.
- h|--token-manage
Token Management permission.
- j|--global-delete
Global Delete permission.
- u|--global-lock
Global Lock permission.
- w|--global-registry
Global Registry permission.
- f|--final-application
Final Application permission.
- z|--global-service
Global Service permission.
- v|--receipt-gen
Receipt generation permission.
- x|--ciphered
Ciphered Load File Data Block permission.
- c|--contactless
Contactless Activation permission.
- g|--contactlessself
Contactless Self-Activation permission.
- q|--install-param params
Install parameters in TLV format (VOP/GP):
Tags: 0xC9 - application specific parameters
0xEF - system specific parameters:
0xC6 - non volatile code space limit
0xC7 - volatile data space limit
0xC8 - non volatile data space limit

Install parameters in TLV format (GSM 03.48):
Tags: 0xEF - system parameters
0xC8 - non volatile memory required for installation

0xC7 - volatile memory required for installation

0xCA - GSM applet specific parameters

0xC9 - applet specific parameters

GSM applet specific parameters are:

1 byte - Length of Access Domain field

1-n bytes - Access Domain:

Either one or three bytes! Possible values:

1 byte : 0x00 - Full access to the GSM file system

1 byte : 0xFF - No access to the GSM file system

3 bytes: 0x01 - APDU access mechanism:

0x0000 - No access

0x0001 - ALWAYS

0x0002 - CHV1

0x0004 - CHV2

0x0010 - ADM

(combinations of these bits are allowed!)

1 byte - Priority level of the Toolkit applet instance

1 byte - Maximum number of timers allowed

1 byte - Maximum text length for a menu entry

1 byte - Maximum number of menu entries

1 byte - Position of the first menu entry ('00' means last)

1 byte - Identifier of the first menu entry ('00' means don't care)

...

1 byte - Position of the last menu entry ('00' means last)

1 byte - Identifier of the last menu entry ('00' means don't care)

params Install parameters (TLV format).

-i|--instance-aid AID

Desired instance AID.

AID Instance AID.

-o|--install-only

Install the applet only, don't make it selectable.

--ip params

Assigned Protocols for Implicit Selection (TLV format):

Tag : 0x80

Length : 0-3

Value : 0x81 (Type A)

0x82 (Type B)

0x84 (Type F)

example :

--ip 80#(818284)

params Assigned Protocols for Implicit Selection (TLV format)

--is params

Initial Contactless Activation State (TLV format) :

Tag : 0x81

Length : 1

Value : 0x00 (DEACTIVATED)

0x01 (ACTIVATED)

example :

--is 81#(01)

params Initial Contactless Activation State (TLV format)

--cpp params

Contactless Protocol Parameters Profile (TLV format) :

Tag : 0xA2

Length : variable

Value : value (please check specifications for value field details)

example :

--cpp A2#(A0#(80#()))

params Contactless Protocol Parameters Profile (TLV format)

--ra params

Recognition Algorithm (TLV format) :

Tag : 0x83

Length : variable

Value : value (please check specifications for value field details)

example :

--ra 83#(01020288888888)

params Recognition Algorithm (TLV format)

--cp params

Continuous Processing (TLV format) :

Tag : 0x84

Length : 1-2

Value : 0x01 - Continuous processing disabled

0x02 - Continuous Processing is enabled

example :

--cp 84#(01)

params Continuous Processing (TLV format)

--cia params

Communication Interface Access Parameters (TLV format) :

Tag : 0xA5

Length : variable

Value : Tag(s)

0x80 - Communication Interface Access Restriction (Length - 1)

0x81 - Communication Interface Access Default (Length - 1)

0x82 - Communication Interface Access Instance (Length - 1)

example :

--cia A5#(80#(01)81#(02)82#(03))

params Communication Interface Access Parameters (TLV format)

--pa params

Protocol Data Type A (TLV format) :

Tag : 0x86

Length : variable

Value : Tag(s)

0xA0 - Protocol Parameter Data

0x80 - Unique Identifier

0x81 - SAK (Select Acknowledge)

0x82 - ATQA

0x83 - ATS LV structure

0x84 - FWI(Frame Waiting Time) & SFGI(Start-up Frame Guard

Time)

0x85 - CID(Card Identifier) Support

0x86 - DATA_RATE_MAX

0xA1 - Protocol Parameter Mandatory Mask

0x80 - Mandatory Mask of LV UID

0x81 - Mask for mandatory bits within SAK

0x82 - Mask for mandatory bits within ATQA

0x83 - Mandatory Mask of LV ATS

0x84 - Operation Indicator for FWI, SFGI

0x85 - Mask for mandatory bits in CID_SUPPORT

0x86 - Operation Indicator for mandatory bits within

DATA_RATE_MAX

example :

```
--pa
86#(A0#(80#()81#(01)82#(0102)83#(7777777777)84#(66)85#(01)86#(123456))A1#(80#(
00111111)81#(01)82#(0101)83#(0011111111)84#(00)85#(01)86#(00FF00)))
```

params Protocol Data Type A (TLV format)

--pb params

Protocol Data Type B (TLV format) :

Tag : 0x87

Length : variable

Value : Tag(s)

0xA0 - Protocol Parameter Data

0x80 - PUPI: Value is LV structure

0x81 - AFI: Application Family Identifier

0x82 - ATQB

0x83 - Higher Layer Response in response to ATTRIB

0x84 - Maximum data rate supported

0xA1 - Protocol Parameter Mandatory Mask

0x80 - Mandatory Mask for LV PUPI

0x81 - Mask for mandatory bits within the AFI.

0x82 - Mask for mandatory bits in ATQB

0x83 - Mandatory Mask for LV Higher Layer Response

0x84 - Operation Indicator for mandatory bits within

DATA_RATE_MAX

example :

```
--pb
87#(A0#(80#(01)81#(02)82#(66666666)83#(1122334455)84#(001122))A1#(80#(01)81#(
01)82#(01010101)83#(0101010101)84#(00FF00)))
```

params Protocol Data Type B (TLV format)

--pf params

Protocol Data Type F (TLV format) :

Tag : 0x88

Length : variable

Value : Tag(s)

0xA0 - Protocol Parameter Data

0xA0 - Type F anti-collision parameters entry(can be present more than once)

0x80 - System Code

0x81 - PICC Identifier

0x82 - Response Time Descriptor

example :

```
--pf
88#(A0#(A0#(80#(0101)81#(0102030405060708)82#(0102030405060708))))
```

params Protocol Data Type F (TLV format)

--dt params

Display Control Template (TLV format) :

Tag : 0x7F20

Length : variable

Value : Tag(s)

0x5F50 - Uniform Resource Locator

0x5F45 - Display Message

0x53 - Image Encoding Format

0x6D - Application Image Template

0x5F44 - Application Image

0x67 - Authentication Data

example :

```
--dt 7F20#(5F50#(01|www.google.com)5F45#(01|welcome)53#(05)
```

params Display Control Template (TLV format)

--ha params

Head Application (TLV format) :

Tag : 0xA0

Length : variable

Value : AID of head application

example :

```
--ha A0#(4F#(1122334455))
```

params Head Application (TLV format)

--ga params

Add to the Group Authorization List (TLV format) :

Tag : 0xA1

Length : variable

Value : AID(s) of Applications

example :

```
--ga A1#(4F#(1111111111)4F#(2222222222))
```

params Add to the Group Authorization List (TLV format)

--gr params

Remove from the Group Authorization List (TLV format) :

Tag : 0xA2
Length : variable
Value : AID(s) of Applications
example :
--gr A2#(4F#(3333333333)4F#(4444444444))
params Remove from the Group Authorization List (TLV format)
--ca params
Add to the CREL List (TLV format) :
Tag : 0xA3
Length : variable
Value : AID(s) of Applications
example :
--ca A3#(4F#(5555555555)4F#(6666666666))
params Add to the CREL List (TLV format)
--cr params
Remove from the CREL List (TLV format) :
Tag : 0xA4
Length : variable
Value : AID(s) of Applications
example :
--cr A4#(4F#(7777777777)4F#(8888888888))
params Remove from the CREL List (TLV format)
--pr params
Policy Restricted Applications (TLV format) :
Tag : 0xA5
Length : variable
Value : AID(s) of Applications
example :
--pr A5#(4F#(5555555555)4F#(6666666666))
params Policy Restricted Applications (TLV format)
--ad params
Application discretionary data (TLV format) :
Tag : 0xA6
Length : variable
Value : Data
example :
--ad A6#(00010203040506070809)

```

    params Application discretionary data (TLV format)
--af params
    Application Family (TLV format) :
        Tag   : 0x87
        Length : 1
        Value  : Application Family Indicator
    example :
        --af 87#(04)
    params Application Family (TLV format)
--dr params
    Display Required Indicator (TLV format) :
        Tag   : 0x88
        Length : 1
        Value  : 00 - Application requires a display
                01 - Application does not require a display
    example :
        --dr 88#(01)
    params Display Required Indicator (TLV format)
--cm params
    Cumulative Granted Volatile Memory (TLV format) :
        Tag   : 0x82
        Length : 2 or 4
        Value  : memory size
    example :
        --cm 82#(1234)
    params Cumulative Granted Volatile Memory (TLV format)
--cn params
    Cumulative Granted Non Volatile Memory (TLV format) :
        Tag   : 0x83
        Length : 2 or 4
        Value  : memory size
    example :
        --cn 83#(1234)
    params Cumulative Granted Non Volatile Memory (TLV format)
-k|--target CAP-file

    If the CAP-file to be authorized shall not be modified this option can be used to
    specify the location and name of the target (signed) CAP-file.
```

CAP-file File name of the authorized CAP-file.

applet-AID AID of the applet in the package to be installed.

CAP-file CAP file

tokenspec

Specify the token holding the private key to be used for signature generation.

Possible token definitions are: "<PKCS#12-file>" | "<PKCS#8-file>" | "<hex-signature>".
You can also pass the signature directly.

For ECDSA algorithm, EC private key has to be provided in a .pem file containing these headers:

```
-----BEGIN EC PRIVATE KEY-----
-----END EC PRIVATE KEY-----
```

Install token generation:

For initializing the EC curve parameters for ECDSA, Initialize the following global variables with hex strings using /set-var command, else 'P-256' curve will be used.

```
ECC_CURVEPARAM_P - The prime p
ECC_CURVEPARAM_A - EC curve coefficient a
ECC_CURVEPARAM_B - EC curve coefficient b
ECC_CURVEPARAM_G - The base point G(Gx,Gy)
ECC_CURVEPARAM_N - The order of G
```

HashAlgorithm

It specifies whether to use SHA1/ SHA-256 digest for signature generation.

Authorize a package in a CAP-file for delegated installation of an applet of this package.
The resulting Install Token information is stored in the CAP-file.

Example:

```
/cap-inst-auth -i 1122334455667788 |myTestApp "S:\test.cap"
"S:\Privatekey.pem" SHA-256
```

```
/cap-inst-auth 1234456789 test_signed.cap
1122334455667788990011223344556611223344556677889900112233445566112233445
5667788990011223344556611223344556677889900112233445566112233445566778899
0011223344556611223344556677889900112233445566112233445566778899001122334
4556611223344556677889900112233445566
```

```
/cap-info test_signed.cap
CAP file name      : test_signed.cap
CAP file version   : 2.2
Java package name  : test_Package
Internal pkg. name : test_Package
```

```

21      ----non relevant parts deleted----
Load Token(s)      :      none
Install Token(s)   :
    Install Token   :      1
    Token           :      11223344556677889900112233445566
                        11223344556677889900112233445566
                        11223344556677889900112233445566
                        11223344556677889900112233445566
                        11223344556677889900112233445566
                        11223344556677889900112233445566
                        11223344556677889900112233445566
                        11223344556677889900112233445566
                        11223344556677889900112233445566
    Package AID     :      746573745061636B616765
    Applet AID      :      1234456789
    Instance AID    :      default (applet AID)
    Install Param.  :      none
    Privileges      :      none

```

Note: Please refer to [Section 8](#) for generation of .pem files.

Refer [Section 11](#) to know about storage of tokens in Cap file.

5.9 /cap-load-auth

usage: /cap-load-auth [-p|--package package-name][-i|--tokenpin PIN][-s|--sdaid AID][-l|--params parameters][-t|--target CAP-file] CAP-file tokenspec HashAlgorithm

-p|--package package-name

Specify the Java package name to search for in the CAP-file. If this option isn't set, the first package found will be authorized for delegated loading.

package-name

Java package name.

-i|--tokenpin PIN

Specify the PIN to open the token. Only used if token is PIN protected.

PIN PIN to open the token referenced by tokenspec.

-s|--sdaid AID

Specify the AID of the Security Domain to be associated with the package and it's applications (Card Manager is default).

AID AID of Security Domain.

-l|--params parameters

Load parameters

parameters Load parameters in raw (TLV) format.

-t|--target CAP-file

If the CAP-file to be authorized shall not be modified this option can be used to specify the location and name of the target (signed) CAP-file.

CAP-file File name of the authorized CAP-file.

CAP-file CAP file

tokenspec

Specify the token holding the private key to be used for signature generation.

Possible token definitions are: "<PKCS#12-file>" | "<PKCS#8-file>" | "<hex-signature>".
You can also pass the signature directly.

For ECDSA algorithm, EC private key has to be provided in a .pem file containing these headers:

-----BEGIN EC PRIVATE KEY-----

-----END EC PRIVATE KEY-----

Load token generation:

For initializing the EC curve parameters for ECDSA, Initialize the following global variables with hex strings using /set-var command, else 'P-256' curve will be used.

ECC_CURVEPARAM_P - The prime p

ECC_CURVEPARAM_A - EC curve coefficient a

ECC_CURVEPARAM_B - EC curve coefficient b

ECC_CURVEPARAM_G - The base point G(Gx,Gy)

ECC_CURVEPARAM_N - The order of G

HashAlgorithm

It specifies whether to use SHA1/ SHA-256 digest for signature generation.

Authorize a package in a CAP-file for delegated loading. The resulting Load Token information is stored in the CAP-file.

Example:

```
/cap-load-auth test_signed.cap
1122334455667788990011223344556611223344556677889900112233445566112233445
5667788990011223344556611223344556677889900112233445566112233445566778899
0011223344556611223344556677889900112233445566112233445566778899001122334
4556611223344556677889900112233445566 SHA-256
/cap-info test_signed.cap
CAP file name      : test_signed.cap
CAP file version   : 2.2
Java package name  : test_Package
Internal pkg. name : test_Package
22      ---non relevant parts deleted---
Load Token(s)      :
    Load Token     : 1
    Token           : 11223344556677889900112233445566
                      11223344556677889900112233445566
                      11223344556677889900112233445566
                      11223344556677889900112233445566
                      11223344556677889900112233445566
```

```

11223344556677889900112233445566
11223344556677889900112233445566
11223344556677889900112233445566
DAP (hash)      : 466266086428143FD8DC93DA0E23E9B1ED673037
Security Domain : default (CardManager)
Load Param.     : none
Package AID     : 746573745061636B616765
Install Token(s) : none

```

Note: Please refer to [Section 8](#) for generation of .pem files.

Refer [Section 11](#) to know about storage of tokens in Capfile.

5.10 /cap-reg-update

usage: /cap-reg-update [-n|--tokenpin PIN][-e|--delegation][-l|--cm-lock][-t|--terminate][-d|--default][-p|--pin-change][-s|--security-domain][-b|--sd-dap][-m|--mandated-dap][-r|--trusted-path][-a|--auth-manage][-h|--token-manage][-j|--global-delete][-u|--global-lock][-w|--global-registry][-f|--final-application][-z|--global-service][-v|--receipt-gen][-x|--ciphered][-c|--contactless][-g|--contactlessself][-q|--regupdate-param params][--privilege][-k|--extradite][-y|--sd sdAID][-i|--ap appAID][-o|--target CAP-file] CAP-file tokenspec Hash Algorithm

- n|--tokenpin PIN
Specify the PIN to open the token. Only used if token is PIN protected.
PIN PIN to open the token referenced by tokenspec.
- e|--delegation
Security Domain with delegated management.
- l|--cm-lock
Card Manager lock permission.
- t|--terminate
Card terminate permission.
- d|--default
Implicit selectable (default) applet.
- p|--pin-change
PIN change permission.
- s|--security-domain
Security Domain privilege.
- b|--sd-dap
Security Domain with DAP verification.
- m|--mandated-dap
Security Domain with mandated DAP verification.
- r|--trusted-path
Trusted Path permission.
- a|--auth-manage
Authorized Management permission.
- h|--token-manage
Token Management permission.
- j|--global-delete
Global Delete permission.
- u|--global-lock
Global Lock permission.
- w|--global-registry
Global Registry permission.
- f|--final-application

Final Application permission.
-z|--global-service
Global Service permission.
-v|--receipt-gen
Receipt generation permission.
-x|--ciphred
Ciphred Load File Data Block permission.
-c|--contactless
Contactless Activation permission.
-g|--contactlessself
Contactless Self-Activation permission.
-q|--regupdate-param params
Install for Registry Update parameters in TLV format (VOP/GP):
0xEF - system specific parameters:
0xCF - Implicit selection parameter
0xCB - Global service parameters
0xD9 - Restrict parameter, lists the functionalities that shall be disabled
0xB6 - Control Reference Template for Digital Signature (Token)
0x42 - Identification Number of the Security Domain with the Token
Verification privilege
0x45 - Image Number of the Security Domain with the Token Verification
privilege
0x5F20 - Application Provider identifier
0x93 - Token identifier/number (digital signature counter)

params Install for Registry Update parameters (TLV format).
--privilege
Include privilege bytes in cmd.
-k|--extradite
Perform extradition.
-y|--sd sdAID
Security Domain to associate the package/application(s) with.
sdAID AID of the Security Domain to which the applet is
to be extradited.
-i|--ap appAID
AID of the applet instance
appAID AID of the applet instance.
-o|--target CAP-file
If the CAP-file to be authorized shall not be modified this option can be used to
specify the location and name of the target (signed) CAP-file.
CAP-file File name of the authorized CAP-file.
CAP-file CAP file
tokenspec Specify the token holding the private key to be used for signature generation.
Possible token definitions are: "<PKCS#12-file>" | "<PKCS#8-file>" | "<hex-signature>".
You can also pass the signature directly.
For ECDSA algorithm, EC private key has to be provided in a .pem file containing
these headers:
-----BEGIN EC PRIVATE KEY-----
-----END EC PRIVATE KEY-----

Registry update token generation:
For initializing the EC curve parameters for ECDSA, Initialize the following global
variables with hex strings using /set-var command, else 'P-256' curve will be used.
ECC_CURVEPARAM_P - The prime p
ECC_CURVEPARAM_A - EC curve coefficient a
ECC_CURVEPARAM_B - EC curve coefficient b

ECC_CURVEPARAM_G - The base point G(Gx,Gy)
 ECC_CURVEPARAM_N - The order of G
 Hash Algorithm
 Specifies whether SHA1/ SHA-256 digest to be used .

Authorize a package in a CAP-file for modifying the GlobalPlatform Registry data associated with an Application, such as Privileges and Implicit Selection parameters. The resulting Registry Update Token information is stored in the CAP-file .

Example:

```
/cap-reg-update -q EF#(A00484022000) -i SHA2 "S:\test.cap"
"S:\WritePrivatekey.pem" SHA-256
```

Note: Please refer to [Section 8](#) for generation of .pem files.

Refer [Section 11](#) to know about storage of tokens in Capfile.

5.11 /cap-sign

usage: /cap-sign [-d|--debug][--p|--package package-name][--i|--tokenpin PIN][--t|--target CAP-file][--a|--algorithm DESVOP201 | DESGP211 | RSAVOP201 | RSAGP211 | ECDSA | RSAGPSE] CAP-file AID tokenspec HashAlgorithm

-d|--debug If this option is set, the DESCRIPTOR/DEBUG components are included in the signature.

--p|--package package-name

Specify the Java package name to search for in the CAP-file. If this option isn't set, the first package found will be signed.

package-name

Java package name.

--i|--tokenpin PIN

Specify the PIN to open the token. Only used if token is PIN protected.

PIN PIN to open the token referenced by tokenspec.

--t|--target CAP-file

If the CAP-file to be signed shall not be modified this option can be used to specify the location and name of the target (signed) CAP-file.

CAP-file File name of the signed CAP-file.

--a|--algorithm DESVOP201|DESGP211|RSAVOP201|RSAGP211|ECDSA|RSAGPSE

Defines whether the DAP generation algorithm defined in VOP 2.0.1 or the one defined in Global Platform 2.1.1 is to be used. The default value is RSAVOP201.

DESVOP201|DESGP211|RSAVOP201|RSAGP211|ECDSA|RSAGPSE

DAP generation algorithm:

DESVOP201 - DES Scheme takes directly input (no hash) for generation of DAP.

DESGP211 - DES Scheme takes hash of input for generation of DAP.

RSVOP201 - RSA scheme as defined in GP Spec 2.0.1 and takes only hash of input for generation of signature.

RSAGP211 - RSA RSA scheme as defined in GP Spec 2.1.1 and takes hash of "hash of input" for generation of signature.

ECDSA - For initializing the EC curve parameters for ECDSA, Initialize the following global variables with hex strings using /set-var command, else 'P-256' curve will be used.

ECC_CURVEPARAM_P - The prime p

ECC_CURVEPARAM_A - EC curve coefficient a

ECC_CURVEPARAM_B - EC curve coefficient b

ECC_CURVEPARAM_G - The base point G(Gx,Gy)

ECC_CURVEPARAM_N - The order of G

RSAGPSE - RSA scheme with SHA-256 as mentioned in GP SE Config spec.

CAP-file CAP file

AID AID of Security Domain to verify this signature.

tokenspec

Specify the token holding the key to be used for signature generation.

In case of PK DAP possible token definitions are: "<PKCS#12-file>" | "<PKCS#8-file>" | "<hex-signature>". For symmetric DAP the tokenspec must be the 16 byte DES key (as HEX string) to be used for DAP generation. You can also pass the signature directly.

JCShell supports PKCS#8 token containing RSA private key in PEM|DER file format. For examples to generate the pkcs8 token file using OpenSSL, refer section [Generate PKCS#8 token file](#).

HashAlgorithm

It specifies whether to use SHA1/ SHA-256 digest for signature generation.

Sign a package in a CAP-file. The resulting DAP information is stored in the CAP-file.
Note: Signing a CAP-file implies invalidating (removing) all Load/Install Tokens in the CAP-file.

Signing of Ciphered CAP-file considers the LFDBH from meta-inf folder.

Example 1:

```
/cap-sign -a ECDSA "S:\test.cap" 112233445567 "S:\WritePrivatekey.pem" SHA-256
```

```
/cap-sign --target test_signed.cap -a GP211 test.cap 1234456789
707172737475767778797a7b7c7d7e7f SHA-256
/cap-info test_signed.cap
----non relevant parts deleted----
DAP block(s) :
DAP block : 1
AID : 1234456789
Signature : 55E23EBAF6FCE3BE
```

```

Algorithm      : GP 2.1.1
Load Token(s)  : none
Install Token(s) : none

```

Note:

The EC private key used in ECDSA DAP generation has to be provided in the PKCS#8 format in a ".pem" file containing the following headers. -----BEGIN EC PRIVATE KEY----- and -----END EC PRIVATE KEY-----.

E.g:

```

-----BEGIN EC PRIVATE KEY-----
V5UlxhwWiCuJpdvLf/frosflxsGkGG8vQtMww1J0hAQ=
-----END EC PRIVATE KEY-----

```

Please refer to [Section 8](#) for generation of .pem files.

Example 2: DAP signing with PKCS#8 token

```

/cap-sign -i tokenpin -t "S:\TARGET_CAP_FILE\testresult.cap" -a RSAGPSE
"S:\PLAIN_CAP_FILE\test.cap" 112233445577 "S:\PKCS8_TOKEN\rsaprivatekey.pem" SHA-256

```

OR

```

/cap-sign -i tokenpin -t "S:\TARGET_CAP_FILE\testresult.cap" -a RSAGPSE
"S:\PLAIN_CAP_FILE\test.cap" 112233445577 "S:\PKCS8_TOKEN\rsaprivatekey.der" SHA-256

```

Refer [Section 11](#) to know about storage of DAP blocks in Capfile.

5.12 /cap-patch

*usage: /cap-patch **[-f|--file CAP-file]****[-t|--target CAP-file]****[-p|--package package-name]****[-c|--component component-name]****[-h|--search search-string]****[-r|--replace replace-string]***

-f|--file CAP-file

Specify the source CAP-file.

CAP-file Source CAP file

-t|--target CAP-file

If the CAP-file to be patched shall not be modified this option can be used to specify the location and name of the target (patched) CAP-file.

CAP-file File name of the patched CAP-file.

-p|--package package-name

Specify the Java package name to search for in the CAP-file. If this option isn't set, the first package found will be patched.

package-name

Java package name.

-c|--component component-name

Specify the CAP component name to search for in the CAP-file. If this option isn't set, the StaticField.cap component will be patched.

component-name

CAP component name.

-h|--search search-string

Specify the search string to search for in the CAP-file.

search-string

Search string.

-r|--replace replace-string

Specify the replace string to patch in the CAP-file.

replace-string

Replace patch string.

Patch a package in a CAP-file. The resulting patch is stored in the CAP-file.

Example 1:

```
/cap-patch -f "S:\test.cap" -t -h 0102030405060708090A -r 1112131415161718191A
```

Example 2:

```
/cap-patch -f "S:\test.cap" -t "S:\testModified.cap" -h 0102030405060708090A -r 1112131415161718191A
```

Example 3:

```
/cap-patch -f "S:\test.cap" -t "S:\testModified.cap" -c "Method.cap" -h 0102030405060708090A -r 1112131415161718191A
```

5.13 /card

usage: /card [-d|--nad <node address>][-t|time time[s|ms]][-c|card-manager class-name][-a|card-manager-aid aid][-v|--verbose][-n|--noatr]

-d|--nad <node address>

Indicates the node address.

NAD is applicable only when Remote Terminal is opened.

<node address>

node address

-t|time time[s|ms]

Wait specified time until card is inserted.

time[s|ms] Time to wait for card.

-c|card-manager class-name

Defines which off-card Card Manager implementation to use.

class-name Class which implements the Card Manager (subclassed
com.nxp.id.jc.CardManager).

-a|card-manager-aid aid

Desired Card Manager AID to be used.

aid Card Manager AID

-v|--verbose

Print some FCI information.

-n|--noatr

No ATR. Only select Card Manager

Issue /atr and select the default card manager via the default logical channel.

The default Card Manager AID is selected as per following table:

Mask Id	AID
0 – 58	0xA000000003000000
59 and above	0xA000000015100000

This command implicitly registers the Card Manager plugin referencing the desired Card Manager implementation with the JCShell under the nickname "cm".

Note: To activate card manager plugin explicitly, type below commands in jcshell terminal-

```
/register cm 00112233445566 Card Manager
/app
```

Example:

```
23  /card
24  resetCard with timeout: 0 (ms)
25  --Waiting for card...
26  ATR=3B FA 13 00 00 81 31 FE 45 4A 43 4F 50 34 31 56 ;.....1.EJCOP41V
27      32 33 31 97 231.
28  => 00 A4 04 00 09 A0 00 00 01 67 41 30 00 FF .....gA0..
29  <= 00 00 01 00 00 00 00 00 74 6F 6F 6C 73 20 00 00 .....tools ..
30      6A 82 j.
31  Status: File not found
32  => 00 A4 04 00 07 A0 00 00 00 03 00 00 00 .....
33  <= 6F 65 84 08 A0 00 00 00 03 00 00 00 A5 59 9F 65 oe.....Y.e
34      01 FF 9F 6E 06 40 51 70 92 00 00 73 4A 06 07 2A ...n.@Qp...sJ..*
35      86 48 86 FC 6B 01 60 0C 06 0A 2A 86 48 86 FC 6B .H..k.'...*.H..k
36      02 02 01 01 63 09 06 07 2A 86 48 86 FC 6B 03 64 ....c...*.H..k.d
```

```

37      0B 06 09 2A 86 48 86 FC 6B 04 02 15 65 0B 06 09      ...*.H..k...e...
38      2B 85 10 86 48 64 02 01 03 66 0C 06 0A 2B 06 01      +...Hd...f...+..
39      04 01 2A 02 6E 01 02 90 00                          ..*.n....
40      Status: No Error

```

5.14 /channel

usage: /channel [<0-3>]

<0-3> Logical channel to switch to.

ISO 7816-4/Global Platform MANAGE CHANNEL command.

Either opens the next available supplementary logical channel or closes the current logical channel.

Upon opening a channel the new channel becomes the current channel.

Upon closing a supplementary channel the default channel (zero) becomes the current channel.

If no channel parameter is provided, this command only lists the logical channels currently open. The channel with the "*" is the one currently active within the JCShell.

Example:

```

41  /channel
42  Channel  Nickname  AID                                Plugin
43  -----  -
44      0                                cm                                A000000003000000
      com.nxp.id.jc.tools.CardManagerPlugin
45  * 1      ----- no applet plugin -----
46  /channel      0
47  Channel  Nickname  AID                                Plugin
48  -----  -
49  *      0                                cm                                A000000003000000
      com.nxp.id.jc.tools.CardManagerPlugin
50      1      ----- no applet plugin -----

```

5.15 /clear-vars

usage: /clear-vars

Clear variable pool

Example:

```
/list-vars
```

```
Global:_____
    path=./scripts
/clear-vars
/list-vars
Global:_____
```

5.16 /close

usage: /close

Close connection to terminal.

5.17 /change-protocol

usage: /change-protocol [-t|--pt *PROTOCOL_TYPE*][-m|--pm *PROTOCOL_MEDIA*][-n|--nad <node address>]

-t|--pt *PROTOCOL_TYPE*

Protocol Type provided in hex, in Range <0x0-0xF>

-m|--pm *PROTOCOL_MEDIA*

Protocol Media provided in hex, in Range <0x0-0xF>

-n|--nad <node address>

Indicates the node address.

NAD is applicable only when Remote Terminal is opened.

<node address>

node address

Changes the Simulator behavior

Changes the return value of APDU.getProtocol() call

Valid Values are as:

PROTOCOL_MEDIA	PROTOCOL_TYPE	Description
0x0	0x0	T=0, contact mode
0x0	0x1	T=1, contact mode, block mode
0x8	0x0	T=CL, TYPE_A, T0
0x8	0x1	T=CL, TYPE_A, T1
0x9	0x0	T=CL, TYPE_B, T0
0x9	0x1	T=CL, TYPE_B, T1

Example:

```
/change-protocol -m 0x0 -t 0x0
```

NOTE: /change-protocol command is supported for remote connection to Simulator

NOTE: T=CL is not supported by the simulator version above 3.0.

Note: Support of protocols and media depends on Simulator version. Error message will be shown, if simulator does not support protocol from above table.

5.18 /echo

usage: /echo[-0|1|2|3] args..

Arguments to be output/echoed.

Example:

```
/echo test
test
```

5.19 /error

usage: /error args..

Abort execution of DEFUN or SCRIPT and set arguments as error string ({last.error} - can be checked in try/catch).

Example:

```
/error test
jcshell: ./scripts/error-test.jcsh[1]: test
```

5.20 /exchange-warmreset

usage: /exchange-warmreset [-t|--time warm reset time] apdu

-t|--time warm reset time

warm reset time option. Warm reset time in usec

apdu Data syntax: HEX[CHARS[HEX...]]

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The 'l' switches modes (HEX/CHARS). Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replaced by the number bytes enclosed in the matching braces.

Examples:

```
"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64
```

```
"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF
```

This command sends an APDU command via the ISO7816 UART to the DUT and waits for an APDU response from the DUT. The time out values are according the previously activated protocol. By means of the additional parameter 'warmreset_time_us' the so-called 'anti-tear' feature can be activated. After reaching the time, defined by 'warmreset_time_us', the RSTN pin is pulled down, triggering a warm reset of DUT.

5.21 /exchange-poweroff

usage: /exchange-poweroff [-g|--gate [0|1|2|3]][-t|--time power-off time][apdu]

-s|--scp Option to use existing SCP session.

-g|--gate [0|1|2|3]

To switch the SWP gates in SCR Terminal. SWP gate values can be either :

- 0 - Card Emulation Type A
- 1 - Wired mode
- 2 - Card Emulation Type B
- 3 - Card Emulation Type F

-t|--time power-off time

power-off time in usec

apdu Data syntax: HEX[|CHARS[|HEX...]]

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The '|' switches modes (HEX/CHARS).

Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replaced by the number bytes enclosed in the matching braces.

Examples:

```
"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64
```

```
"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF
```

This command sends an APDU command via the SWP block to the DUT and waits for an APDU response from the DUT. The time out values are according the previously activated protocol. By means of the additional parameter "power_off_time_us" the so-called "anti-tear" feature can be activated. Supports both SWP and DWP Interface.

Example:

```
/term SCR:DWP:2:5.0:2:0:1:180000:1.8:0.5
```

```

/atr
/card
/exchange-poweroff -g 1 -t 3000 00A4040000
>> 00 A4 04 00 00          .....
jcshell: Error code: -6 (Card terminal error)
jcshell: Communication problems: No response after sending APDU;
[Terminal error. Exchange failed;]
/atr
/exchange-poweroff -g 1 -t 30000 00A4040000
>> 00 A4 04 00 00          .....
(139684 usec)
<< 6F 10 84 08 A0 00 00 01 51 00 00 00 A5 04 9F 65   o.....Q.....e
01 FF 90 00          ....

```

5.22 /exec

*usage: /exec **[[**-a**]**--async**]**[-s|--silent**]**[-t|--store**]**[-l|--label *label*][-i|--in *label param*][-k|--kill]**]****[***command* *[argument..]****]***

-a|--async Start process asynchronously

-s|--silent

Suppress process output

-t|--store *s*Tores process output in variables *exec.output.1*, *exec.output.2* .. using space as delimiter

-l|--label *label*

Attach this label to identify process

-i|--in *label param*

pass the parameter to asynchronous process identified by this label.

-k|--kill Kill process - pass label as command name.

Omitted command kills all processes.

argument.. Command and list of arguments to pass on to process

Omitted command/args kills or lists all processes

Execute/kill/wait a subprocess. Path of subprocess will be taken from *exec.dir* variable. *exec.output* variable will contain the text output of the process. *exec.errorlog* will contain the error stream of the process if any.

Example:

```

/exec test.bat
P:\JCShell_Test>set jahr=2007

```

```
P:\JCShell_Test>set monat=10
P:\JCShell_Test>set tag=29
P:\JCShell_Test>echo 2007-10-29
2007-10-29
```

5.23 /expr

usage: /expr args..

Arguments are evaluated as expression and the result is set as return value. E.g. X=\$(/expr \${X} + 1)

Example:

```
/echo $(/expr 1 + 1)
2
```

5.24 /get-transaction-time

usage: /get-transaction-time

Returns the elapsed time in us since the transaction timer has been reset last. If supported by the currently connected device, a device timer is used. If not, a transaction timer based on the system clock is taken instead. The transaction time will additionally be stored in the variable 'timer.execution.time'. Please note that on the JCOP Simulator the number of Java byte codes is returned if supported.

Example:

```
> /get-transaction-time
Transaction time: 54822 JavaCard byte codes
> /echo ${timer.execution.time}
54822
```

5.25 /glob

usage: /glob pattern..

Collect all file name matching given patterns and as return value. E.g. X[*]=\$(/glob *.jcsch)

Example:

```
/echo $(/glob jc*.rc)
jcshell.rc
```

5.26 /identify

usage: identify [-q]

Example:

```
/identify
=> 00 A4 04 00 09 A0 00 00 01 67 41 30 00 FF .....gA0..
<= 00 00 01 00 00 00 00 00 74 6F 6F 6C 73 20 00 00 .....tools ..
    6A 82 j.
Status: File not found
FABKEY ID: 0x00
PATCH ID: 0x00
TARGET ID: 0x01 (generic)
MASK ID: 0x00 (0)
CUSTOM MASK: 00000000
MASK NAME: tools
FUSE STATE: not fused
ROM INFO: -
COMBO NAME: generic-m00-tools
```

5.27 /list-readers

usage: /list-readers

Display the list of the Readers currently connected to System.

Example:

```
/list-readers
-----
Name ( Type )
-----
Broadcom Corp Contacted SmartCard 0 ( PCSC )
MF RD 70x (USB) ( T=CL )
```

5.28 /list-vars

usage: /list-vars [[pattern]!pattern]..]

[pattern]!pattern]..

List only variables matching the listed patterns.

A leading ! means the pattern must not match the variable name.

List matching/all variables.

Example:

```
/list-vars
Global:-----
      argv=
      path=./scripts
      script.dir=.
      script.name=jcshell.rc
      test=12345
Script Level 1:-----
      argv=
      path=./scripts
      script.dir=.
      script.name=jcshell.rc
      test=12345
```

5.29 /manage-channel

usage: /manage-channel open|close

open|close Channel management operation.

ISO 7816-4/Global Platform MANAGE CHANNEL command.

Either opens the next available supplementary logical channel or closes the current logical channel.

Upon opening a channel the new channel becomes the current channel.

Upon closing a supplementary channel the default channel (zero) becomes the current channel.

Example:

```
/manage-channel open
=> 00 70 00 00 01      .p...
<= 01 90 00           ...
Status: No Error
/channel
Channel  Nickname  AID      Plugin
-----  -
0              cm      A0000000003000000
com.nxp.id.jc.tools.CardManagerPlugin
* 1          ----- no applet plugin -----
```

5.30 /mode

usage: /mode [*internal=on|off*][*trace=on|off|num*][*echo=on|off*][*remote-echo=on|off*][*verbose=on|off*][*debug=on|off*][*commDetails=on|off*]

internal=on|off

Print debug messages.

trace=on|off|num

Turn APDU tracing on/off or log at most

num lines of each APDU

echo=on|off

Echo command lines.

remote-echo=on|off

Send echo output to the remote server.

verbose=on|off

Turn verbose mode on/off

debug=on|off

Turn debug mode on/off

commDetails=on|off

Turn communication details mode on/off

Example:

```
/mode echo=on trace=on
/mode
    trace=on
    echo=on
    remote-echo=on
    debug=off
    commDetails=off
    verbose=off
    internal=off
```

5.31 /printf

usage: /printf [-0|1|2|3] *format args..*

Format args according to format specification (like C printf).

%x %X : format argument as lowercase/uppercase hexadecimal number

%d %u : format argument as decimal number (signed/unsigned)

%s : format argument as a string

%Ns : N is a decimal number or '*', specifies padding width
* reads padding from respective argument list

%.Ms : M is a decimal number or '*', specifies maximum item length.
* reads max length from respective argument list

%N.Ms : truncate string to M chars and pad to length N

%-Ns : string is left aligned, padding to the right

%+Ns : string is right aligned, padding to the left (default if +/- is missing)

%|Ns : string is centered, padding on both sides

%0Ns : pad with zero characters

%[C]Ns : pad with C character, C can be any character

\[rnt] : replace with carriage return, line feed, and tab

Example:

```
/printf "%[-]|9.*s\n" 3 abcdef
---abc---
```

5.32 /r-echo

usage: /r-echo args..

Echo arguments to be output (remotely).

Note: /r-echo command is supported only for remote simulator.

Example:

```
/r-echo foo
foo
```

5.33 /register

usage: /register [name aid [plugin]]

name Nick name of the applet

aid AID data (syntax see :send)

plugin A fully qualified class name specifying a class

implementing com.nxp.id.jc.tools.JCShellPlugin interface.

If omitted com.nxp.id.jc.tools.OPAppletPlugin is used.

The class com.nxp.id.jc.tools.<plugin>Plugin is also checked

if plugin does not contain any dots.

List or introduce an applet to the shell.

Example:

```
/register ah A000000003000000
/register
Nickname      AID                                     Plugin
-----      -
cm                                                     A000000003000000
com.nxp.id.jc.tools.CardManagerPlugin
ah                                                     A000000003000000
com.nxp.id.jc.tools.OPAppletPlugin
```

Example 2:

```
/register ah 0011223344556677
/register
Nickname      AID                                     Plugin
-----      -
cm                                                     A0000000151000000
com.nxp.id.jc.tools.CardManagerPlugin
ah                                                     0011223344556677
com.nxp.id.jc.tools.OPAppletPlugin
```

5.34 /reset

usage: /reset [[-i|interfaceReset [1|2|3]][-w|warmReset][-c|powerOnReset]][-b|boardReset][-t|time time[s|ms]][-n|--nad <node address>][-p|--seqNumI2C][-s|--softI2C][-f|nfcReset][-m|nfccPowerMode [0|1]]]

-i|interfaceReset [1|2|3]

Perform warm reset of secondary interface from contact interface.

[1|2|3] 01 - Contact interface
02 - Contactless Interface
03 - Both contact and contactless

-w|warmReset

Warm Reset

-c|powerOnReset

Power on reset (Cold ATR)

-b|boardReset

Perform board reset

-t|time time[s|ms]

Wait specified time until card is inserted.

time[s|ms] Time to wait for card.
 -n|--nad <node address>
 Indicates the node address.
 NAD is applicable only when Remote Terminal is opened.
 <node address>
 node address
 -p|--seqNumI2C
 T1I2C-Protocol Reset
 -s|--softI2C
 T1I2C-Soft Reset
 -f|nfcReset
 Send nfc reset interrupt
 -m|nfccPowerMode [0|1]
 Configure the NFCC in low or full power mode
 [0|1] 0 - Low Power Mode
 1 - Full Power Mode

Reset inserted card and get ATR.
 This command should be executed between opening a terminal and sending other card-related commands.

Example:

```
> /reset
resetCard with timeout: 0 (ms)
--Waiting for card...
ATR=3B 8F 80 01 4A 43 4F 50 33 20 50 37 33 20 56 36 ;...JCOP3 P73 V6
    20 20 20 3F ?
IOCTL().
ATR:
    T = 0
    T = 1
```

Please refer to section 5.2.1 for the behavior of /atr and /reset.

5.35 /reset-transaction-time

usage: /reset-transaction-time

Resets the transaction timer. If possible with the currently connected device, the device transaction timer will be reset. If not, a transaction timer based on the system clock is used instead. To get the current value of the transaction timer, use /get-transaction-time

Example:

```
> /reset-transaction-time
```

5.36 /select

usage: /select AID [<0-3>]

AID AID of applet to be selected

<0-3> Target logical channel (current channel is default).

Selects an applet either on the current logical channel or optionally on another, possibly new, logical channel. If the target channel is already open the assigned applet plugin, if any, is reused.

If the select FCI returned by the applet includes a remote object reference and there is no applet plugin selected or the current applet plugin is the Card Manager plugin, then it is automatically switched to the JCRMI plugin.

Example:

```
/select 1234456789
=> 00 A4 04 00 05 12 34 45 67 89 00          .....4Eg..
<= 90 00                                     ..
Status: No Error
```

5.37 /send

usage: /send [-r|--raw][nad=NAD][-t--timeout timeout][-l--le explength][-q|--queryhcuparam][-s|--sethcuparam <gateParam>][apdu [pattern..]]

-r|--raw data is sent in raw mode (not encapsulated in any protocol)

nad=NAD Indicates the node address and APDU is sent to this node address.

NAD is applicable only when Remote Terminal is opened.

NAD node address

-t|--timeout timeout

max time to wait for response data in raw mode.

timeout option is applicable for RAW mode only

timeout time unit is device-dependent

-l|--le explength

length of expected response for raw data sent on I2C

explength expected length of response for raw data sent on I2C

-q|--queryhcuparam

Query HCI parameters

-s|--sethcuparam <gateParam>

Set HCI parameter

<gateParam>

Possible gate parameters across different RF Gates

Supported values for <gateParam> are :

Type A :

mode

uid_reg

sak

atqa

application_on_data

fwi_sfgi

cid_support

clt_support

datarate_max

Type B :

mode

pupi_reg

afi

atqb

higher_layer_response

datarate_max

Type F :

mode

speed_cap

clt_support

apdu Data syntax: HEX[|CHARS[|HEX...]]

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The '|' switches modes (HEX/CHARS). Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replaced by the number bytes

enclosed in the matching braces.

Examples:

"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64

"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF

pattern.. Description of expected responses. Each argument is pattern describing response as sequence of hexdigits with embedded wild cards (*,?,[.]). A leading ! char means NOT. Arguments '&' and '|' are interpreted as the respective logical operators. The response is ok if it matches the pattern and the logical expression.

Send APDU data (using negotiated protocol/channel or raw).

NOTE: For FeliCa, only data shall be provided.

(excluding length and CRC).

The data shall be provided as follows:

/send 0DF402FE010101010101000000

The length and CRC is computed and appended to frame by JCShell.

The reader receives the complete frame, i.e.

0DF402FE0101010101010000004777

Example:

/send 00A4040000

NAD=FF

=> 00 A4 04 00 00

(2376 usec [SYS], 695 JavaCard byte codes [DEV])

<= 6F 10 84 08 A0 00 00 01 51 00 00 00 A5 04 9F 65 o.....Q.....e

01 FF 90 00 Status: No Error

/send -q

/send -s mode FF

5.38 /set-channel

usage: /set-channel [<0-3>]

<0-3> Desired logical channel.

Sets the logical channel number used in any subsequent APDU communication.

The CLA byte of any APDU header sent to the card is automatically modified to reflect the logical channel number as defined in ISO 7816-4. Modification only takes place if the original CLA byte indicates channel zero, otherwise the given channel info is maintained.

By default the logical channel number is zero.

Note: Normally the current channel number is implicitly modified by the commands `/channel` or `/manage-channel` and thus it is not necessary to use this command under normal circumstances.

Example:

```
/channel
Channel  Nickname  AID                               Plugin
-----  -
      0                               cm                               A0000000003000000
com.nxp.id.jc.tools.CardManagerPlugin
*  1          ----- no applet plugin -----
/set-channel 0
```

5.39 /set-var

usage: `/set-var [-d|--def][-g|--global][-q|--quote] varname [expr..]`

`-d|--def` Update/create variable in defining scope. If var not defined use local scope.

`-g|--global`

Update/create variable in global scope

`-q|--quote` Do not evaluate arguments as expression. If multiple arguments create an array.

`expr..` Shell expression

Set specified variable.

Example:

```
/list-vars
Global: _____
      path=./scripts
/set-var test 12345
/list-vars
Global: _____
```

```
path=./scripts
test=12345
```

Note:

There are 2 scopes maintained at JCShell.

- **Global scope**
- **Script level scope**

If the /set-var command is used with the -g option, then the values are stored in **GLOBAL_SCOPE** stack.

If the /set-var command is used without the -g option in an interactive mode (command line input), then the values are stored in **GLOBAL_SCOPE** stack.

If the /set-var command is used without the -g option in JCShell scripts file, then the values are stored in **Script level** stack.

The JCShell maintains a separate stack to store the values for these 2 scopes. One stack is used per JCShell script file.

Variables defined using /set-var without -g option in a JCShell script, then all the variables share the same stack.

For example.

In an interactive mode (command line input).

```
- /set-var -g var1 'global 1'
- /list-var
Global:_____
    var1=global 1
- /set-var var2 'global 2'
- /list-var
Global:_____
    var1=global 1
    var2=global 2
-
```

With scripts execution.

```
script1.jcsh
    /echo 'In script1'
    /set-var -g global 'global in s1'
    /set-var local 'local in s1'
```

```
/list-var
```

```
script2
```

```
/list-var
```

```
script2.jcsh
```

```
/echo 'In script2'
```

```
/set-var -g global 'global in s2'
```

```
/set-var local 'local in s2'
```

```
/list-var
```

```
script3
```

```
script3.jcsh
```

```
/echo 'In script3'
```

```
/set-var -g global 'global in s3'
```

```
/set-var local 'local in s3'
```

```
/list-var
```

Output:

```
- /set-var path "D:\scripts"
```

```
- script1.jcsh
```

```
In script1
```

```
Global:_____
```

```
global=global in s1
```

```
Script Level 1:_____
```

```
local=local in s1
```

```
In script2
```

```
Global:_____
```

```
global=global in s2
```

```
Script Level 1:_____
```

```
local=local in s1
```

```
Script Level 2:_____
```

```
local=local in s2
```

```
In script3
```

```
Global:_____
```

```
global=global in s3
```

```
Script Level 1:_____
```

```

        local=local in s1
Script Level 2: _____
        local=local in s2
Script Level 3: _____
        local=local in s3
In script1
Global: _____
        global=global in s3
Script Level 1: _____
        local=local in s1
-

```

5.40 /set-random-seed

usage: /set-random-seed [[-s|--rseed rseed]]

-s|--rseed rseed

Initializes the internally used random number generators to the given seed

rseed random seed

Command to set the random seed.

Note: This command shall be used only for the purpose of analyzing the issues and so to allow for reproduction.

Example:

```
/set-random-seed -s 1234567890
```

```
Random number generators RNG initialization from --rseed option initialized with seed
00000000499602D2
```

```
/set-random-seed
```

```
Random number generators No --rseed option found, RNG initialized with random seed
EBA8F0E3FF7A62FD
```

5.41 /sleep

usage: /sleep time[s|ms]

time[s|ms] Time to sleep.

Delay execution for given amount of time.

Example:

```
/sleep 1000ms
/sleep 1      ----seconds is default----
```

5.42 /start-timer

usage: /start-timer *[[-i|--id timer id]]*

-i|--id timer id

<1-F> Unique Id for timer, provided in hex, Optional

Starts a single timer

If no id is given, a default id will be taken and stored in '*timer.id*' variable and corresponding timer will be started.

Total **time elapsed** in microseconds will be stored in '*timer.execution.time*' and '*timer.system.time*' variables in decimal format. '*timer.execution.time*' is deprecated and will be removed in future release. This total time elapsed is the system time, elapsed between the /start-timer and /stop-timer commands.

Total **processing time** will be stored in '*timer.device.time*' variable in decimal format. This total processing time is the processing time of all the APDUs between the /start-timer and /stop-timer commands.

Example:

```
/start-timer -i 0xE
```

5.43 /stop-timer

usage: /stop-timer *[[-i|--id timer id]]*

-i|--id timer id

<1-F> Unique Id for timer, provided in hex, optional

Stops a single timer

If no id is given, id stored in '*timer.id*' will be taken and corresponding timer will be stopped.

Total **time elapsed** in microseconds will be stored in '*timer.execution.time*' and '*timer.system.time*' variables in decimal format. '*timer.execution.time*' is deprecated and will be removed in future release. This total time elapsed is the system time, elapsed between the /start-timer and /stop-timer commands.

Total **processing time** will be stored in '*timer.device.time*' variable in decimal format. This total processing time is the processing time of all the APDUs between the /start-timer and /stop-timer commands.

Example:

```
/stop-timer -i 0xE
```

5.44 /start-scr-timer

usage: /start-scr-timer SWP|SPI|ISO7816|SCI2C
SWP|SPI|ISO7816|SCI2C SCR Interface ID

To start the timer and to get the execution time of the multiple commands sent via SWP or SPI or ISO7816 or SCI2C in SCR Terminal.
Interface IDs can be either DUAL_COMM_SPI or DUAL_COMM_SWP or ISO7816 or SCI2C

Possible values: SWP|SPI|ISO7816|SCI2C

Example: start-scr-timer SWP

5.45 /stop-scr-timer

usage: /stop-scr-timer

To stop the SCR timer and to get the execution time of the multiple commands sent via SWP or SPI or ISO7816 or SCI2C in SCR Terminal.

Execution time will be stored in global variable '*scrtimer.exec.time*'.

5.46 /start-mp500-timer

usage: /start-mp500-timer

Starts the MP500 timer and initializes the variable to zero. The variable is used to store the execution time of each command sent to card on MP500 terminal.

Example:

```
/set-var mptimer.exec.time.unit usec
```

```
/start-mp500-timer
```

5.47 /stop-mp500-timer

usage: /stop-mp500-timer

Stops the mp500 timer and displays the time measured for execution of APDU sent to card. The measured time will be stored in special variable "*mp500timer.exec.time*". By default the unit of time measured is in micro seconds. The unit can be modified through special variable "*mptimer.exec.time.unit*".

Example:

```
/stop-mp500-timer
```

```
/echo ${mp500timer.exec.time}
```

5.48 /start-mp300-timer

usage: /start-mp300-timer

Starts the MP300 timer.

Example:

```
/term MP300:USB:MP3.13.15.01:122:1:1
/set-var mptimer.exec.time.unit usec

/start-mpreader-timer
```

5.49 /stop-mp300-timer

usage: /stop-mp300-timer

Stops the MP300 timer based on the reader terminal opened and reads the time measured for execution of APDU sent to card. The measured time will be stored in special variable "*mp300timer.exec.time*". By default the unit of time measured is in micro seconds. The unit can be modified through special variable "*mptimer.exec.time.unit*".

Example:

```
/term MP300:USB:MP3.13.15.01:122:1:1
/start-mpreader-timer
..... Send APDU to card...
/stop-mpreader-timer
/echo ${mptimer.exec.time}
```

5.50 /start-mpreader-timer

usage: /start-mpreader-timer

Starts the MP300/MP500 timer based on the reader terminal opened.

Example:

```
/term MP300:USB:MP3.13.15.01:122:1:1
/set-var mptimer.exec.time.unit usec

/start-mpreader-timer
```

5.51 /stop-mpreader-timer

usage: /stop-mpreader-timer

Stops the MP300/MP500 timer based on the reader terminal opened and reads the time measured for execution of APDU sent to card. The measured time will be stored in special variable "*mptimer.exec.time*". By default the unit of time measured is in micro seconds. The unit can be modified through special variable "*mptimer.exec.time.unit*".

Example:

```
/term MP300:USB:MP3.13.15.01:122:1:1
/start-mpreader-timer
..... Send APDU to card...
/stop-mpreader-timer
/echo ${mptimer.exec.time}
```

5.52 /terminal

usage: /terminal [[-p|--pop]][[term]]]

-p|--pop Pop top most filter

term Specify either the name of a terminal or a name plus some terminal parameter.
The parameter must be separated by a ':' character.

If <term> is defined as 'SCComm', the following parameters

may be given:

- 1 for PCSC reader,
- 2 for TCL reader,

This reader may have following sub-parameters

/term SComm:2[:<reader name>][:<card type>]

<card type> = A 'Card Type A'

B 'Card Type B'

AB 'Card Type AB'

Default card type value is 'AB'

- 3 for OEM reader,
- 4 for the (Keil) simulation

This reader may have the following sub-parameters

/term SComm:4:<simu-type>:<ip-addr>:<portNo>:<Debug>

<simu-type> = 0 TCL (old interface via MFGeneric.dll)

1 P60-CONTACT_TRANS

2 P60-CONTACT_T1

3 P60-CONTACTLESS_TRANS

4 P60-CONTACTLESS_L3

5 P60-CONTACTLESS_TCL

6 I2C

7 SPI

8 Remote

<ip-addr> = This parameter is required.

String containing either 'localhost',
an ip-address like '127.0.0.1' or the
computer name.

Default is 'localhost'.

<portNo> = Number of the port to be used

Default is 8080.

<Debug> = On means that all timeouts are set
to 'infinite'.

This will cause the system not to
throw exceptions even if the
server doesn't respond in time.

Off Use command timeouts

Default is 'Off'.

Example:

SCComm:4:6:localhost:8080:off

5 for the I2C reader.

This reader may have the following sub-parameters

/term SCComm:5:<subType>:<I2C speed>:<I2C Addr (8bit)>

<subType> with the following possible values:
 0 USB Bird with external power supply
 1 USB Bird powering the target
 2 Intel SMB (requires additional DLL)
 default is 1
 <I2C speed> from 1 to 255, defining the communication
 bit rate in kBit/sec
 default is 50kBit/sec
 <I2C Addr (8bit)> I2C slave address in hex/decimal-notation
 default is 0x90

Example:

SCComm:5:0:60:0x84 to select the I2C reader with external
 power, 60kBit/sec and I2C Slave address is 0x84

6 for the SPI reader.

This reader may have the following sub-parameters

/term SCComm:6:<SPI speed>

<SPI speed> from 0 to 6, defining the communication
 bit rate
 0 500kBit/sec
 1 1MBit/sec
 2 2MBit/sec
 3 5MBit/sec
 4 8MBit/sec
 5 10MBit/sec
 6 12MBit/sec
 default is 500kBit/sec

Example:

SCComm:6:1 to select the SPI reader with 1MBit/sec

 SCR ::

If <term> is defined as 'SCR', the following parameters may be given:

Interface	DWP, DWP_IO, SWP, SPI, SWP_SPI, DWP_SPI, ISO7816, I2C, SCI2C or SPI_I2C, VIRTUALISO	Interface type to be used for communication
Protocol	0, 1 or 2	Value 0 selects T=0. Value 1 selects T=1. Value 2 uses an AUTOMATIC detection.
Frequency	F	f is a numeric float value to represent the frequency in MHz (e.g. 5.0 for 5 [MHz])
activation-mode	0, 1 or 2	Value 0 selects BUS. Value 1 selects ACT. Value 2 selects HCI.
power-cycle-enable	0 or 1	Value 0 to disable. Value 1 to enable
target-gate	0 or 1	Value 0 to select card-emulation. Value 1 to select wired-mode.
waiting-time	wt	wt is a integer value to set the waiting-time. Default values: ISO7816 interface – 20000000us Other interfaces - 1000 ms
SWP voltage	1.8	value in V
SWP frequency	e.g. 0.5	value in MHz
SPI bit rate	e.g. 1000	value in kbps
polling mode	0 1 2 3	NAD(TX)=0x00, continuous polling. NAD(TX)=0x00, interval polling (every 1 msec). NAD(TX)=0x5A, continuous polling. NAD(TX)=0x5A, interval polling

	4 5	(every 1 msec). This setting is used as default after DLL startup. NAD(TX)=0x5A, wait on ready line high NAD(TX)=0x5A, wait on ready line low
Block Guard Time (BGT) option (Specific to SPI I2C)	1-250 (Any other value shall set polling to default 1msec)	1-250 Polling Time
Control byte (Specific to I2C)	0-127	RAW I2C control byte value between 0 and 127
IO Mode (Specific to I2C SPI)	0 1 2 3 4 5 6 7 8 9 12 13 14 15	SECIF_NONE SECIF_DWP_IO3IO4 SECIF_SPI_CLKIO1IO2IO3 SECIF_I2C_IO2IO3 SECIF_S2C_IO2IO3 SECIF_I2C_CLKIO1 SECIF_SPI_CLKIO1IO2IO3IO4 SECIF_DWP_TP1TP2 SECIF_RAWI2C_IO3IO4 SECIF_RAWI2C_CLKIO1 SECIF_T1I2C_IO2IO3 SECIF_T1I2C_IO3IO4 SECIF_SPI_CLKIO1IO2TP1TP2 SECIF_RAWI2C_IO2IO3
Reset mode (Specific to I2C SPI)	0 1 2	RST_MODE_ISO7816 RST_MODE_HIGH RST_MODE_LOW
Clock mode (Specific to I2C SPI)	0 1 2 3	CLK_MODE_ISO7816 CLK_MODE_HIGH CLK_MODE_LOW CLK_MODE_NONE

ISS (Specific to T1I2C)	true false	ISS enabled ISS disabled
ISS server (Specific to T1I2C)	server IP address	
ISS port (Specific to T1I2C)	server IP port	
Checksum (Specific to T1I2C)	NAD_LRC NAD_CRC	
GPIO state (Specific to SPI)	HIGH LOW	GPIO state set to HIGH GPIO state set to LOW
T1TYPE	0 – P73 Automotive 1 – SN1XX 2 – SN2XX 3 – SN300 (EOS) 4 – SE050	This option shall choose the corresponding T1 version
RST	SFRAME	Setting this option shall send respective S-Frame for reset command (Warm Cold). C4 – Warm Interface Reset C6 – Cold OS Reset C7 – Get ATR
NAD values (Specific to SPI & I2C)	SCR01_T1I2C_PROTOCOL_SEND_NAD_VALUE SCR01_SPI5_PROTOCOL_SEND_NAD_VALUE	NAD byte values which are used during communication
log file option	1 or 0	1 - enable log, disabled otherwise

Usage ::

```
/terminal
SCR:interface=<value>;protocol=<value>;frequency=<value>;activation_mode=<value>;
power_cycle_enable=<value>;target_gate=<value>;waiting_time=<value>;voltage=<value>;
e::frequency=<value>;bit_rate=<value>;log=<value>
```

ISO7816 ::

Usage ::

```
/terminal
SCR:interface=ISO7816:protocol=<value>:waiting_time=<value>:voltage=<value>:frequency=<value>:log=<value>
```

Example :

```
SCR:interface=ISO7816:protocol=2:waiting_time=2000000:voltage=1.8:frequency=3.0:log=1
```

DWP ::

Usage ::

```
/terminal
SCR:interface=DWP:protocol=<value>:frequency=<value>:activation_mode=<value>:power_cycle_enable=<value>:target_gate=<value>:waiting_time=<value>:voltage=<value>:swp_frequency=<value>:log=<value>
```

Example :

```
SCR:interface=DWP:protocol=2:frequency=5.0:activation_mode=2:power_cycle_enable=0:target_gate=1:waiting_time=180000:voltage=1.8:swp_frequency=0.5:log=1
```

DWP_IO ::

Usage ::

```
/terminal
SCR:interface=DWP_IO:protocol=<value>:frequency=<value>:activation_mode=<value>:power_cycle_enable=<value>:target_gate=<value>:waiting_time=<value>:voltage=<value>:swp_frequency=<value>:log=<value>
```

Example :

```
SCR:interface=DWP_IO:protocol=2:frequency=5.0:activation_mode=2:power_cycle_enable=0:target_gate=1:waiting_time=180000:voltage=1.8:swp_frequency=0.5:log=1
```

SWP ::

Usage ::

```
/terminal
SCR:interface=SWP:protocol=<value>:frequency=<value>:activation_mode=<value>:power_cycle_enable=<value>:target_gate=<value>:waiting_time=<value>:voltage=<value>:swp_frequency:log=<value>
```

Example :

```
SCR:interface=SWP:protocol=2:frequency=5.0:activation_mode=2:power_cycle_enable=0:target_gate=1:waiting_time=180000:voltage=1.8:swp_frequency=0.5:log=1
```

SPI ::

Usage ::

```
/terminal
SCR:interface=SPI:protocol=<value>:bit_rate=<value>:voltage=<value>:swp_frequency=<value>:data_available_line=<value>:gpio=<value>:bgt=<value>:io_mode=<value>:reset_mode=<value>:clock_mode=<value>:nad=<value>:log=<value>
```

Example :

```
SCR:interface=SPI:protocol=2:bit_rate=1000.0:voltage=1.8:swp_frequency=
0.5:data_available_line=4:log=1

SCR:interface=SPI:protocol=2:bit_rate=1000.0:voltage=1.8:swp_frequency=
0.5:data_available_line=4:gpio=HIGH:bgt=0:log=1

SCR:interface=SPI:protocol=2:bit_rate=1000.0:voltage=1.8:swp_frequency=
0.5:data_available_line=4:gpio=HIGH:log=1

SCR:interface=SPI:protocol=2:bit_rate=1000.0:voltage=1.8:swp_frequency=
0.5:data_available_line=4:gpio=LOW:log=1

SCR:interface=SPI:protocol=2:bit_rate=1000.0:voltage=1.8:swp_frequency=
0.5:data_available_line=4:bgt=0:log=1

SCR:interface=SPI:protocol=2:bit_rate=1000.0:voltage=1.8:swp_frequency=
0.5:data_available_line=4:bgt=0:io_mode=8:reset_mode=1:clock_mode=3:log
=1

SCR:interface=SPI:protocol=1:bit_rate=100:voltage=1.8:swp_frequency=1.0
:tltype=0:rst=SFRAME:log=1
```

VIRTUALISO ::

Usage ::

/terminal

SCR:interface=VIRTUALISO:protocol=<value>:bit_rate=<value>:voltage=<value>:swp_frequency=<value>:data_available_line=<value>:gpio=<value>:bgt=<value>:io_mode=<value>:reset_mode=<value>:clock_mode=<value>:log=<value>

Example :

```
SCR:interface=VIRTUALISO:protocol=0:bit_rate=100:voltage=1.8:frequency=
1.0:log=1
```

SWP_SPI ::

Usage ::

/terminal

SCR:interface=SWP_SPI:protocol=<value>:frequency=<value>:activation_mode=<value>:power_cycle_enable=<value>:target_gate=<value>:waiting_time=<value>:voltage=<value>:swp_frequency=<value>:bit_rate=<value>:nad=<value>:log=<value>

Example :

```
SCR:interface=SWP_SPI:protocol=2:frequency=5.0:activation_mode=2:power_
cycle_enable=0:target_gate=1:waiting_time=180000:voltage=1.8:swp_freque
ncy=0.5:bit_rate=1000.0:log=1
```

DWP_SPI ::

Usage ::

/terminal

SCR:interface=DWP_SPI:protocol=<value>:frequency=<value>:activation_mode=<value>:power_cycle_enable=<value>:target_gate=<value>:waiting_time=<value>:voltage=<value>:swp_frequency=<value>:bit_rate=<value>:nad=<value>:log=<value>

Example :

```
SCR:interface=DWP_SPI:protocol=2:frequency=5.0:activation_mode=2:power_
cycle_enable=0:target_gate=1:waiting_time=180000:voltage=1.8:swp_freque
ncy=0.5:bit_rate=1000.0:log=1
```

I2C ::

Usage ::

/terminal

```
SCR:interface=I2C:protocol=<value>:bit_rate=<value>:voltage=<value>:frequency=<valu
e>:control_byte=<value>:waiting_time=<value>:io_mode=<value>:reset_mode=<value>:
clock_mode=<value>:iss=<value>:iss_server=<value>:iss_port=<value>:checksum=<val
ue>:bgt=<value>:nad=<value>:log=<value>
```

Example :

```
SCR:interface=I2C:protocol=0:bit_rate=100:voltage=3.0:frequency=1.0:con
trol_byte=0x20:waiting_time=1000:io_mode=8:reset_mode=1:clock_mode=3:lo
g=1
```

```
SCR:interface=I2C:protocol=1:bit_rate=100:voltage=3.0:frequency=1.0:con
trol_byte=0x20:waiting_time=1000:iss=true:iss_server=localhost:iss_port
=15200:io_mode=8:reset_mode=1:clock_mode=3:log=1
```

```
SCR:interface=I2C:protocol=1:bit_rate=100:voltage=3.0:frequency=1.0:con
trol_byte=0x20:waiting_time=1000:iss=true:iss_server=localhost:iss_port
=15200:checksum=NAD_LRC:io_mode=8:reset_mode=1:clock_mode=3:log=1
```

```
SCR:interface=I2C:protocol=1:bit_rate=100:voltage=3.0:frequency=1.0:con
trol_byte=0x20:waiting_time=1000:checksum=NAD_LRC:io_mode=8:reset_mode=
1:clock_mode=3:log=1
```

```
SCR:interface=I2C:protocol=1:bit_rate=100:voltage=3.0:frequency=1.0:con
trol_byte=0x20:waiting_time=1000:checksum=NAD_CRC:io_mode=8:reset_mode=
1:clock_mode=3:log=1
```

```
SCR:interface=I2C:protocol=1:bit_rate=1000:voltage=1.8:frequency=1.0:co
ntrol_byte=0x20:waiting_time=10000:checksum=NAD_CRC:
tltype=0:rst=SFRAME:log=1
```

NOTE:

1. <IO mode Option>, <reset mode option>, <clock mode option>, <ISS>, <ISS_Server>, <ISS_Port>, <checksum> are optional inputs.
2. By default IO Mode = 8, reset mode = 1 and clock mode = 3 for raw I2C communication on SCR reader.
3. <ISS>, <ISS_Server>, <ISS_Port> shall be provided only in case of T=1I2C
4. <checksum=NAD_LRC> is set only if NAD byte is to be included with LRC for T=1I2C
5. <checksum=NAD_CRC> is set only if NAD byte is to be included with CRC for T=1I2C.
6. <BGT=TIME_MS> takes value (in msec) ranging from 1-250 (default is 1msec)

SCI2C ::

Usage ::

/terminal

SCR:interface=SCI2C:protocol=<value>:bit_rate=<value>:voltage=<value>:frequency=<value>:power_cycle_enable=<value>:waiting_time=<value>:io_mode=<value>:reset_mode=<value>:clock_mode=<value>:nad=<value>:log=<value>

Example :

```
SCR:interface=SCI2C:protocol=0:bit_rate=100:voltage=3.0:frequency=1.0:power_cycle_enable=0:waiting_time=1000:io_mode=8:reset_mode=1:clock_mode=3:log=1
```

NOTE: <IO mode Option>, <reset mode option>, <clock mode option> are optional inputs. By default IO Mode = 3, reset mode = 1 and clock mode = 3 for SC I2C communication on SCR reader.

SPI_I2C ::

Usage ::

/terminal

SCR:interface=SPI_I2C:protocol=<value>:bit_rate=<value>:voltage=<value>:frequency=<value>:control_byte=<value>:waiting_time=<value>:io_mode=<value>:reset_mode=<value>:clock_mode=<value>:nad=<value>:log=<value>

Example :

```
SCR:interface=SPI_I2C:protocol=1:bit_rate=100:voltage=1.8:frequency=1.0:control_byte=0x20:waiting_time=1000:io_mode=7:reset_mode=1:clock_mode=3:log=1
```

NOTE: <IO mode Option>, <reset mode option>, <clock mode option> are optional inputs.

Micropross Readers:

If <term> is defined as 'Micropross Reader', the following parameters:

1. For MP300 reader:

Usage:

/terminal MP300:<Connection Port>:<Serial Number>:<Resource ID>:<Coupler Number>:<Card Type>:<Tx Data Rate>:<Rx Data Rate>

- i. <Connection Port> = USB (Default)
- ii. <Serial Number> = Serial number of Micropross Reader
- iii. <Resource ID> = 122
- iv. <Coupler ID> = 1
- v. <Card Type>

Card Type A	= 1
Card Type B	= 2
Card Type TYPE_MIFARE	= 3
Card Type TYPE_VICINITY	= 4

Card Type TYPE_FELICA = 5

vi. <Tx Data Rate> : Transmission Data Rate

Transmission Data Rate	Value
TX_106KB	106
TX_212KB	212
TX_424KB	424
TX_848KB	848
TX_1695KB	1695
TX_3390KB	3390
TX_6780KB	6780

vii. <Rx Data Rate> : Receiving Data Rate

Receiving Data Rate	Value
RX_106KB	106
RX_212KB	212
RX_424KB	424
RX_848KB	848
RX_1695KB	1695
RX_3390KB	3390
RX_6780KB	6780

NOTE: For Card Type A and B supported Tx Data Rate and Rx Data Rate are 106KB/212KB/424KB/848KB. By Default Tx and Rx Data Rate are set to 106KB.

Example:

```
/term MP300:USB:9050:122:1:1
```

2. For MP500 reader:

Usage:

```
/terminal MP500:<Connection Port>:<Serial Number>:<Resource ID>:<Coupler  
Number>:<Card Type>:<Tx Data Rate>:<Rx Data Rate>
```

- i. <Connection Port> = USB (Default)
- ii. <Serial Number> = Serial number of Micropross Reader
- iii. <Resource ID> = 246
- iv. <Coupler ID> = 0
- v. <Card Type>

Card Type A = 1
Card Type B = 2
Card Type TYPE_MIFARE = 3
Card Type TYPE_VICINITY = 4
Card Type TYPE_FELICA = 5

vi. <Tx Data Rate> : Transmission Data Rate

Transmission Data Rate	Value
TX_106KB	106
TX_212KB	212
TX_424KB	424
TX_848KB	848
TX_1695KB	1695
TX_3390KB	3390
TX_6780KB	6780

vii. <Rx Data Rate> : Receiving Data Rate

Receiving Data Rate	Value
RX_106KB	106
RX_212KB	212
RX_424KB	424
RX_848KB	848
RX_1695KB	1695
RX_3390KB	3390
RX_6780KB	6780

NOTE: For Card Type A and B supported Tx_Data Rate and RX Data Rate are 106KB/212KB/424KB/848KB. By Default Tx and Rx Data Rate are set to 106KB.

Example:

```
/term MP500:USB:9050:246:0:1
```

3. For MP65 reader:

Usage:

```
/term MP65:<Protocol>:<Voltage>:<Frequency>:<Port>:<Serial Number>
```

i.<Protocol> = SWP (Default)

ii.<Voltage> = 1800mV (Default)

iii.<Frequency> = 5000000MHz (Default)

iv.<Connection Port> = USB (Default)

v. <Serial Number> = Serial number of Micropross Reader

NOTE: Protocol values supported for MP65 reader are SWP and ISO7816.

Default voltage value supported via ISO7816 protocol is 5000mV

Examples:

```
/term MP65:SWP:1800:5000000:USB:MCQ.05.27.18
```

```
/term MP65:ISO7816:5000:5000000:USB:MCQ.05.27.18
```

Remote ::

If <term> is defined as 'Remote', the following parameters

can be given:

1. /term Remote[[<address>][:<port>]]
2. /term Remote[[<address>][:<port>][:<protocol>][:<terminal-initialization-string>]]

a. PC Simulator

Usage ::

```
/term Remote[[[:<address>][:<portC>][:<portCL>]]
```

```
/term Remote[[[:<address>][:<portC>][:<portCL>]][:<protocol>]
```

<address>: string containing either 'localhost', an ip-address like '127.0.0.1'.

<portC> : Number of the port to be used for contact based interface.

Default is 8050.

<portCL> : Number of the port to be used for contactless based interface.

<protocol>: Type of protocol supported by Remote Terminal:

Value	Description
T0	T=0 byte oriented transmission protocol
T0_TPDU	T=0 protocol , displays only SW1SW2, does not send GET RESPONSE command automatically
T1	T=1 block oriented transmission protocol
SWPDWP	SWP/DWP protocol

NOTE :: SWPDWP protocol would be supported in future. Current behavior would be same as like T=1 protocol.

EXAMPLE ::

```
/term Remote|localhost:8050:T0
```

```
/term Remote|localhost:8050:T0_TPDU
```

```
/term Remote|T1
```

b. PCSC Reader over TCP interface

Usage ::

```
/term Remote[<address>][:<port>][:<protocol>]:<PCSC[reader name]>]]
```

<protocol> : Type of protocol supported by Remote Terminal:

Value	Description
T0	T=0 byte oriented transmission protocol
T1	T=1 block oriented transmission protocol

<PCSC[reader name]>: string starting with 'PCSC' followed by '|' and reader name
reader name is optional, if reader name not provided,
JCShell checks for all connected PCSC readers for card
present. if card is not present on any reader, throws error.

<port> : Number of the port to be used for TCP interface. Default is 8060

EXAMPLE ::

```
/term Remote|8070:T0:"PCSC|OMNIKEY CardMan 5x21 0"  
/term Remote|8060:"PCSC|OMNIKEY CardMan 5x21 0"  
/term Remote|PCSC
```

Setup terminal or print info about current terminal.

NOTE : For connection to PCSC card readers in Linux/Mac, only “/term” command
will be enough.

For PCSC readers in Linux, make sure that pcsc daemon is running in background.

Example:

```
/terminal "Remote|localhost:1435"  
--Opening terminal  
>  
/terminal "SCComm:1:Broadcom Corp Contacted SmartCard 0"  
--Opening terminal
```

NutTerminal ::

If <term> is defined as 'Nut', the following parameters
can be given:

```
/term Nut|nfc:<MW_NFC_DLL>  
/term Nut|nfc:<MW_NFC_DLL>[:ftdi:<FTDI_NAME_BOARD>]  
/term Nut|nfc:<MW_NFC_DLL>[:ftdi:<FTDI_NAME_BOARD>][:<verbose>]
```

EXAMPLES ::

```
/term Nut|nfc:"Q:\phNfcMwDllApi.dll"  
/term Nut|nfc:"Q:\phNfcMwDllApi.dll";ftdi:A5Z08QGG
```

```
/term Nut|nfc:"Q:\phNfcMwDllApi.dll";ftdi:A5Z08QGG;verbose
```

Ashling Calypso Board ::

If <term> is defined as 'Calypso', the following pattern can be given:

1. /term Calypso:<interfacetype>:<interfaceInfo>
<interfacetype>: Supported communication interfaces
<interfaceInfo> : communication interface information
COM port or IP address

InterfaceType	InterfaceInfo	Description
0	COM port number	Serial interface
1	IP address	Ethernet interface

EXAMPLE ::

```
/term Calypso:0:4
```

RC663 ::

To load and use the RC663 Terminal :

```
/term RC663
```

If no parameter provided the default configuration is Type A with baudrate 106

In order to use different configuration the RC663 Terminal offers few parameters:

```
/term RC663|mode:<Optional A|B|F >;baudrate:<Optional 106|212|424|848>;ftdi:<D2XX Optional >;nocolor<Optional>;verbose<Optional>
```

Examples start type A polling

```
/term RC663|mode:A;baudrate:106
```

```
/term RC663|mode:A;baudrate:212;ftdi:<FTDI_NAME_BOARD>;nocolor
```

```
/term RC663|mode:A;baudrate:424;ftdi:<FTDI_NAME_BOARD>;nocolor;verbose
```

Examples start type B polling

```
/term RC663|mode:B;baudrate:106
```

```
/term RC663|mode:B;baudrate:424;ftdi:<FTDI_NAME_BOARD>;nocolor
```

Examples start type F polling

```
/term RC663|mode:F;baudrate:212
```

```
/term RC663|mode:F;baudrate:424;ftdi:<FTDI_NAME_BOARD>;nocolor
```

RC663Plus ::

Mac: the device ID is fixed:

```
/term RC663|deviceid:/dev/cu.usbmodemNXP_71
```

Windows: You need to check on which COM the board is connected and provide:

```
/term RC663|deviceid:COMXX
```

5.53 ?

usage: ? [topic]

List all commands or print usage information
about a specific command. (Not as verbose as *help [topic]*)

Example:

```
? /send
```

```
usage: /send [-r|--raw][nad=NAD][-t|--timeout timeout] apdu [pattern..]
```

5.54 >

usage: > [list] [{0|1|2|3} [-[+file]]

list	list of open log files.
0 1 2 3	select default output channels (0=no output, 1=stdout, 2=logfile, 3=stdout and log file
-	close recently opened log file
file	create log file
+file	append to log file
-file	close specified log file if in use.

Redirect shell output.

Example:

```
> 3 test.txt          ----set log mode----
/echo test           ----echo test to stdout and logfile----
test
> 1                  ----turn off logging mode----
```

5.55 auth

usage: auth [plain|mac|enc] [keydata]

Authenticate using the given (or default) initial key (key set version 255). JCShell will take the first key for all 3 operations if key version is the initial (0xFF).

Example:

```
auth
=> 80 50 00 00 08 67 5B 9B EF 29 33 92 71 00 .P...g[...]3.q.
<= 00 00 0D B9 90 A3 51 8E 15 4C FF 02 00 01 6C 7F .....Q..L....1.
    FC 11 3F B9 D4 B1 6C EB 10 9D AB EC 90 00 ..?...1.....
Status: No Error
=> 84 82 00 00 10 92 96 EB 5F 60 B9 39 6B 11 BD 78 .....'_'.9k..x
    14 73 F9 EA C5 .s...
<= 90 00 ..
Status: No Error
```

5.56 card-info

usage: card-info [[-o|--old-format]][-q|--quiet]][-x|--exclude]][-e|--exists]][-n|--not-exists]][-a|--applets]][-p|--packages]][AID..]]

-o|--old-format

Chose old output format

-q|--quiet Don't print anything, just setup list of AIDs as return value

-x|--exclude

Exclude specified AIDs from listing/return value

-e|--exists

List/include in return value only specified AIDs if existing

-n|--not-exists

List all but specified AIDs, return value are not existing AIDs

-a|--applets

List only installed applets

-p|--packages

List only packages (load files)

AID.. List of AIDs

AID Data syntax: HEX[[CHARS[[HEX...]]]

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The ']' switches modes (HEX/CHARS). Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces [], {},

<>. The '#' element is replace by the number bytes enclosed in the matching braces.

Examples:

"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64

"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF

Displays card information (Card Manager state/AID and registry info).

The privileges are coded as follows:

- S - Security Domain
- V - Security Domain with DAP verification
- E - Security Domain with delegated management
- L - Card Manager lock privilege
- T - Card Manager terminate privilege
- D - Implicit selectable (default applet)
- P - PIN change privilege
- M - Security Domain with mandated DAP verification

Note: This command corresponds to the GET STATUS Command in GP Spec.

Example:

```
card-info
=> 80 F2 80 00 02 4F 00 00          .....0..
<= 08 A0 00 00 00 03 00 00 00 01 9E 90 00      .....
Status: No Error
=> 80 F2 40 00 02 4F 00 00          ..@..0..
<= 05 12 34 45 67 89 07 00 90 00              ..4Eg.....
Status: No Error
=> 80 F2 10 00 02 4F 00 00          .....0..
<= 07 A0 00 00 00 03 53 50 01 00 01 08 A0 00 00 00      .....SP.....
    03 53 50 41 0B 74 65 73 74 50 61 63 6B 61 67 65      .SPA.testPackage
    01 00 01 05 12 34 45 67 89 90 00              .....4Eg...
Status: No Error
```

Card Manager AID : A000000003000000

Card Manager state : OP_READY

```
Application: SELECTABLE (-----) 1234456789
Load File :      LOADED (-----) A0000000035350 (Security Domain)
Module :      A000000003535041
```

```
Load File :      LOADED (-----) "testPackage"
Module    :                               1234456789
```

5.57 delete

usage: delete [-k|--delete-key KVN] [--r|--delete-related][--c|--delete-cumulative] AID]

-k|--delete-key KVN

-r|--delete-related

Also delete related objects (related instances of a package).

-c|--delete-cumulative

Cumulative delete (Delete a root Security Domain and all associated Applications).

AID AID of object to be deleted.

Global Platform DELETE command. Delete a uniquely identifiable object such as an Executable Load File, an Application or an Executable Load File and its related Applications.

Delete a root Security Domain and all associated Applications.

NOTE: DELETE COMMAND supports -k option to delete keys for specified key version number (KVN). Options -r, -c are supported to delete the Executable Load File applications and related objects.

Deleting SCP keys and deleting executable load file applications or related objects features are supported as mutually exclusive in DELETE command.

Example 1:

```
delete 1234456789
=> 80 E4 00 00 07 4F 05 12 34 45 67 89 00      .....0..4Eg..
<= 00 90 00                                     ...
Status: No Error
```

Example 2:

```
-/term Remote
>/atr
>/card
cm>set-key          1/1/DES-ECB/505152535455565758595A5B5C5D5E5F          1/2/DES-
ECB/606162636465666768696A6B6C6D6E6F 1/3/DES-ECB/707172737475767778797A7B7C7D7E7F
cm>auth
cm>put-keyset 1
```

```
cm>set-key          33/1/DES-ECB/505152535455565758595A5B5C5D5E5F          33/2/DES-
ECB/606162636465666768696A6B6C6D6E6F 33/3/DES-ECB/707172737475767778797A7B7C7D7E7F

cm> auth

cm> put-keyset 33

cm> delete -k 33

=> 80 E4 00 00 03 D2 01 21          .....!

(4575 usec)

<= 00 90 00          ...

Status: No Error
```

5.58 delete-key

usage: delete-key [keyref..]

keyref.. List of key references.

keyref Key reference syntax: SET/ID.

Delete keys currently stored in the off-card repository.

Example usage: delete-key 1/1 1/2 1/3. By default all keys are removed from the repository.

Example:

```
print-key

255/1/DES-ECB/404142434445464748494A4B4C4D4E4F

255/2/DES-ECB/404142434445464748494A4B4C4D4E4F

255/3/DES-ECB/404142434445464748494A4B4C4D4E4F

delete-key 255/1

print-key

255/2/DES-ECB/404142434445464748494A4B4C4D4E4F

255/3/DES-ECB/404142434445464748494A4B4C4D4E4F
```

5.59 ext-auth

usage: ext-auth [plain|mac|enc|rmac|crrmac|crrmacenc|crrmacencrrenc]

plain|mac|enc|rmac|crrmac|crrmacenc|crrmacencrrenc

Desired security level for subsequent APDUs:

plain : no secure messaging (default)

mac : Command MAC (C-MAC) generation

enc : Command data encryption (C-ENC) and C-MAC generation
 rmac : Response MAC (R-MAC) generation
 crmac : C-MAC and R-MAC generation
 crmacenc: C-MAC, R-MAC and C-ENC
 crmacencrenc: C-MAC, R-MAC ,C-ENC and R-ENC

Global Platform EXTERNAL AUTHENTICATE command.

Example:

```
init-update
=> 80 50 00 00 08 12 65 5D F1 14 00 86 34 00 .P....e]....4.
<= 00 00 0D B9 90 A3 51 8E 15 4C FF 02 00 00 3D 02 .....Q..L....=.
    9C 31 C7 89 06 C6 7E 1C CB 9B FE 74 90 00 .1....~....t..
Status: No Error

ext-auth plain
=> 84 82 00 00 10 C6 EC 49 A3 AC 8E FB 33 A2 E4 5C .....I....3..\
    E6 25 91 3F 9B .%.?.
<= 90 00 ..
Status: No Error
```

5.60 flush

usage: flush

Flush Global Platform session info (close secure channel).

Example:

```
session-info
Session State      : Established & authenticated
Version            : Global Platform 2.1.1
Secure Channel Protocol : SCP 02 option 15
Security Level     : plain

Flush
session-info
Session State      : No session
Version            : unknown
Secure Channel Protocol : unknown
Security Level     : plain
```

5.61 get-cplc

usage: get-cplc

Get Card Production life cycle data. Dates for years after 2010, will be shown as 20x0

Example:

```
get-cplc
=> 80 CA 9F 7F 00          .....
(31703 usec)
<= 9F 7F 2A 47 90 50 40 47 91 81 02 31 00 81 96 00    ..*G.P@G...1....
    23 14 91 39 78 48 12 82 03 00 00 00 00 11 1D 37    #..9xH.....7
    30 32 33 31 34 00 00 00 00 00 00 00 00 90 00      02314.....
Status: No Error
IC Fabricator           : 4790
IC Type                 : 5040
Operating System ID     : 4791
Operating System release date : 8102 (12.4.2008)
Operating System release level : 3100
IC Fabrication Date     : 8196 (15.7.2008)
IC Serial Number        : 00231491
IC Batch Identifier      : 3978
IC Module Fabricator     : 4812
IC Module Packaging Date  : 8203 (22.7.2008)
ICC Manufacturer         : 0000
IC Embedding Date       : 0000
IC Pre-Personalizer      : 111D
IC Pre-Perso. Equipment Date : 3730
IC Pre-Perso. Equipment ID  : 32333134
IC Personalizer          : 0000
IC Personalization Date   : 0000
IC Perso. Equipment ID    : 00000000
```

5.62 get-data

usage: get-data tag

tag Two byte tag value (HEX string) defining the data object
to be retrieved. Example usage: get-data 9F7F.

Global Platform GET DATA command used to retrieve data objects.

Example:

```
get-data 9F7F
```

```
=> 80 CA 9F 7F 00
<= 9F 7F 2A 47 90 50 17 40 51 70 92 00 00 0D B9 90
    A3 51 8E 15 4C 47 90 46 B4 00 00 00 00 48 F6 29
    50 26 F5 E2 23 00 00 00 00 00 00 00 90 00
Status: No Error
```

```
.....
..*G.P.@Qp.....
.Q..LG.F.....H.)
P&..#.....
```

5.63 help

usage: help [topic]

List all commands or print detailed information
about a specific command.

Example:

```
help
List of currently active commands:
/applet      /cap-sign   /error      /manage-channel /send        ?
/atr         /card       /exec       /mode          /set-channel help
/cap-clear   /channel    /expr       /printf        /set-var     quit
/cap-info    /clear-vars /glob       /r-echo        /sleep
version
/cap-inst-auth /close     /identify   /register      /terminal
/cap-load-auth /echo      /list-vars  /select        >
```

5.64 init-update

usage: init-update [key-set [scp]]

key-set	Key version number to be used. Zero is default and means that the card dictates, which key set to be used.
scp	Global Platform 2.1.1 Secure Channel Protocol (SCP) to be used. Valid values are: SCP_UNDEFINED SCP01 :: SCP_01_05, SCP_01_15 SCP02 :: SCP_02_04, SCP_02_05, SCP_02_0A, SCP_02_0B, SCP_02_14, SCP_02_15, SCP_02_1A, SCP_02_1B SCP03 :: SCP_03_00, SCP_03_10, SCP_03_20, SCP_03_30, SCP_03_60, SCP_03_70 SCP03 :: SCP_11_A_00, SCP_11_A_01, SCP_11_A_02, SCP_11_A_03, SCP_11_B_00, SCP_11_B_01, SCP_11_B_02, SCP_11_B_03 SCP_UNDEFINED is the default value and means that SCP_02_15 will be used if the card indicates that it supports SCP 02. SCP_03_10 is used if the card indicates that it supports SCP 03. Otherwise SCP_01_05 will be used (that's the Global (Open) Platform 2.0.1' compatible protocol.

Global Platform INITIALIZE-UPDATE command.

Example:

```
init-update
=> 80 50 00 00 08 5B FF 35 42 B1 1B 94 A6 00      .P...[.5B.....
<= 00 00 0D B9 90 A3 51 8E 15 4C FF 02 00 00 3D 02  .....Q..L....=.
    9C 31 C7 89 D9 6A 2A E1 18 DC 2C 7B 90 00      .1...j*...,{..
Status: No Error
ext-auth plain
=> 84 82 00 00 10 43 70 D2 23 DE 79 E4 9C F2 5A F9  ....Cp.#.y...Z.
    20 97 5E B9 59      .^.Y
<= 90 00      ..
```

5.65 install

usage: install [-e|--delegation][--l|--cm-lock][--t|--terminate][--d|--default][--p|--pin-change][--s|--security-domain][--b|--sd-dap][--m|--mandated-dap][--r|--trusted-path][--a|--auth-manage][--h|--token-manage][--j|--global-delete][--u|--global-lock][--w|--global-registry][--f|--final-application][--z|--global-service][--v|--receipt-gen][--x|--ciphered][--c|--contactless][--g|--contactlessself][--q|--install-param params][--i|--instance-aid AID][--o|--install-only][--ip params][--is params][--cpp params][--ra params][--cp params][--cia params][--pa params][--pb params][--pf params][--dt params][--ha params][--ga params][--gr params][--ca params][--cr params][--pr params][--ad params][--af params][--dr params][--cm params][--cn params] pkgAID appAID

-e|--delegation

Security Domain with delegated management.

--l|--cm-lock

Card Manager lock privilege.

--t|--terminate

Card terminate privilege.

--d|--default

Implicit selectable (default) applet.

--p|--pin-change

PIN change privilege.

--s|--security-domain

Security Domain privilege.

--b|--sd-dap

Security Domain with DAP verification.

--m|--mandated-dap

Security Domain with mandated DAP verification.

--r|--trusted-path

Trusted Path privilege.

-a|--auth-manage

Authorized Management privilege.

-h|--token-manage

Token Management privilege.

-j|--global-delete

Global Delete privilege.

-u|--global-lock

Global Lock privilege.

-w|--global-registry

Global Registry privilege.

-f|--final-application

Final Application privilege.

-z|--global-service

Global Service privilege.

-v|--receipt-gen

Receipt generation privilege.

-x|--ciphered

Ciphered Load File Data Block privilege.

-c|--contactless

Contactless Activation privilege.

-g|--contactlessself

Contactless Self-Activation privilege.

-q|--install-param params

Install parameters in TLV format (VOP/GP):

Tags: 0xC9 - application specific parameters

0xEF - system specific parameters:

0xC6 - non volatile code space limit

0xC7 - volatile data space limit

0xC8 - non volatile data space limit

Install parameters in TLV format (GSM 03.48):

Tags: 0xEF - system parameters

0xC8 - non volatile memory required for installation

0xC7 - volatile memory required for installation

0xCA - GSM applet specific parameters

0xC9 - applet specific parameters

GSM applet specific parameters are:

1 byte - Length of Access Domain field

1-n bytes - Access Domain:

Either one or three bytes! Possible values:

1 byte: 0x00 - Full access to the GSM file system

1 byte: 0xFF - No access to the GSM file system

3 bytes: 0x01 - APDU access mechanism:

0x0000 - No access

0x0001 - ALWAYS

0x0002 - CHV1

0x0004 - CHV2

0x0010 - ADM

(Combinations of these bits are allowed!)

1 byte - Priority level of the Toolkit applet instance

1 byte - Maximum number of timers allowed

1 byte - Maximum text length for a menu entry

1 byte - Maximum number of menu entries

1 byte - Position of the first menu entry ('00' means last)

1 byte - Identifier of the first menu entry ('00' means don't care)

...

1 byte - Position of the last menu entry ('00' means last)

1 byte - Identifier of the last menu entry ('00' means don't care)

Install parameters in TLV format (GP 2.2.1):

Tags: 0xC9 - Application specific parameters

0xEF - System specific parameters

0xC7 - Volatile Memory quota

0xC8 - Non Volatile Memory Quota

0xCB - Global Service Parameters

0xD7 - Volatile Reserved Memory

0xD8 - Non Volatile Reserved Memory

0xCA - TS 102 226 specific parameter

0xCF - Implicit selection parameter

0xEA - TS 102 226 specific template

0xB6 - Control reference template for Digital Signature (Token)

0x42 - Identification number of SD with Token verification privilege.

0x45 - Image number of SD with Token verification privilege.

0x5F20 - Application provider identifier.

0x93 - Token identifier/number.

params Install parameters (TLV format).

-i|--instance-aid AID

Desired instance AID.

AID Instance AID.

-o|--install-only

Install the applet only, don't make it selectable.

--ip Assigned Protocols for Implicit Selection (TLV format) :

Tag : 0x80

Length : 0-3

Value : 0x81 (Type A)

0x82 (Type B)

0x84 (Type F)

example :

--ip 80#(818284)

--is Initial Contactless Activation State (TLV format):

Tag : 0x81

Length : 1

Value : 0x00 (DEACTIVATED)

0x01 (ACTIVATED)

example :

--is 81#(01)

--cpp Contactless Protocol Parameters Profile (TLV format):

Tag : 0xA2

Length : variable

Value : value (please check specifications for value field details)

example :

--cpp A2#(A0#(80#()))

--ra Recognition Algorithm (TLV format) :

Tag : 0x83

Length : variable

Value : value (please check specifications for value field details)

example :

--ra 83#(01020288888888)

```
--cp Continuous Processing (TLV format) :
    Tag   : 0x84
    Length : 1-2
    Value  : 0x01 - Continuous processing disabled
            0x02 - Continuous Processing is enabled
example  :
    --cp 84#(01)

--cia Communication Interface Access Parameters (TLV format) :
    Tag   : 0xA5
    Length : variable
    Value  : Tag(s)
            0x80 - Communication Interface Access Restriction (Length - 1)
            0x81 - Communication Interface Access Default (Length - 1)
            0x82 - Communication Interface Access Instance (Length - 1)
example  :
    --cia A5#(80#(01)81#(02)82#(03))

--pa Protocol Data Type A (TLV format) :
    Tag   : 0x86
    Length : variable
    Value  : Tag(s)
            0xA0 - Protocol Parameter Data
                    0x80 - Unique Identifier
                    0x81 - SAK (Select Acknowledge)
                    0x82 - ATQA
                    0x83 - ATS LV structure
                    0x84 - FWI(Frame Waiting Time) & SFGI(Start-up Frame Guard
                        Time)
                    0x85 - CID(Card Identifier) Support
                    0x86 - DATA_RATE_MAX
            0xA1 - Protocol Parameter Mandatory Mask
                    0x80 - Mandatory Mask of LV UID
                    0x81 - Mask for mandatory bits within SAK
                    0x82 - Mask for mandatory bits within ATQA
                    0x83 - Mandatory Mask of LV ATS
                    0x84 - Operation Indicator for FWI, SFGI
                    0x85 - Mask for mandatory bits in CID_SUPPORT
```

0x86 - Operation Indicator for mandatory bits within
DATA_RATE_MAX

example :

```
--pa
86#(A0#(80#(01)81#(01)82#(0102)83#(7777777777)84#(66)85#(01)86#(123456))A1#(80#(
00111111)81#(01)82#(0101)83#(0011111111)84#(00)85#(01)86#(00FF00)))
```

--pb Protocol Data Type B (TLV format) :

Tag : 0x87

Length : variable

Value : Tag(s)

0xA0 - Protocol Parameter Data

0x80 - PUPI: Value is LV structure

0x81 - AFI: Application Family Identifier

0x82 - ATQB

0x83 - Higher Layer Response in response to ATTRIB

0x84 - Maximum data rate supported

0xA1 - Protocol Parameter Mandatory Mask

0x80 - Mandatory Mask for LV PUPI

0x81 - Mask for mandatory bits within the AFI.

0x82 - Mask for mandatory bits in ATQB

0x83 - Mandatory Mask for LV Higher Layer Response

0x84 - Operation Indicator for mandatory bits within

DATA_RATE_MAX

example :

```
--pb
87#(A0#(80#(01)81#(02)82#(66666666)83#(1122334455)84#(001122))A1#(80#(01)81#(
01)82#(01010101)83#(0101010101)84#(00FF00)))
```

--pf Protocol Data Type F (TLV format) :

Tag : 0x88

Length : variable

Value : Tag(s)

0xA0 - Protocol Parameter Data

0xA0 - Type F anti-collision parameters entry(can be present more
than once)

0x80 - System Code

0x81 - PICC Identifier

0x82 - Response Time Descriptorexample :

```
--pf
88#(A0#(A0#(80#(0101)81#(0102030405060708)82#(0102030405060708))))

--dt Display Control Template (TLV format) :
    Tag   : 0x7F20
    Length : variable
    Value  : Tag(s)
            0x5F50 - Uniform Resource Locator
            0x5F45 - Display Message
            0x53   - Image Encoding Format
            0x6D   - Application Image Template
            0x5F44 - Application Image
            0x67   - Authentication Data

example :
    --dt 7F20#(5F50#(01|www.google.com)5F45#(01|welcome)53#(05)

--ha Head Application (TLV format) :
    Tag   : 0xA0
    Length : variable
    Value  : AID of head application

example :
    --ha A0#(4F#(1122334455))

--ga Add to the Group Authorization List (TLV format) :
    Tag   : 0xA1
    Length : variable
    Value  : AID(s) of Applications

example :
    --ga A1#(4F#(1111111111)4F#(2222222222))

--gr Remove from the Group Authorization List (TLV format) :
    Tag   : 0xA2
    Length : variable
    Value  : AID(s) of Applications

example :
    --gr A2#(4F#(3333333333)4F#(4444444444))

--ca Add to the CREL List (TLV format) :
    Tag   : 0xA3
    Length : variable
    Value  : AID(s) of Applications

example :
```

```

--ca A3#(4F#(5555555555)4F#(6666666666))

--cr Remove from the CREL List (TLV format) :
    Tag   : 0xA4
    Length : variable
    Value  : AID(s) of Applications
example  :
    --cr A4#(4F#(7777777777)4F#(8888888888))

--pr Policy Restricted Applications (TLV format) :
    Tag   : 0xA5
    Length : variable
    Value  : AID(s) of Applications
example  :
    --pr A5#(4F#(5555555555)4F#(6666666666))

--ad Application discretionary data (TLV format) :
    Tag   : 0xA6
    Length : variable
    Value  : Data
example  :
    --ad A6#(00010203040506070809)

--af Application Family (TLV format) :
    Tag   : 0x87
    Length : 1
    Value  : Application Family Indicator
example  :
    --af 87#(04)

--dr Display Required Indicator (TLV format) :
    Tag   : 0x88
    Length : 1
    Value  : 00 - Application requires a display
             01 - Application does not require a display
example  :
    --dr 88#(01)

--cm Cumulative Granted Volatile Memory (TLV format) :
    Tag   : 0x82
    Length : 2 or 4
    Value  : memory size
example  :

```

```
--cm 82#(1234)

--cn Cumulative Granted Non Volatile Memory (TLV format) :
    Tag   : 0x83
    Lenght : 2 or 4
    Value  : memory size
example  :
    --cn 83#(1234)

pkgAID AID of the package (Load File) to install from.
appAID AID of the applet (Executable Module within the Load File) to be installed.
```

Install an applet (via the Card Manager) with certain privileges and, if desired, make it selectable.

Example:

```
install -i 1234456789 -q C9#() |testPackage 1234456789
=> 80 E6 0C 00 1E 0B 74 65 73 74 50 61 63 6B 61 67      .....testPackag
    65 05 12 34 45 67 89 05 12 34 45 67 89 01 00 02      e..4Eg...4Eg....
    C9 00 00 00                                          ....

<= 90 00                                              ..
Status: No Error
```

5.66 ls

usage: ls [[-o|--old-format][--f|--new-format][--q|--quiet][--x|--exclude][--e|--exists][--n|--not-exists][--a|--applets][--p|--packages][AID..]]

-o|--old-format

Choose old output format

-f|--new-format

Choose new output format which is according to GP2.2 specification.

-q|--quiet Don't print anything, just setup list of AIDs as return value

-x|--exclude

Exclude specified AIDs from listing/return value

-e|--exists

List/include in return value only specified AIDs if existing

-n|--not-exists

List all but specified AIDs, return value are not existing AIDs

-a|--applets

List only installed applets

-p|--packages

List only packages (load files)

AID.. List of AIDs

AID Data syntax: HEX[[CHARS[[HEX...]]

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The ']' switches modes (HEX/CHARS). Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replaced by the number bytes enclosed in the matching braces.

Examples:

"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64

"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF

Displays card information (Card Manager state/AID and registry info).

The privileges are coded as follows:

- S - Security Domain
- V - Security Domain with DAP verification
- E - Security Domain with delegated management
- L - Card Manager lock privilege
- T - Card Manager terminate privilege
- D - Implicit selectable (default applet)
- P - PIN change privilege
- M - Security Domain with mandated DAP verification

Note: This command corresponds to the GET STATUS Command in GP Spec.

Example:

```
ls
=> 80 F2 80 00 02 4F 00 00          .....0..
<= 08 A0 00 00 00 03 00 00 00 01 9E 90 00      .....
Status: No Error
=> 80 F2 40 00 02 4F 00 00          ..@..0..
<= 6A 88                                     j.
Status: Reference data not found
=> 80 F2 10 00 02 4F 00 00          .....0..
<= 07 A0 00 00 00 03 53 50 01 00 01 08 A0 00 00 00      .....SP.....
    03 53 50 41 90 00                .SPA..
```

Status: No Error

Card Manager AID : A000000003000000

Card Manager state : OP_READY

Load File : LOADED (-----) A0000000035350 (Security Domain)

Module : A000000003535041

5.67 print-key

usage: print-key [keyref..]

keyref.. List of key references.

keyref Key reference syntax: SET/ID.

Print information about keys currently stored in the off-card repository.

Example usage: print-key 1/1 1/2 1/3. By default information about all keys is printed.

Example:

```
print-key
255/1/DES-ECB/404142434445464748494A4B4C4D4E4F
255/2/DES-ECB/404142434445464748494A4B4C4D4E4F
255/3/DES-ECB/404142434445464748494A4B4C4D4E4F
```

5.68 put-key

usage: put-key [-m|--mode replace|add][--replace-keySet key-set]keydef|keyref..

-m|--mode replace|add

PutKey operation mode. Defines whether the key set to which the keys belong to is to be replaced or added.

replace|add

Replace or add key set.

--replace-keySet key-set

Defines the key set to be replaced. Only applicable if the operation mode is set to 'replace'.

key-set Key set to be replaced.

keydef|keyref..

List of keys to be put onto the card.

keydef|keyref

Either a key reference or definition. Syntax see [set-key](#)

Add/replace one or more DES/AES keys.

If the mode option is not set the default mode is add.

All keys must belong to the same key set and must be ordered (lowest index first). If the operation mode is 'replace' a replace key set must be defined.

Example:

```
set-key 24/1/DES-ECB/101112131415161718191A1B1C1D1E1F
put-key 24/1
=> 80 D8 00 01 17 18 80 10 22 82 AF 69 FE 46 31 A1      .....".i.F1.
    DE 7C CA C0 D4 6B 98 33 03 FE 8A 09 00                .|...k.3.....

<= 6A 80                                                  j.
Status: Wrong data
Add new key set didn't work, try modify ...
=> 80 D8 18 01 17 00 80 10 22 82 AF 69 FE 46 31 A1      .....".i.F1.
    DE 7C CA C0 D4 6B 98 33 03 FE 8A 09 00                .|...k.3.....

<= 18 FE 8A 09 90 00                                      .....
Status: No Error
```

5.69 put-keyset

usage: put-keyset [-m|--mode replace|add][--replace-keySet key-set] key-set..

-m|--mode replace|add

PutKey operation mode. Defines whether the key set to which the keys belong to is to be replaced or added.

replace|add

Replace or add key set.

--replace-keySet key-set

Defines the key set to be replaced. Only applicable if the operation mode is set to 'replace'.

key-set Key set to be replaced.

key-set.. List of key sets to be put onto the card.

key-set Key set to be put. Key 1,2 and 3 must be present in this key set.

Add/replace one DES/AES key set (keys at index 1,2 and 3).

If the mode option is not set the default mode is 'add'.

If the operation mode is 'replace' a replace key set must be defined.

Example:

```
set-key 24/1/DES-ECB/101112131415161718191A1B1C1D1E1F
set-key 24/2/DES-ECB/101112131415161718191A1B1C1D1E1F
set-key 24/3/DES-ECB/101112131415161718191A1B1C1D1E1F
put-keyset 24      # put keyset to card
=> 80 D8 00 81 43 18 80 10 22 82 AF 69 FE 46 31 A1      ....C..."..i.F1.
    DE 7C CA C0 D4 6B 98 33 03 FE 8A 09 80 10 22 82      .|...k.3.....".
    AF 69 FE 46 31 A1 DE 7C CA C0 D4 6B 98 33 03 FE      .i.F1..|...k.3..
    8A 09 80 10 22 82 AF 69 FE 46 31 A1 DE 7C CA C0      ...."..i.F1..|..
    D4 6B 98 33 03 FE 8A 09 00                          .k.3.....
    <= 18 FE 8A 09 FE 8A 09 FE 8A 09 90 00              .....
Status: No Error
```

5.70 put-pub-key

usage: put-pub-key [-i|--tokenpin PIN][--token tokenspec][--d|--keyID key-ID][--r|--replace-keyset replace-keyset][--m|--mod key-modulus][--e|--exp key-exponent] key-set

-i --tokenpin	PIN Specify the PIN to open the token. Only used if token is PIN protected.
PIN	PIN to open the token referenced by tokenspec.
--token	tokenspec Certificate file containing RSA token
tokenspec	Specify the token holding the key pair including the public key to be put. Possible token definitions are: "pkcs11:<dllname>" "windows" "<PKCS#12-file>" "<PKCS#8-file>". The string "windows" means to search your Windows system (CAPI) for a key pair.
--keyID	key-ID Define key index in the key set version of the public key to be put (default is 1).
key-ID	Key index in the key set version.
--replace-keyset	replace-keyset Replace kvn. KVN to be replaced with should be same.
replace-keyset	keyset to be replaced
--mod	key-modulus RSA Public Key Modulus
key-modulus	RSA Public key modulus
--exp	key-exponent RSA Public Key Exponent
key-exponent	RSA Public key exponent

key-set Key set version to be added.

Add a RSA public key to the desired key set at the desired key index.
The specified token will be searched for key pairs
(private certs) holding public keys which can be put onto the card.

5.71 quit

usage: quit [[exit-value] msg..]

Quit shell process with the given exit status (defaults to 0).

5.72 registry-update

usage: registry-update [-s|--security-domain][-b|--sd-dap][-e|--delegation][-l|--cm-lock][-t|--terminate][-r|--reset][-p|--pin-change][-m|--mandated-dap][-i|--trusted-path][-a|--auth-manage][-h|--token-manage][-d|--global-delete][-o|--global-lock][-w|--global-registry][-f|--final-application][-z|--global-service][-v|--receipt-gen][-x|--ciphered][-c|--contactless][-g|--contactlessself][-q|--install-param params][-y|--sd sdAID][-n|--ap appAID][-k|--extradite][-j|--privilege]

-s|--security-domain

A Security Domain.

-b|--sd-dap

Security Domain with DAP verification.

-e|--delegation

Security Domain with delegated management.

-l|--cm-lock

Card Manager lock privilege.

-t|--terminate

Card terminate privilege.

-r|--reset Card reset privilege.

-p|--pin-change

PIN change privilege.

-m|--mandated-dap

Security Domain with mandated DAP verification.

-i|--trusted-path

Trusted Path privilege.

-a|--auth-manage

Authorized Management privilege.

-h|--token-manage

Token Management privilege.

- d|--global-delete
Global Delete privilege.
- o|--global-lock
Global Lock privilege.
- w|--global-registry
Global Registry privilege.
- f|--final-application
Final Application privilege.
- z|--global-service
Global Service privilege.
- v|--receipt-gen
Receipt generation privilege.
- x|--ciphpered
Ciphpered Load File Data Block privilege.
- c|--contactless
Contactless Activation privilege.
- g|--contactlessself
Contactless Self-Activation privilege.
- q|--install-param params
registry-update parameters in TLV format GP 2.2.1 (11.5.2.3.7):

params Install parameters (TLV format).
- y|--sd sdAID
Security Domain to associate the package/application(s) with.
sdAID AID of the Security Domain to which the applet is
to be extradited.
- n|--ap appAID
AID of the applet instance
appAID AID of the applet instance.
- k|--extradite
Perform extradition.
- j|--privilege
Include privilege bytes in cmd.

Corresponds with the Global Platform Install [for registry update] command.

Modify GlobalPlatform Registry data associated with an Application.

This process also allows the restricting of Card Content Management functionality of a specific Security Domain or OPEN itself (if AppID is not provided)

The Application AID shall be present except when modifying the functionality of the OPEN.

For updating or revoking privileges the Privileges options shall be present.

Privileges not specified at command line will be unset if --privilege option is present.

An extradition may also be performed.

5.73 select

usage: select

Select the Issuer Security Domain (CardManager) using GlobalPlatform specific SELECT format.

Example:

```
select
=> 00 A4 04 00 08 A0 00 00 00 03 00 00 00 00 .....
<= 6F 65 84 08 A0 00 00 00 03 00 00 00 A5 59 9F 65 oe.....Y.e
    01 FF 9F 6E 06 40 51 70 92 00 00 73 4A 06 07 2A ...n.@Qp...sJ..*
    86 48 86 FC 6B 01 60 0C 06 0A 2A 86 48 86 FC 6B .H..k.`...*.H..k
    02 02 01 01 63 09 06 07 2A 86 48 86 FC 6B 03 64 ....c...*.H..k.d
    0B 06 09 2A 86 48 86 FC 6B 04 02 15 65 0B 06 09 ...*.H..k...e...
    2B 85 10 86 48 64 02 01 03 66 0C 06 0A 2B 06 01 +...Hd...f...+..
    04 01 2A 02 6E 01 02 90 00 ..*.n....

Status: No Error
```

5.74 send

usage: send [-e|--ext-len]apdu [pattern..]

-e|--ext-len

Extended-Length command data is determined based on the APDU case. This option is ignored.

apdu Data syntax: HEX[|CHARS[|HEX...]]

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The '|' switches modes (HEX/CHARS).

Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replaced by the number bytes enclosed in the matching braces.

Examples:

"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64

"01#(lab#{FE FF})" ==> 01 05 61 62 02 FE FF

pattern.. Description of expected responses. Each argument is pattern describing response as sequence of hexdigits with embedded wild cards (*,?,[.]). A leading ! char means NOT. Arguments '&' and '|' are interpreted as the respective logical operators. The response is ok if it matches the pattern and the logical expression.

Send APDU via secure channel, if established.

Example:

```
send 00A4040005123445678900
=> 00 A4 04 00 05 12 34 45 67 89 00
<= 90 00
```

.....4Eg..

5.75 session-info

usage: session-info

Print Global Platform session (secure channel) status.

Following table specifys the possible session status for each session of session-info command.

Session	Variable Names	Possible Session Status
Session State	session.session.state	No session Established Established & authenticated
Version	session.version	Global Platform 2.0.1 Global Platform 2.1.1 Global Platform 2.2
Secure Channel Protocol	session.secure.channel.protocol	SCP 01 option 05, SCP 01 option 15 SCP 02 option 04, SCP 02 option 05, SCP 02 option 0A, SCP 02 option 0B, SCP 02 option 14, SCP 02 option 15, SCP 02 option 1A SCP 03 option 00, SCP 03 option 10, SCP 03 option 20, SCP 03 option 30, SCP 03 option 60, SCP 03 option 70 SCP 11 A option 00, SCP 11 A option 01, SCP

		11_A option 02, SCP 11_A option 03 SCP 11_B option 00, SCP 11_B option 01, SCP 11_B option 02, SCP 11_B option 03
Security Level	session.security.level	Plain C-MAC C-MAC and C-ENC R-MAC C-MAC and R-MAC C-MAC, R-MAC and C-ENC C-MAC, R-MAC, C-ENC and R-ENC

Note: Possible combination of security level for different SCP session are:

SCP01: C-MAC, C-DEC

SCP02: Plain

C-MAC, R-MAC and C-DEC

SCP03: C-MAC, R-MAC, C-DEC, R-ENC

Example:

```

session-info
Session State      : Established & authenticated
Version           : Global Platform 2.1.1
Secure Channel Protocol : SCP 02 option 15
Security Level     : plain
/l-v
Global:
SESSION.ID=18B965B4793AE5566E99DCFCF3D3469F
exec.dir=D:\jcop_tools\trunk\nxp-offcard
jcop.v242=false
jrcp.version=2+
last.apdu.executiontime=
last.apdu.executiontime.unit=msec
last.executiontime=4.0
last.executiontime.unit=msec
last.response.status=9000
logfile.list=
path=./scripts
session.secure.channel.protocol=SCP 02 option 15
session.security.level=plain
session.session.state=Established & authenticated
session.version=Global Platform 2.1.1
version=${project.version}

```

5.76 set-applet

usage: set-applet AID installed|selectable|personalized|blocked|locked

installed|selectable|personalized|blocked|locked

Set application life cycle state.

Set the life cycle state of a given application.

Example:

```
51  card-info
      ----non          relevant          parts          deleted----
Card      Manager      AID              :              A000000003000000
Card Manager state :  OP_READY

Application:  SELECTABLE (-----) 1234456789
Load File   :              LOADED (-----) A0000000035350      (Security
Domain)
Module      :              A000000003535041
Load File   :              LOADED (-----) "testPackage"
Module      :              1234456789

set-applet 1234456789 locked
=> 80 F0 40 FF 05 12 34 45 67 89          ..@...4Eg.
<= 90 00                                  ..
Status: No Error
card-info
52      ----non relevant parts deleted----
Card Manager AID   :  A000000003000000
Card Manager state :  OP_READY

Application:      LOCKED (-----) 1234456789
Load File   :      LOADED (-----) A0000000035350      (Security
Domain)
Module      :      A000000003535041
Load File   :      LOADED (-----) "testPackage"
Module      :      1234456789
```

5.77 set-key

usage: set-key keydef..

keydef.. List of keys to be set in off-card repository, separated by slashes.

keydef Key syntax: SET/ID/TYPE/HEX with:

SET

Key set version in decimal representation.

ID

Key identifier in decimal representation.

TYPE

TYPE specifies the key type is must be one of:

AES, DES, DES-ECB, DES-CBC, RSA-PRIV, RSA-PUB,
RSA-CRT, DSA-PRIV, DSA-PUB.

HEX

HEX is the representation of the key material.

White space is allowed in HEX between pair of hex digits

Any slash can be substituted by the following chars: '.,|;::

Example: 255/1/DES-ECB/404142434445464748494A4B4C4D4E4F

Example:

```
set-key 255/1/DES-ECB/404142434445464748494A4B4C4D4E4F
```

5.78 set-scp

usage: set-scp scp

scp Global Platform 2.2.1 Secure Channel Protocol (SCP) to be set.

Valid values are: SCP_02_04, SCP_02_05, SCP_02_0A, SCP_02_0B, SCP_02_14, SCP_02_15, SCP_02_1A, SCP_02_1B, SCP_03_00, SCP_03_10, SCP_03_20, SCP_03_30, SCP_03_60, SCP_03_70, SCP_11_A_00, SCP_11_A_01, SCP_11_A_02, SCP_11_A_03, SCP_11_B_00, SCP_11_B_01, SCP_11_B_02, SCP_11_B_03.

Allows defining the secure channel protocol to be used in subsequent implicit channel setup. If no secure channel exists and a SCP has been set using this command, implicit channel setup takes place automatically as soon as an APDU is sent.

Example:

```
set-scp SCP_02_1B
session-info
Session State          : Established & authenticated
Version                : Global Platform 2.1.1
Secure Channel Protocol : SCP 02 option 1B
Security Level         : plain
```

5.79 set-sd-state

usage: set-sd-state sdID installed|selectable|personalized|locked [[-a|--associated]]

installed|selectable|personalized|locked

Set Security Domain life cycle state.

-a|--associated

Apply to all applications which are associated with specified Security Domain

Set the life cycle state of a given Security Domain.

Example:

```
set-sd-state 0102030405 personalized
```

5.80 set-security

usage: set-security plain|mac|enc|rmac|crmac|crmacenc|crmacencrenc

plain|mac|enc|rmac|crmac|crmacenc

Desired security level:

plain : no secure messaging

mac : Command MAC (C-MAC) generation

enc : Command data encryption (C-ENC) and C-MAC generation

rmac : Response MAC (R-MAC) generation

crmac : C-MAC and R-MAC generation

crmacenc: C-MAC, R-MAC and C-ENC

crmacencrenc: C-MAC, R-MAC, C-ENC and R-ENC

Sets the security level of the current secure channel.

Example:

```
set-security mac
session-info
Session State      : Established & authenticated
Version            : Global Platform 2.1.1
Secure Channel Protocol : SCP 02 option 1B
Security Level     : C-MAC
```

5.81 set-state

usage: set-state ready|initialized|secured|locked|terminated

ready|initialized|secured|locked|terminated

Life cycle state.

Set the Card Manager life cycle state.

Example:

```
set-state initialized
=> 80 F0 80 07 08 A0 00 00 00 03 00 00 00 .....
<= 90 00 ..
Status: No Error
card-info
53      ----non relevant parts deleted----
Card Manager AID   : A000000003000000
Card Manager state : INITIALIZED

Application: SELECTABLE (-----) 1234456789
Load File   :          LOADED (-----) A0000000035350 (Security
Domain)
Module      :          A000000003535041
Load File   :          LOADED (-----) "testPackage"
Module      :          1234456789
```

5.82 store-aid

usage: store-aid AID

AID Desired AID.

Change the Card Manager AID via a Global Platform 2.1.1 STORE DATA command.

Example:

```
store-aid A000000004000000
=> 80 E2 80 00 0C 00 70 4F 08 A0 00 00 00 04 00 00 .....p0.....
00 .
<= 6A 80 j.
Status: Wrong data
=> 80 E2 80 00 0A 4F 08 A0 00 00 00 04 00 00 00 .....0.....
<= 90 00 ..
Status: No Error
card-info
54      ----non relevant parts deleted----
Card Manager AID   : A000000004000000
```

Card Manager state : OP_READY

```

Load File      :          LOADED (-----) A0000000035350      (Security
Domain)
Module        :          A000000003535041

```

5.83 store-data

*usage: store-data [-m|--more-blocks][-b|--block-number number][-e|--encrypt]
[-f|--data-format] data*

-m|--more-blocks

Indicates that this is not the last block (more store-data commands to follow).

-b|--block-number number

Block number in case of command chaining.

Default value is zero indicating that this is the first and only block.

number Block number (0-255)

-e|--encrypt hex-byte

Encryption used for data field

Valid Options(hex values) :

00 - No general encryption information or non-encrypted data(default value)

20 - Application dependent encryption of the data

60 - Encrypted data

-f|--data-format hex-byte

Format used for data field

Valid Options(hex values) :

00 - No general data structure information(default value)

08 - DGI format of the command data field

10 - BER-TLV format of the command data field

data Data (HEX string) to be stored.

Global Platform STORE DATA command used to transfer data to an application or Security Domain.

Example:

```

store-data 110400112233
=> 80 E2 80 00 06 11 04 00 11 22 33      .....f..."3
<= 90 00                                  ..
Status: No Error

```

5.84 store-keyset

usage: store-keyset [-r|--replace-keySet replace-key-set] key-set

-r|--replace-keySet replace-key-set

Defines the key set to be replaced (value 1-111). If a new key set version is to be added this option shall not be set.

replace-key-set

Key set to be replaced.

key-set New key set version to be stored.

Add or replace one complete key set version (keys at index 1,2 and 3)
via a Global Platform STORE DATA command.

Example:

```
set-key 25/1/DES-ECB/101112131415161718191A1B1C1D1E1F
set-key 25/2/DES-ECB/101112131415161718191A1B1C1D1E1F
set-key 25/3/DES-ECB/101112131415161718191A1B1C1D1E1F
store-keyset 25
=> 80 E2 80 00 42 8F 01 30 22 82 AF 69 FE 46 31 A1      ....B..0"...i.F1.
    DE 7C CA C0 D4 6B 98 33 22 82 AF 69 FE 46 31 A1      .|...k.3"...i.F1.
    DE 7C CA C0 D4 6B 98 33 22 82 AF 69 FE 46 31 A1      .|...k.3"...i.F1.
    DE 7C CA C0 D4 6B 98 33 7F 01 0C 00 19 80 FE 8A      .|...k.3.....
    09 FE 8A 09 FE 8A 09                                  .....
<= 90 00                                                  ..
Status: No Error
```

5.85 upload

usage: upload [-p|--progress][--c|--components][--r|--random][--t|--load-custom][--l|--package package-name][--s|--sd SD-AID][--m|--params parameters][--b|--block_length length][--a|--auto][--d|--load-debug][--f|--clfdb][--x|--explicit][--j|--mode algorithm][--e|--pad algorithm][--k|--key key][--v|--licv][--i|--iv vector][--h|--hashAlgo Hash Algorithm][--g|--save-to-file save-to-file] [-u|--upload-ciphered-cap] CAP-file

-p|--progress

Display upload progress.

-c|--components

Load CAP-file component wise.

-r|--random

Load with random APDU length.

-t|--load-custom

Load custom components if available.

-l|--package package-name

Package name to look for in the CAP-file

package-name

Java package name.

-s|--sd SD-AID

Security Domain to associate the package/application(s) with.

SD-AID Security Domain AID

-m|--params parameters

System specific load parameters in TLV format (VOP/GP):

Tags: 0xEF - system specific parameters:

0xC6 - non volatile code space limit

0xC7 - volatile data space limit

0xC8 - non volatile data space limit

System specific load parameters in TLV format (GSM 03.48):

Tags: 0xEF - system specific parameters:

0xC6 - non volatile memory space required for package loading

0xC8 - non volatile memory requirements for installation

0xC7 - volatile memory requirements for installation

parameters Load parameters in raw (TLV) format.

-b|--block_length length

Max. block (APDU) length sent to the card during package upload.

length Max. block (APDU) length.

-a|--auto Auto-install/select applet.

The instance AID is generated by appending 'i' to the applet AID. If the applet AID consists of 16 bytes already, the last character is replaced by 'i' (or 'x' if already equals 'i').

-d|--load-debug

Load descriptor/debug component if available.

-f|--clfdb

Send encrypted load file data block.

Mandatory Option(s): -k key

Optional Option(s): -j & -e.

-x|--explicit

Load DAP block and load file data blocks separately.

`-j|--mode` algorithm

To be used only if `-f` option is set i.e., required only for ciphered load file data block.

The mode of operation. The size of the provided key determines if single DES or triple DES with 2 or 3 keys has to be performed.

The following modes are supported:

DES - Equivalent with 'DES/ECB'

DES/ECB - Perform an encryption/decryption task in ECB mode (Electronic Code Book), i.e. each input block is encrypted/decrypted on it's own.

DES/CBC - Perform an encryption/decryption task in CBC mode (Cipher Block Chaining), i.e. the input of any subsequent block depends on the cryptographic output of the previous block.

AES - Equivalent with 'AES/ECB'

AES/ECB - Perform an encryption/decryption task in ECB mode(Electronic Code Book), i.e. each input block is encrypted/decrypted on it's own.

AES/CBC - Perform an encryption/decryption task in CBC mode (Cipher Block Chaining), i.e. the input of any subsequent block depends on the cryptographic output of the previous block.

Note: Only 'NOPAD' and 'M4' are valid padding modes for AES algorithm.

RSA - Perform an encryption/decryption task in RSA Mode

Note : Only 'NOPAD' and 'PKCS1' are valid padding modes for RSA algorithm.

`-e|--pad` algorithm

To be used only if `-f` option is set i.e., required only for ciphered load file data block

Defines the padding algorithm to be applied.

If this mode is used in encipherment operation, the provided plaintext data will be padded accordingly. If this mode is used in decipherment operation, the deciphered plaintext data will be checked if the correct

padding has been applied.

The following padding algorithms are available:

'NOPAD' - No padding will be performed, i.e. the provided data will be provided unaltered to the cipher engine and thus it must be blocksize aligned.

'M2' - The data is right padded with a single '1' bit plus as few as possible '0' bits necessary to obtain a data string whose length is a positive integer multiple of the blocksize.

'M4' - Append an 0x80 to the right of the data block, If the resultant data block length is a multiple of BLOCK SIZE, no further padding is required. Append binary zeroes to the right of the data block until the data block length is a multiple of BLOCK SIZE
NOTE : It shall be used only with AES.

'PKCS1' - Padding according to PKCS#1 v1.5 std. It will be used ONLY in case of RSA

This parameter is optional, if it is not present, it defaults to M2.

-k|--key key

To be used only if -f option is set i.e., required only for ciphered load file data block

A sequence of symbols that controls the operation of a cryptographic transformation (e.g., encipherment, decipherment, cryptographic check function computation, signature generation or signature verification)

When mode is RSA, the key will contain the RSA key modulus and exponent.

[len modulus][modulus][len exponent][exponent]

Syntax: HEX[[CHARS[[HEX...]]]

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The ']' switches modes (HEX/CHARS). Additionally, nested elements of the form '#(...)' are

parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replaced by the number bytes enclosed in the matching braces.

Examples:

"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64

"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF

-v|--licv

When set, loads the ICV to the card (Either from cap or the user provided value). User value is used only if the ICV block is absent in cap file

-i|--iv vector

To be used only if -f option is set i.e., required only for ciphered load file data block

The initialization vector to seed the initial transformation function which is applied at the beginning of the MAC algorithm.

Syntax: HEX[|CHARS[|HEX...]]

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The '|' switches modes (HEX/CHARS).

Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replaced by the number bytes enclosed in the matching braces.

Examples:

"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64

"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF

-h| hashAlgorithm Hash Algorithm

Option is used to decide what SHA digest to be used.

Hash Algorithm

Specifies whether SHA1/ SHA-256 digest to be used.

-g|--save-to-file save-to-file

Option is for name of the file where APDU list is to be saved.

save-to-file

Name of the file where APDU list is to be saved.

-u|--upload-ciphered-cap

Validation of Security Domain to associate the package/application(s)

with META-Inf.

CAP-file Name of the CAP-file to be loaded.

Load a package onto the card via the Card Manager .

NOTE:

1. -c|--components and -f|--clfdb options are mutually exclusive.
2. If -f|--clfdb opt is set, if algorithm is not provided by the user - algorithm DES/CBC with M2 padding shall be used.
3. SHA2 indicates the hashing algorithm SHA-256.
4. Based on JCOP version, JCShell will decide on which hash algorithm should be used to generate digest over the load file data
For JCOP version 2.4.x, JCShell supports only 'SHA1' hashing algorithm.
5. If block length provided by user is greater than 255 bytes, extended length APDUs shall be used. The block length is fixed when security level for auth is plain or mac. For auth=enc, the parameter poses an upper limit for the total block length which contains the encrypted data and MAC. The length of the encrypted data is a multiple of 16 for AES, and a multiple of 8 for DES. MAC length is 8 in both cases.
For more details, refer the Global Platform Specification.
6. -f|--clfdb option for ciphered cap file is not allowed.
7. Loading ciphered cap files with -c|--components option or -d|--load-debug is disallowed.
8. Options -t|--load-custom, -j|--mode algorithm, -e|--pad algorithm, -k|--key key, -i|--iv vector are not supported for ciphered cap file loading.
9. If given Ciphered CAP file has 'CLSD AID' stored in Meta-Inf, upload command reads 'CLSD AID' from Meta-Inf and loads data to specified CLSD.
10. Option -u|--upload-ciphered-cap can be used only if -s|--sd option is set.

Example:

```
upload -d -b 250 test.cap
=> 80 E6 02 00 18 0B 74 65 73 74 50 61 63 6B 61 67      .....testPackag
    65 08 A0 00 00 00 03 00 00 00 00 00 00 00
<= 00 90 00      e.....
                ...
Status: No Error
55      ----actual data transmission deleted----
Load report:
814 bytes
effective code size on card:
+ package AID      11
+ applet AIDs      12
+ classes          17
+ methods          71
+ statics          0
```

```

+ exports          0
-----
overall          111 bytes

```

5.86 upload-install-make-selectable

usage: upload-install-make-selectable [-p|--progress][-c|--components][-r|--random][-l|--package package-name][-s|--sd SD-AID][-m|--params parameters][-b|--block_length length][-d|--load-debug][-e|--delegation][-l|--cm-lock][-t|--terminate][-d|--default][-p|--pin-change][-s|--security-domain][-b|--sd-dap][-m|--mandated-dap][-q|--install-param params][-h|--hashAlgorithm][-i|--instance-aid AID] appAID CAP-file

-p|--progress

Display upload progress.

-c|--components

Load CAP-file component wise.

-r|--random

Load with random APDU length.

-l|--package package-name

Package name to look for in the CAP-file

package-name

Java package name.

-s|--sd SD-AID

Security Domain to associate the package/application(s) with.

SD-AID Security Domain AID

-m|--params parameters

System specific load parameters in TLV format (VOP/GP):

Tags: 0xEF - system specific parameters:

0xC6 - non volatile code space limit

0xC7 - volatile data space limit

0xC8 - non volatile data space limit

System specific load parameters in TLV format (GSM 03.48):

Tags: 0xEF - system specific parameters:

0xC6 - non volatile memory space required for package loading

0xC8 - non volatile memory requirements for installation

0xC7 - volatile memory requirements for installation

parameters Load parameters in raw (TLV) format.

-b|--block_length length

Max. block (APDU) length sent to the card during package upload.

length Max. block (APDU) length.

- d|--load-debug
Load descriptor/debug component if available.
- e|--delegation
Security Domain with delegated management.
- l|--cm-lock
Card Manager lock permission.
- t|--terminate
Card terminate permission.
- d|--default
Implicit selectable (default) applet.
- p|--pin-change
PIN change permission.
- s|--security-domain
A Security Domain.
- b|--sd-dap
Security Domain with DAP verification.
- m|--mandated-dap
Security Domain with mandated DAP verification.
- q|--install-param params
Install parameters in TLV format (VOP/GP):
 - Tags: 0xC9 - application specific parameters
 - 0xEF - system specific parameters:
 - 0xC6 - non volatile code space limit
 - 0xC7 - volatile data space limit
 - 0xC8 - non volatile data space limit
Install parameters in TLV format (GSM 03.48):
 - Tags: 0xEF - system parameters
 - 0xC8 - non volatile memory required for installation
 - 0xC7 - volatile memory required for installation
 - 0xCA - GSM applet specific parameters
 - 0xC9 - applet specific parameters
GSM applet specific parameters are:
 - 1 byte - Length of Access Domain field
 - 1-n bytes - Access Domain:
 - Either one or three bytes! Possible values:
 - 1 byte : 0x00 - Full access to the GSM file system

1 byte : 0xFF - No access to the GSM file system

3 bytes: 0x01 - APDU access mechanism:

0x0000 - No access

0x0001 - ALWAYS

0x0002 - CHV1

0x0004 - CHV2

0x0010 - ADM

(combination of these bits are allowed!)

1 byte - Priority level of the Toolkit applet instance

1 byte - Maximum number of timers allowed

1 byte - Maximum text length for a menu entry

1 byte - Maximum number of menu entries

1 byte - Position of the first menu entry ('00' means last)

1 byte - Identifier of the first menu entry ('00' means don't care)

...

1 byte - Position of the last menu entry ('00' means last)

1 byte - Identifier of the last menu entry ('00' means don't care)

params Install parameters (TLV format).

-h|--hashAlgorithm

It specifies whether to use SHA1/ SHA-256 digest for signature generation.

-i|--instance-aid AID

Desired instance AID.

AID Instance AID.

appAID AID of the applet (Executable Module within the Load File) to be installed.

CAP-file Name of the CAP-file to be loaded.

GLOBAL PLATFORM INSTALL [for load, install and make selectable] cmd.

Load a package onto the card via the Card Manager.

NOTE:

1. -c|--components and -f|--clfdb options are mutually exclusive.
2. If -f|--clfdb opt is set, if algorithm is not provided by the user - algorithm DES/CBC with M2 padding shall be used.
3. SHA2 indicates the hashing algorithm SHA-256.
4. Based on JCOP version, JCShell will decide on which hash algorithm should be used to generate digest over the load file data
For JCOP version 2.4.x, JCShell supports only 'SHA1' hashing algorithm.

5. If block length provided by user is greater than 255 bytes, extended length APDUs shall be used. The block length is fixed when security level for auth is plain or mac. For auth=enc, the parameter poses an upper limit for the total block length which contains the encrypted data and MAC. The length of the encrypted data is a multiple of 16 for AES, and a multiple of 8 for DES. MAC length is 8 in both cases.

For more details, refer the Global Platform Specification.

6. Loading ciphered cap files with `-c|--components` option or `-d|--load-debug` is disallowed.
7. If given Ciphered CAP file has 'CLSD AID' stored in Meta-Inf, upload command reads 'CLSD AID' from Meta-Inf and loads data to specified CLSD.

5.87 manage-upgrade-elf-start

usage: manage-upgrade-elf-start [-o|--oldElfAid <Old Elf AID>][--n|--newElfAid <New Elf AID>][--p|--preserve][--t|--trigger][--r|--resume][--e|--minElfVersion Minimum Old ELF Version][--v|--volatile Volatile Minimum Memory][--l|--nonvolatile Non-Volatile Minimum Memory][--s|--token Tocken][--i|--image image][--a|--api api][--d|--digital digital][--u|--uTocken uTocken]

`-o|--oldElfAid <Old Elf AID>`

AID of old ELF version (i.e. the one being upgraded).

`<Old Elf AID>`

AID of old ELF version

`--n|--newElfAid <New Elf AID>`

AID of new ELF version (i.e. replacing the old one).

`<New Elf AID>`

AID of new ELF version

`--p|--preserve`

Preserve old ELF version for Recovery Procedure.

`--t|--trigger`

Trigger Restore Phase automatically.

`--r|--resume`

Automatically resume after power loss.

`--e|--minElfVersion Minimum Old ELF Version`

Minimum ELF version number for the old ELF version.

Minimum Old ELF Version

Minimum ELF version number for the old ELF version

`--v|--volatile Volatile Minimum Memory`

Volatile data Minimum Memory Requirement.

Volatile Minimum Memory

Volatile data Minimum Memory Requirement

-l|--nonvolatile Non-Volatile Minimum Memory

Non-volatile data Minimum Memory Requirement.

Non-Volatile Minimum Memory

Non-volatile data Minimum Memory Requirement

-s|--token Token

Identification Number of the SD with Token Verification privilege.

Token Identification Number of the SD with Token Verification privilege

-i|--image image

Image number of the SD with Token Verification privilege.

image Image number of the SD with Token Verification privilege

-a|--api api

Application Provider identifier.

api Application Provider identifier

-d|--digital digital

Token identifier/number (digital signature counter).

digital Token identifier/number (digital signature counter)

-u|--uToken uToken

Upgrade Token.

uToken Upgrade Token

Start New Upgrade Session.

This command may only be issued within a Secure Channel Session.

The security level for the command depends on the security level defined in the EXTERNAL AUTHENTICATE command.

5.88 manage-upgrade-elf-resume

usage: manage-upgrade-elf-resume

Resume Current Upgrade Session.

This command may only be issued within a Secure Channel Session.

The security level for the command depends on the security level defined in the EXTERNAL AUTHENTICATE command.

5.89 manage-upgrade-elf-abort

usage: manage-upgrade-elf-abort

Abort Current Upgrade Session.

This command may only be issued within a Secure Channel Session.

The security level for the command depends on the security level defined in the EXTERNAL AUTHENTICATE command.

5.90 manage-upgrade-elf-status

usage: manage-upgrade-elf-status

Get Upgrade Session Status.

This command may only be issued within a Secure Channel Session.

The security level for the command depends on the security level defined in the EXTERNAL AUTHENTICATE command.

5.91 version

usage: version

Display the current program version.

Example:

```
version
current version is 2.3.0.3 [1]
```

5.92 get-crs-state

usage: get-crs-state [-s|--search][-t|--tag] AID

-s|--search

Amd C - Other Search criteria TLVs

Below are valid search tags :

Tag	Length	Description
9F70	2	Application Lifecycle State
81	1	Selection Priority
A2	Var	Group Head Application
A5	Var	Policy Restricted Applications
87	Var	Application Family
88	1	Display Required Indicator

-t|--tag Amd C - Tag List for Get Status command

Below are valid filter tags :

Tag	Description
4F	Application AID
9F70	Application Lifecycle State
7F20	Display Control Template
5F50	Uniform Resource Locator
6D	Application Image Template
5F45	Display Message
80	Application Update Counter
81	Selection Priority
A2	Group Head Application
A3	Group Member's Application
A4	CREL Application AID List
A5	Policy Restricted Applications

A6	Application Discretionary Data
87	Application Family
88	Display Required Indicator
8A	Continuous Processing
8B	Recognition Algorithm for Implicit Selection
8C	Assigned Protocols for Implicit Selection

AID AID of application

Get the state of given CRS registered contactless application.

5.93 set-crs-state

usage: *set-crs-state* [-s|--state *activated|deactivated|nonactivatable|groupactivated|groupdeactivated|individualactivated|individualdeactivated*][-p|--priority *high|low|volatile|resetvolatile*][-c|--commaccess *ON|OFF*]*CII|AID..*

-s|--state *activated|deactivated|nonactivatable|groupactivated|groupdeactivated|individualactivated|individualdeactivated*
CRS Application States
activated|deactivated|nonactivatable|groupactivated|groupdeactivated|individualactivated|individualdeactivated
CRS Application States
-p|--priority *high|low|volatile|resetvolatile*
CRS Application Priorities
high|low|volatile|resetvolatile
CRS Application Priorities
-c|--commaccess *ON|OFF*
Communication Interface Access
ON|OFF Communication Interface State
CII|AID.. CII or List of AID(s)
CII|AID AID of application OR Communication Interface Identifier(CII)

Set the state of given CRS registered contactless application.

Valid values for the CRS state are as follows:

CRS STATE	CRS VALUES
state	activated deactivated nonactivatable groupactivated groupdeactivated individualactivated individualdeactivated
priority	high low volatile

	resetvolatile	

commaccess	ON	
	OFF	
+-----+		

Example:

```
set-crs-state -s activated AID
set-crs-state -p low AID1 AID2
set-crs-state --priority low AID
set-crs-state -c ON CII
set-crs-state -r AID
```

5.94 put-key-ecc

usage: put-key-ecc [-s|--keyset keyset][--i|--keyid keyid][--t|--keytype ECC-PUB|ECC-PRIV][--p|--paramp arg][--a|--parama arg][--b|--paramb arg][--g|--paramg arg][--n|--paramn arg][--c|--paramk arg] arg [-k|--tokenpin PIN][--o|--token tokenspec]

-s|--keyset keyset
Key Set for ECC key
keyset KeySet for ECC key

--i|--keyid keyid
Key id for ECC key
keyid Key Id for ECC key

--t|--keytype ECC-PUB|ECC-PRIV
Public(ECC-PUB) or Private(ECC-PRIV) key for ECC
ECC-PUB|ECC-PRIV
Public(ECC-PUB) or Private(ECC-PRIV) key for ECC

--p|--paramp arg
ECC curve parameter P : field specification

--a|--parama arg
ECC curve parameter A : first coefficient

--b|--paramb arg
ECC curve parameter B : second coefficient

--g|--paramg arg
ECC curve parameter G : generator

--n|--paramn arg
ECC curve parameter N : order of generator

--c|--paramk arg
ECC curve parameter k : cofactor of order of generator
arg Key Data

--k|--tokenpin PIN
Specify the PIN to open the token. Only used if token is PIN protected.
PIN PIN to open the token referenced by tokenspec.

--o|--token tokenspec
Certificate file containing ECC token
tokenspec Specify the token holding the key pair including the public key to be put.
Possible token definitions are: "pkcs11:<dllname>" | "windows" | "<PKCS#12-file>" |
"<PKCS#8-file>". The string "windows" means to search your Windows system (CAPI) for a key pair.

Add ECC key with its own ECC curve parameters or input the .p12 file containing the certificate and the public key.

NOTE : Only Public key is supported!!

5.95 /get-screxec-time

usage: /get-screxec-time SWP|SPI|ISO7816|SCI2C
SWP|SPI|ISO7816|SCI2C SCR Interface ID

To query the execution time of the last command sent via SWP or SPI or ISO7816 or SCI2C in SCR Terminal.

Interface IDs can be either DUAL_COMM_SPI or DUAL_COMM_SWP or ISO7816 or SCI2C

Possible values: SWP|SPI| ISO7816|SCI2C

Example: scr-getexec-time SWP

5.96 /get-scr-serialnum

usage: /get-scr-serialnum

This command provides the serial number of currently installed SCR01 reader.

5.97 /switch-swp-gate

usage: /switch-swp-gate [0|1|2|3|4]
[0|1|2|3|4] SWP gates

To switch the SWP gates in SCR Terminal and Simulator.

SWP gate values can be either:

- 0 - Card Emulation Type A
- 1 - Wired mode
- 2 - Card Emulation Type B
- 3 - Card Emulation Type F
- 4 - APDU Card ETSI Gate

Example: /switch-swp-gate 2

5.98 /get-clf-param

usage: /get-clf-param [A|B|F]
[A|B|F] RF gate type.

Displays settings of CLF protocol parameters.

RF type gate values can be:

- A - Type A gate*
- B - Type B gate*
- F - Type F gate*

Example: /get-clf-param A

5.99 /select-scr01

usage: /select-scr01 SerialNo
SerialNo SCR01 serial number

Select SCR01 reader by serial number provided.
If two or more SCR01 readers are connected, to select a particular SCR01 reader, serial number (present at the bottom of SCR01 reader) has to be provided.

Note: This command should be used before calling the /term command.
Example: /select-scr01 2042

5.100 /set-scr01-icctrigger

usage: /set-scr01-icctrigger iccValue
iccValue Reference value set for the internal ICC Comparator block

Set reference 'icc' value for the internal ICC comparator block (in mA). The current consumption (on VCC pin) is measured continuously and is compared with the 'icc' reference value.
The reference value should be in range of 0.000 to 40.000 (mA).

Example: /set-scr01-icctrigger 38.000

5.101 /set-scr01-vtptrigger

usage: /set-scr01-vtptrigger vtpValue
vtpValue Reference value set for the internal VTP Comparator block

Set reference 'vtp' value for the internal VTP comparator block (in V). The voltage value is measured continuously and is compared with the 'vtp' reference value.
The reference value should be in range of 0.000 to 2.500 (V).

Example: /set-scr01-vtptrigger 2.000

5.102 /set-scr01-trigger1

usage: /set-scr01-trigger1 index
index value used to switch the selected internal signal to TRIG1 port at rear side of SCR01 reader.

NOTE: Refer the following table for index value. Consider NOTATION value as input the command /set-scr01-trigger1 as stated in example.

TABLE: SIGNAL INDEX VALUE

INDEX NUMBER (decimal)	NOTATION	DESCRIPTION
0	LOGICAL_0	Logical '0' (default value)
1	LOGICAL_1	Logical '1'
2	CLK_SIGNAL	1 MHz clock signal (for testing purpose).
3	ICC_TRIGGER	ICC trigger signal
4	VTP_TRIGGER	VTP trigger signal

For ISO7816/I2C Interface		
100	ISO7816_IO1	IO1 input signal, shows the logical value seen on IO1 pin
101	ISO7816_CLK	CLK input signal, shows the logical value seen on CLK pin
102	ISO7816_UART_WAIT	UART is in 'WAIT' state, waiting for first start bit to be sent to DUT
103	ISO7816_UART_IDLE	UART is in 'IDLE' state, neither sending any data nor waiting for an incoming frame
104	ISO7816_UART_RECV	UART is in 'RECEIVE' state waiting for first start bit
105	ISO7816_UART_TRANSMIT	UART is in 'TRANSMIT' state sending a frame to DUT
For SWP/DWP Interface		
200	SWP_OUT_S1	SWP out signal(S1)
201	SWP_IN_S2	SWP in signal (S2)
202	SWP_IN_S2_FILTERED	Filtered version of SWP in signal(S2)
203	SWP_RECV_SOF	SWP receive block is in 'RECV_SOF' state, SOF(start of frame) of a SWP frame is currently received
204	SWP_RECV_EOF	SWP receive block is in 'RECV_EOF' state, EOF(end of frame) of a SWP frame is currently received
205	SWP_SEND_SOF	SWP send block is in 'SEND_SOF' state, SOF(start of frame) of a SWP frame is currently sent
206	SWP_SEND_EOF	SWP send block is in 'SEND_EOF' state, EOF(end of frame) of a SWP frame is currently sent
For SPI/SCSPI Interface		
300	SPI_MOSI	SPI MOSI signal(Master Out/Slave In)
301	SPI_MISO	SPI MISO signal(Master In/Slave Out)
302	SPI_SCLK	SPI SCLK signal
302	SPI_NSS	SPI NSS signal
For I2C/SCI2C Interface		
400	I2C_SCL_OUT	SCL out signal, driven by the I2C block. This value is not necessarily the same value as on real SCL pin
401	I2C_SCL_IN	SCL in signal, seen by the I2C block
402	I2C_SDA_OUT	SDA out signal, driven by the I2C block. This value is not necessarily the same value as on real SDA pin
403	I2C_SDA_IN	SDA in signal, seen by the I2C block
404	I2C_SCL	SCL port signal

405	I2C_SDA	SDA port signal
-----	---------	-----------------

Example: /set-scr01-trigger1 LOGICAL_0

5.103 /set-scr01-trigger2

usage: /set-scr01-trigger2 index

index value used to switch the selected internal signal to TRIG2 port at rear side of SCR01 reader.

NOTE: Please refer the above table SIGNAL INDEX VALUE for index value. Consider NOTATION value as input the command /set-scr01-trigger2 as stated in example.

Example: /set-scr01-trigger2 LOGICAL_0

5.104 /set-scr01-vcc

usage: /set-scr01-vcc voltage

Voltage value for VCC can be set in range from 0 to 6 (V) for SCR01 reader.

Example: /set-scr01-vcc 3

5.105 personalize

usage: personalize appAID

appAID AID of the applet instance to be personalized.

Initiate that the currently selected Security Domain shall personalize one of its associated applets (via subsequent STORE DATA commands). Corresponds with the Global Platform Install [for personalization] command.

5.106 get-sd-cert

usage: get-sd-cert [-k|--PK-CA-KLCC-ECDSA PK.CA-KLCC.ECDSA] key-set key-id [-k|--PK-CA-KLCC-ECDSA PK.CA-KLCC.ECDSA]

Public key 'PK.CA-KLCC.ECDSA' of the CA-KLCC used for verifying SD certificates

PK.CA-KLCC.ECDSA

Public key 'PK.CA-KLCC.ECDSA' of the CA-KLCC used for verifying SD certificates

key-set Key version number of SD Certificate.

key-id Key identifier of SD Certificate.

GET-DATA command to get SD Certificate for SCP11.

Example : get-sd-cert 48 17

5.107 create-ecka-certificate

usage: create-ecka-certificate [-m|--casd-img-no casd-img-no][--d|--discretionary-data discretionary-data][--f|--discretionary-data-format][--e|--effective-date effective-date][--s|--signature signature][--o|--sk-ecdsa sk-ecdsa][--n|--cert-serial cert-serial][--c|--ca-id ca-id][--i|--subject-id subject-id][--u|--key-usage key-usage][--x|--expiration-date expiration-date][--k|--

```

public-keyQ public-keyQ][--key-param-ref-curve key-param-ref-curve][--private-key
private-key]
-m|--casd-img-no casd-img-no
    CA Security Domain Image Number
casd-img-no
    CA Security Domain Image Number
-d|--discretionary-data discretionary-data
    Discretionary Data
discretionary-data
    Discretionary Data
-f|--discretionary-data-format
    Discretionary Data BER-TLV encoded format
-e|--effective-date effective-date
    Effective Date
effective-date
    Effective Date
-s|--signature signature
    Signature
signature
    Signature
-o|--sk-ecdsa sk-ecdsa
    key to generate sign for certificate
sk-ecdsa
    key to generate sign for certificate
-n|--cert-serial cert-serial
    Certificate Serial Number
cert-serial
    Certificate Serial Number
-c|--ca-id ca-id
    CA Identifier
ca-id
    CA Identifier
-i|--subject-id subject-id
    Subject Identifier
subject-id
    Subject Identifier
-u|--key-usage key-usage
    Key Usage
key-usage
    Key Usage
-x|--expiration-date expiration-date
    Expiration Date
expiration-date
    Expiration Date
-k|--public-keyQ public-keyQ
    Public Key Q
public-keyQ
    Public Key Q
-r|--key-param-ref-curve key-param-ref-curve
    Key Parameter Reference (Curve)
key-param-ref-curve
    Key Parameter Reference (Curve)
-p|--private-key private-key
    Certificate Private Key
private-key
    Certificate Private Key

```

Command to create TLV structured ECKA certificate using provided certificate data.

Example :

```
create-ecka-certificate -d 0102030405060708 -e 20100401 -o
034B87170A255DD4D23078B0FA6EAA6C233E1824C9312F20FAD9EF5886F85963 -n
43657274696669636174655F534E3031 -c 43415F4964656E7469666965725F3031 -i
5375626A6563745F4964656E745F -u 80 -x 20300401 -k
04263F3CB70AEE43FEC8FB927EE127BA3ECEBA09B218208B5E63DD4A27FC34E08
25A7C2B545B6C4987556EDFA1B34FDC01E5AB7CF751458D06CF9FF12532EDF135
-r 00 -p
1D79101E8282ED270B9953B6A8682499CCA5B4095A192282A1D4035BB728B7C0
```

5.108 perform-security-operation

usage: *perform-security-operation* [-s|--key-set *key-set*][-i|--key-id *key-id*][-k|--oce-cert-sk
oce-cert-sk][-c|--cert *ecka-certificate(s)*..]

-s|--key-set *key-set*

Key version number of PK.CA-KLOC.ECDSA

key-set Key version number of PK.CA-KLOC.ECDSA.

-i|--key-id *key-id*

Key identifier of PK.CA-KLOC.ECDSA

key-id Key identifier of PK.CA-KLOC.ECDSA.

-k|--oce-cert-sk *oce-cert-sk*

OCE Certificate SK

oce-cert-sk

OCE Certificate SK.

-c|--cert *ecka-certificate(s)*..

TLV structured list of certificate(s)

ecka-certificate(s)..

List of certificate(s)

ecka-certificate(s)

ECKA Certificate

Global Platform PERFORM-SECURITY-OPERATION command (AMD F).

Example: *perform-security-operation* -s 48 -i 16 -c \${certificate01} \${certificate02}
\${certificate03} \${certificate04} ...

5.109 mutual-authenticate

usage: *mutual-authenticate* [-h|--host-id *host-id*][-s|--sin *sin*][-d|--sdin *sdin*] *scpMode*

crmac|*crmacencrnc*

-h|--host-id *host-id*

Host ID

host-id Host ID

-s|--sin *sin*

SIN

sin SIN

-d|--sdin *sdin*

SDIN

sdin SDIN

scpMode Secure Channel Protocol 11 Mode to be used.

Valid values are:

SCP11 :: SCP_11_A_00, SCP_11_A_01, SCP_11_A_02, SCP_11_A_03

SCP_11_A_00 is the default value.

crmac|*crmacencrnc*

Desired security level for subsequent APDUs:

crmac : C-MAC and R-MAC generation
crmacencrenc : C-MAC, R-MAC ,R-ENC and C-ENC

Global Platform Amd F SCP11a MUTUAL AUTHENTICATE command.

SCP Mode	Description
SCP_11_A_00	- Protocol : SCP11a - Does not include Host and Card ID in key derivation - Key Usage : 0x34 (Secure messaging with MAC only)
SCP_11_A_01	- Protocol : SCP11a - Includes Host and Card ID in key derivation - Key Usage : 0x34 (Secure messaging with MAC only)
SCP_11_A_02	- Protocol : SCP11a - Does not include Host and Card ID in key derivation - Key Usage : 0x3C (Secure messaging with MAC & ENCRYPTION)
SCP_11_A_03	- Protocol : SCP11a - Includes Host and Card ID in key derivation - Key Usage : 0x3C (Secure messaging with MAC & ENCRYPTION)

Example : mutual-authenticate SCP_11_A_00 crmac

5.110 internal-authenticate

usage: internal-authenticate [-h|--host-id host-id][--s|--sin sin][--d|--sdin sdin] scpMode
crmac|crmacencrenc

-h|--host-id host-id

Host ID

host-id Host ID

-s|--sin sin

SIN

sin SIN

-d|--sdin sdin

SDIN

sdin SDIN

scpMode Secure Channel Protocol 11 Mode to be used.

Valid values are:

SCP11 :: SCP_11_B_00, SCP_11_B_01, SCP_11_B_02, SCP_11_B_03

SCP_11_B_00 is the default value.

crmac|crmacencrenc

Desired security level for subsequent APDUs:

crmac : C-MAC and R-MAC generation

crmacencrenc : C-MAC, R-MAC ,R-ENC and C-ENC

Global Platform Amd F SCP11b INTERNAL AUTHENTICATE command.

SCP Mode	Description
SCP_11_B_00	- Protocol : SCP11b - Does not include Host and Card ID in key derivation - Key Usage : 0x34 (Secure messaging with MAC only)
SCP_11_B_01	Protocol : SCP11b

	- Includes Host and Card ID in key derivation Key Usage : 0x34 (Secure messaging with MAC only)
SCP_11_B_02	- Protocol : SCP11b - Does not include Host and Card ID in key derivation Key Usage : 0x3C (Secure messaging with MAC & ENCRYPTION)
SCP_11_B_03	- Protocol : SCP11b - Includes Host and Card ID in key derivation Key Usage : 0x3C (Secure messaging with MAC & ENCRYPTION)

Example : internal-authenticate SCP_11_B_00 crmac

5.111 notifications

usage: notifications [-o|--operation operation][--seqNo seq-no]

-o|--operation operation

The following operations are supported:

list - list all available pending notifications from an eUICC

remove - remove a notification from the pending notification list

listandremove - list and remove all available pending notifications from an eUICC

--seqNo seq-no

Sequence number of the notification to be removed

seq-no Sequence number of notification

NOTIFICATIONS command is used by the LPA to either list all available pending notifications from an eUICC or remove all/single notification from the pending notification list.

5.112 /set-swp-activation-time

usage: /set-swp-activation-time SWP-Activation-Time

SWP-Activation-Time SWP Activation time

Command to set SWP Activation Time

5.113 /set-swp-actsync-time

usage: /set-swp-actsync-time SWP-ACTSync-Time

SWP-ACTSync-Time SWP ACTSync Time

Command to set SWP ACTSync Time

5.114 /get-swp-activation-time

usage: /get-swp-activation-time

Displays SWP Activation Time of initialized SCR01 reader.

5.115 /get-swp-actsync-time

usage: /get-swp-actsync-time

Displays SWP Activation Sync Time of initialized SCR01 reader.

5.116 /set-scr01-baudrate

usage: /set-scr01-baudrate <0-15> [<0-15>]

<0-15> Transmission factor FI
<0-15> Transmission factor DI

Sets the baud rate of the ISO7816 UART. According ISO7816 specification the baud rate is defined by the parameters FI and DI and is noted in clocks per ETU.

5.117 /send-hci-event

usage: /send-hci-event [-f|--frametype <frame type>][-d|--data <event data>][-n|--nad <node address>] <gate> <event type>

-f|--frametype <frame type>

Indicates the frame type of "send HCI" request

<frame type> The frame type of the event. Default value is 'i-frame'.

Supported values for <frame type> are:

clt-frame
i-frame

-d|--data <event data>

Indicates the event data of HCI event request

<event data> Data syntax: HEX[[CHARS[[HEX...]]]

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The '|' switches modes (HEX/CHARS). Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replaced by the number bytes enclosed in the matching braces.

Examples:

"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64
"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF

-n|--nad <node address>

Indicates the node address.

NAD is applicable only when Remote Terminal is opened.

<node address>

node address

<gate>

Specifies the gate that to be selected, if gate is not specified then by default typeA gate is selected.

Supported gates for Remote terminal are:

typeA
typeB
typeF
connectivity
apdu

<event type> Indicate the HCI event type.

Supported values for <event type> for Remote Terminal are in the range 0x00– 0xFF as specified in HCI standards.

Below table provides information about some of the HCI events supported by Remote terminal:

Gate ID	Event type	Event Description
Card Emulation Gates (typeA, typeB and	0x10	EVT_SEND_DATA
	0x11	EVT_FIELD_ON
	0x12	EVT_CARD_DEACTIVATED

typeF)	0x13	EVT_CARD_ACTIVATED
	0x14	EVT_FIELD_OFF
connectivity	0x10	EVT_CONNECTIVITY
	0x12	EVT_TRANSACTION
	0x13	EVT_OPERATION_ENDED
apdu	0x10	EVT_C_APDU

For more information on Host Controller Interface(HCI) please refer the ETSI specification.

Example:

Remote Terminal,

```
/term Remote
/reset
/atr
/card
/send-hci-event -d 00 typeA 0x14
SEND-HCI-EVENT DATA = 23 14 02 00
```

SCR Terminal,

```
/term SCR:DWP:2:5.0:2:0:1:100000:1.8:0.5:1
/reset
/atr
/card
auth
/send-hci-event -d 01 APDUCard 0x0A
SEND-HCI-EVENT DATA = F0 0A 02 01
```

NOTE: Events buffered before sending hci event request are cleared in case of SCRTerminal.

5.118 /get-hci-event

usage: /get-hci-event *[[-n|--nad <node address>][-t|--wtime wtime]]*

-n|--nad <node address>

Indicates the node address.

NAD is applicable only when Remote Terminal is opened.

<node address>

node address

-t|--wtime wtime

Defines the waiting time (milliseconds) to get event data in decimal notation.

wtime waiting time

Command to get the HCI events.

Example:

Remote Terminal

```
/term Remote
/reset
/atr
/card
auth
/send-hci-event -d 00A4040000 apdu 0x10
SEND-HCI-EVENT DATA = 30 10 02 00 A4 04 00 00
0.....
/get-hci-event
```

```

Number of Events received : 2
Event Received 1:
Pipe Id      : 45
Gate Id      : F0
Event Type   : 10
Frame Id     : 02
Data Length  : 98
Event        Data
6F5E8408A000000151000000A552734C06072A864886FC6B01600B06092A86488
6FC6B020
202630906072A864886FC6B03640B06092A864886FC6B040255650E060C2A8648
86FC6B050601000001660C060
A2B060104012A026E01039F6501FF9000

```

```

Event Received 2:
Pipe Id      : 45
Gate Id      : F0
Event Type   : 10
Frame Id     : 02
Data Length  : 98
Event        Data
6F5E8408A000000151000000A552734C06072A864886FC6B01600B06092A86488
6FC6B020
202630906072A864886FC6B03640B06092A864886FC6B040255650E060C2A8648
86FC6B050601000001660C060
A2B060104012A026E01039F6501FF9000

```

SCR Terminal:

```

/term SCR:DWP:2:5.0:2:0:1:100000:1.8:0.5:1
/reset
/atr
/card
auth
/send-hci-event -d 01 APDUCard 0x0A
SEND-HCI-EVENT DATA = F0 0A 02 01
/get-hci-event
Number of Events received : 1
Event Received 1:
Pipe Id      : 45
Gate Id      : F0
Event Type   : 12
Frame Id     : 00
Data Length  : 0

```

5.119 /clear-hci-event

usage: /clear-hci-event [[-n|--nad <node address>]][-t|--wtime wtime]]

-n|--nad <node address>

Indicates the node address.

NAD is applicable only when Remote Terminal is opened.

<node address>

node address

-t|--wtime wtime

Defines the waiting time (milliseconds) to clear event data in decimal notation.

wtime waiting time

Command to clear the HCI events.

5.120 **/get-last-event**

usage: /get-last-event [[-n|--nad <node address>]]

-n|--nad <node address>

Indicates the node address.

NAD is applicable only when Remote Terminal is opened.

<node address>

node address

Query the JRCP server for the last non-HCI events queued. If no event is present on the remote side, nothing will be printed to the terminal, otherwise the raw event data is displayed.

5.121 **/connect-device**

usage: /connect-device terminal string

terminal string

Connects a reader attached to the remote controller and assigns a node address (NAD) to the reader. In case this operation was successful, the node address is returned. Please note that not all JRCP servers support this command. Use */list-readers* to obtain a list of already mapped readers from the remote side. NADs in the range of 0x00 and 0x7F are fixed and cannot be changed. In this case, */connect-device* is not supported.

A terminal string interpreted by the remote controller for establishing the connection to a node device. Please refer to the user manual of the remote controller for more information.

5.122 **/disconnect-device**

usage: /disconnect-device NAD

NAD

The node address (NAD) of the device to be removed from the remote controller.

Disconnects the specified device from the remote controller. Please note that not all JRCP servers support this command. Use */list-readers* to obtain a list of already mapped readers from the remote side. NADs in the range of 0x00 and 0x7F are fixed and cannot be changed. In this case, */disconnect-device* is not supported.

5.123 **/felica-polling**

usage: /felica-polling [-s|--system-code <systemCode>][--r|--request-code

<requestCode>][--t|--time-slot <timeSlot>]

-s|--system-code <systemCode>

<systemCode>

The system code (SC) of the device to be polled or FFFFh to poll any device.

If no value is specified, 0xFFFFh is taken

--r|--request-code <requestCode>

<requestCode>

The request code (RC) specifying the polled information.

Valid values are:

00h ... No system code information

01h ... Request system code information

02h ... Request supported advance protocol features

By default, 01h is taken.

--t|--time-slot <timeSlot>

<timeSlot> Time Slot Number (TSN), valid values are 00h, 01h, 03h, 07h, 0Fh.
Default is 01h.

Command for executing FeliCa Polling

5.124 /set-scr01-rf-error-indicator

usage: /set-scr01-rf-error-indicator SWP-RF-Error-Indicator
SWP-RF-Error-Indicator
SWP RF Error Indicator

Command to set RF Error Indicator for SCR reader.

5.125 /prepare-tearing

usage: /prepare-tearing [-w|--warm][-c|--cold][-d|--device-specific <device-specific action>][--n|--nad <node address>] timeout
-w|--warm Warm Reset.
-c|--cold Cold Reset.
-d|--device-specific <device-specific action>
Performs device specific operations .
<device-specific action>
Data syntax: HEX[[CHARS[[HEX...]]]
where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The 'l' switches modes (HEX/CHARS). Additionally, nested elements of the form '#{...}' are parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replaced by the number bytes enclosed in the matching braces.
Examples:
"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64
"01#{ab|#{FE FF}}" ==> 01 05 61 62 02 FE FF
--n|--nad <node address>
Indicates the node address.
NAD is applicable only when Remote Terminal is opened.
<node address>
node address
timeout Timeout before tearing is performed in milliseconds

Command to prepare for tearing operation.

5.126 /set-io-pin

usage: /set-io-pin [--n|--nad <node address>][--i|--id <ID>][--v|--value <VAL>]
--n|--nad <node address>
Specify the node address NAD to which the command is sent.
This option is only supported when connected to JRCP servers such as the JCOP simulator and is ignored otherwise.
<node address>
node address
--i|--id <ID>
Specify the one byte pin identifier ID as a hexadecimal value.
Please refer to the user manual of the remote device for allowed values of ID.
--v|--value <VAL>
Specify the value the the IO pin shall be set to.
Allowed values are 0 and 0 in either decimal or hexadecimal notation.

Set the IO pin of a device to a specified value. This is not supported by all readers.

5.127 /set-default-nad

usage: /set-default-nad [-n|--nad nad][--r|--reader_string <reader string>]

-n|--nad nad

Indicates the node address NAD to which the command is sent.

nad Specify node address value as hexadecimal number.

Input range allowed is 0x00 - 0xFF.

--r|--reader_string <reader string>

Indicates reader string or reader interface name

<reader string>

Specify the reader interface name

Sets the default node address used for JRCPC communication, if it is not specified directly to JCShell commands. If a reader string is provided, it is matched against the list of available readers.

Only in case a unique match is found, the corresponding NAD is taken and set as a default value. In case an NAD is provided no checks are performed

5.128 /send-jrcp-frame

usage: /send-jrcp-frame <jrcp frame data>

<jrcp frame data>

Data syntax: HEX[|CHARS[|HEX...]] where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The '|' switches modes (HEX/CHARS).

Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replaced by the number bytes enclosed in the matching braces.

Examples:

"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64

"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF

Sends JRCPC frames or raw data to JCOP Simulator or JRCPC Server via JRCPC communication. Applicable only for RemoteJCTerminal. The "<jrcp frame data>" input shall follow "JRCPC extended message" format defined in AN_JCOP_Remote_Communication.docx specification.

5.129 /modify-scr01-iso7816-timeout

usage: /modify-scr01-iso7816-timeout <timeout>

<timeout> timeout in usecs

Command to modify ISO7816 timeout value for SCR terminal.

Timeout value provided is in microseconds.

5.130 /modify-scr01-protocol-params

usage: /modify-scr01-protocol-params <protocol control value>

<0, 1> are the supported protocol control values

This command is used to control (some) protocol parameters of ISO7816 interface

(Currently supported protocol control values <0, 1>) where

0 -> ISO7816_PROTOCOL_DEFAULT_SETTINGS

1 -> ISO7816_PROTOCOL_DISABLE_GETRESPONSE_CASE4.

5.131 /scr01-update-clpp

usage: /scr01-update-clpp [-r|--registry Gate registry entry] [A|B|F]

-r|--registry Gate registry entry

Gate registry entry as bytestream

Gate registry entry

Data syntax: HEX[|CHARS[|HEX...]]

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The '|' switches modes (HEX/CHARS).

Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replaced by the number bytes enclosed in the matching braces.

Examples:

"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64

"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF

[A|B|F] RF gate type.

Sets settings of CLF protocol parameters with SCR reader.

RF type gate values can be :

A - Type A gate

B - Type B gate

F - Type F gate

Example: /scr01-update-clpp A

(/scr01-update-clpp is deprecated. Please use /update-clpp instead)

Note: If the registry data is empty, then CLF protocol parameters will be cleared for the given RF gate type.

5.132 /set-scr01-clockmode

usage: /set-scr01-clockmode <clock mode >

<0, 1, 2> are the supported clock modes

This command is used to control the state of the CLK pin of ISO7816 interface

(Currently supported protocol control values <0, 1, 2>) where

0 -> SCR01_ISO7816_CLOCK_AUTO

1 -> SCR01_ISO7816_CLOCK_FORCE_0

2 -> SCR01_ISO7816_CLOCK_FORCE_1.

5.133 store-certificate

usage: store-certificate [-m|--more-store-data-cmd][--block-number block-number][--dgi cert-dgi cert-dgi][--cert-data cert-data]

-m|--more-store-data-cmd

More STORE-DATA commands to follow.

```
-b|--block-number block-number
    STORE-DATA command block number, default is '0'
block-number
    Block Number
-d|--cert-dgi cert-dgi
    Certificate DGI (7F21, 00AE/00DE, 80AE/80DE)
cert-dgi Certificate DGI (7F21, 00AE/00DE, 80AE/80DE)
-c|--cert-data cert-data
    Certificate Data
cert-data Certificate Data
```

STORE-DATA command to store Certificate with DGI.

5.134 store-rsa-pub-key

```
usage: store-rsa-pub-key [-m|--more-store-data-cmd][--b|--block-number block-number][--k|--dgi-00B9 dgi-00B9][--d|--dgi-0010 dgi-0010][--e|--dgi-0011 dgi-0011]
-m|--more-store-data-cmd
    More STORE-DATA commands to follow.
-b|--block-number block-number
    STORE-DATA command block number, default is '0'
block-number
    Block Number
-k|--dgi-00B9 dgi-00B9
    Contents for DGI 00B9
dgi-00B9 Contents for DGI 00B9
-d|--dgi-0010 dgi-0010
    Contents for DGI 0010
dgi-0010 Contents for DGI 0010
-e|--dgi-0011 dgi-0011
    Contents for DGI 0011
dgi-0011 Contents for DGI 0011
```

STORE-DATA command to store Certificate with DGI.

5.135 /Update-clpp

```
usage: /update-clpp [-r|--registry Gate registry entry] [A|B|F]
-r|--registry Gate registry entry
    Gate registry entry as bytestream
Gate registry entry
    Data syntax: HEX[CHARS[HEX...]]
    where HEX is a possibly empty sequence of pairs of hex
    digits forming a byte sequence. CHARS is a possibly empty
    sequence of charaters. Their ASCII encoding forms a
    sequence of bytes. The 'l' switches modes (HEX/CHARS).
    Additionally, nested elements of the form '#(...)' are
    parsed. Instead of () you can also use the braces [], {},
    <>. The '#' element is replace by the number bytes
    enclosed in the matching braces.
    Examples:
    "0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64
    "01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF
[A|B|F] RF gate type.
```

Sets settings of CLF protocol parameters with SCR reader.
RF type gate values can be :
A - Type A gate

B - Type B gate

F - Type F gate

Example: /update-clpp A

Note: If the registry data is empty, then CLF protocol parameters will be cleared for the given RF gate type.

5.136 /viso-switch-interface

Command to switch interface from SPI to Virtual ISO and vice-versa.

usage: /viso-switch-interface [0|1]

0 SPI.

1 VIRTUALISO

/viso-switch-interface 0

5.137 /viso-reset

Command to Soft reset a logical interface. eSE will return ATR bytes of the logical interface in INF field of the response S-Block

usage: /viso-reset

5.138 /viso-end-of-apdu-session

Command to indicate length of APDU session. INF file of response S-Block will hold DS timer info. eSE will reset the I-Block sequence to zero on receiving END OF APDU session command.

usage: /viso-end-of-apdu-session

5.139 /viso-get-atr

Command to retrieve ATR bytes. INF bytes of the response S-Block will hold the ATR bytes.

usage: /viso-get-atr

5.140 /viso-get-allowed-power-state

Command to get allowed power state of eSE. INF byte indicates the following:

0x60 - one or more interfaces are active or context save in progress.

0x80 - Ready for power OFF

0x40 - Ready to enter DPD (keep eSE active)

usage: / viso-get-allowed-power-state

5.141 /viso-get-protocol-version

Command to read NXP T=1 protocol version.

usage: / viso-get-protocol-version

5.142 /viso-eSeReset

Command to Soft reset a logical interface. eSE will return ATR bytes of the logical interface in INF field of the response S-Block

usage: / viso-eSeReset

5.143 /set-ifsc-ifsd-size

Command to set the IFSC and IFSD size to support extended I-block for SPI interface.

*usage: /set-ifsc-ifsd-size [<10-1000>]
<10-1000> IFSC and IFSD Size*

5.144 /switch-logical-interface

Command to switch between logical interfaces

usage: /switch-logical-interface [0|1|2|3]

0 SPI

1 VIRTUALISO

2 SPI TEE(GPIO HIGH)

3 SPI REE(GPIO LOW)

/switch-logical-interface 1

Currently the switch is supported between SPI(0) and VIRTUAL ISO(1) and also between SPI TEE (2) and SPI REE(3)

5.145 /configure-se-gpio

Command to configure the SE GPIO direction

usage: /configure-se-gpio [1|2] [1|2]

[1|2] GPIO Pin Identifier.

Indicates SE GPIO1 = 01

Indicates SE GPIO2 = 02

[1|2] Direction.

Input = 01

Output = 02

5.146 /set-se-gpio

Command to set the SE GPIO state.

usage: /set-se-gpio [0|1]

SPI TEE 1

SPI REE 0

SPI TEE :

GPIO_HIGH -> value 1.

SPI REE :

GPIO_LOW -> value 0.

Example: `/set-se-gpio 1`

5.147 `/get-se-gpio`

Command to retrieve the SE GPIO state.

usage: `/get-se-gpio [[1|2]]`

[1|2] GPIO Pin Identifier.

Indicates SE GPIO1 = 01

Indicates SE GPIO2 = 02

Note:

The optional values `[[1|2]]` are not required for SCR terminal.

Example:

`/get-se-gpio 0`

`/get-se-gpio 1`

5.148 `/end-of-apdu`

Command to indicate end of APDU session. INF file of response S-Block will hold DS timer info.

eSE will reset the I-Block sequence to zero on receiving END OF APDU session command.

usage: `/end-of-apdu`

Example: `/end-of-apdu`

5.149 `/set-nfcc-mode`

Command to set the NFC mode.

usage: `/set-nfcc-mode [virtual|off|apdu|poweroff]`

[virtual|off|apdu|poweroff]

Nfc modes:

Virtual - eSE acts as a card in this mode. eSE can communicate with external reader being in this mode.

Apdu - This mode allows to do Apdu exchange with eSE.

Off - eSE deactivation. Only Standby enable in case of SN100/SN200 Bx onwards.

Poweroff - eSE deactivation. Applicable only for SN100/SN200 Bx onwards.

Example: `/set-nfcc-mode virtual`

5.150 /simu-get-device-state

Command to query if JCOP is in DPD mode or not. Response shall indicate the JCOP device state

If the payload data has the value:

1 - JCOP is in DPD state

0 - JCOP is not in DPD state.

Example:

```
cm> /end-of-apdu
cm> /simu-get-device-state
cm> /echo ${last.response.data}
01
cm>
```

5.151 /swp-control-recv

Command to set SWP control receive acknowledge.

usage: /swp-control-recv-ack [[0|1]]
[0|1] SWP Control Recv Ack

To set the SWP Control Receive Acknowledge.

SWP control receive acknowledge can be either:

0 – scr01 swp control receive acknowledge ignored

1 – scr01 swp control receive acknowledge delivered

Example:

```
cm> /swp-control-recv-ack 0
```

5.152 /src01-set-ifsc

usage: /src01-set-ifsc [[-p|--protocol [T1I2C]]][[-v|--ifsc-value num]]

-p|--protocol [T1I2C]

Protocol, valid values = T1I2C

[T1I2C] Protocol

-v|--ifsc-value num

IFSC value

num ifsc value

Command to set ifsc for SCR01 reader

Example :

```
> /src01-set-ifsc -p T1I2C -v 1000
```

5.153 /scr01-set-protocol-parameter

usage: /scr01-set-protocol-parameter [[-p|--protocol [ISO7816|T1I2C|RAWI2C|SPI5]]][[-m|--parameter [1|2|3|4|5|6|7|8]]][[-v|--value num]]*

-p|--protocol [ISO7816|T1I2C|RAWI2C|SPI5]

-m|--parameter [1|2|3|4|5|6|7|8]

-v|--value num

Protocol, valid values = ISO7816, T1I2C, RAWI2C, SPI5
[ISO7816|T1I2C|RAWI2C|SPI5]

Protocol

-m|--parameter [1|2|3|4|5|6|7|8]

Protocol parameter

[1|2|3|4|5|6|7|8]

Protocol parameter	code
ISO7816_PROTOCOL_GETRESPONSE_CASE4	1
ISO7816_PROTOCOL_ATR_WAIT_CLOCKS	2
ISO7816_PROTOCOL_CLOCKSTOP_COUNT1	3
ISO7816_PROTOCOL_CLOCKSTOP_COUNT2	4
ISO7816_PROTOCOL_T1_DISABLE_ERROR_HANDLING	5
ISO7816_PROTOCOL_COLD_RESET_TIME	6
SCR01_T1I2C_PROTOCOL_NAD_LRC	1
SCR01_T1I2C_PROTOCOL_DIS_HS_STRETCH	2
SCR01_T1I2C_PROTOCOL_USE_CRC	3
SCR01_T1I2C_PROTOCOL_T1_DISABLE_ERROR_HANDLING	4
SCR01_T1I2C_PROTOCOL_DIS_HS_MODE	5
SCR01_T1I2C_PROTOCOL_DIS_ALL_STRETCH	6
SCR01_RAWI2C_PROTOCOL_DIS_HS_STRETCH	1
SCR01_RAWI2C_PROTOCOL_ERROR_MODE	2
SCR01_RAWI2C_PROTOCOL_ERROR_START	3
SCR01_RAWI2C_PROTOCOL_ERROR_LENGTH	4
SCR01_RAWI2C_PROTOCOL_ERROR_PATTERN	5
SCR01_RAWI2C_PROTOCOL_IGNORE_NACK	6
SCR01_RAWI2C_PROTOCOL_DIS_HS_MODE	7
SCR01_RAWI2C_PROTOCOL_DIS_ALL_STRETCH	8
SCR01_SPI5_PROTOCOL_T1_DISABLE_ERROR_HANDLING	1
SCR01_SPI5_PROTOCOL_NAD_LRC	2
SCR01_SPI5_PROTOCOL_POLLING_TIME_MS	3
SCR01_SPI5_PROTOCOL_USE_CRC	4
SCR01_SPI5_PROTOCOL_SEND_NAD_VALUE	5
SCR01_SPI5_PROTOCOL_RECV_NAD_VALUE	6
SCR01_SPI5_PROTOCOL_GP_MODE	7

-v|--value num

Protocol parameter value

num parameter value

Command to modify SCR01 protocol parameters.

6. Crypto Plugin Commands

Crypto Plugin for JCShell handles cryptographic algorithms.

To activate crypto plugin, type below commands in jcshell terminal -

/register crypto 0000000000 Crypto

/app crypto

6.1 hash

*usage: hash [-m|--mode algorithm][-p|--print][data [pattern..]]**

-m|--mode *algorithm*

The mode of operation.

The following modes are supported:

- SHA - Same as SHA-1
- SHA-1 - The Secure Hash Algorithm, as defined in Secure Hash Standard, NIST FIPS 180-1
- SHA-224 - Secure hash Algorithm SHA-2 as of NIST FIPS 180-3
- SHA-256 - Secure hash Algorithm SHA-2 as of NIST FIPS 180-3
- SHA-384 - Secure hash Algorithm SHA-2 as of NIST FIPS 180-3
- SHA-512 - Secure hash Algorithm SHA-2 as of NIST FIPS 180-3
- MD2 - The MD2 message digest algorithm as defined in RFC 1319
- MD5 - The MD5 message digest algorithm as defined in RFC 1321
- SM3 - SM3 Hash Function as published by Chinese Commercial Cryptography Administration Office

-p|--print Print the message digest

data *Data syntax: HEX[|CHARS[|HEX...]]*

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The '|' switches modes (HEX/CHARS). Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replaced by the number bytes enclosed in the matching braces.

Examples:

"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64

"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF

pattern..

Description of expected result. Each argument is pattern describing result as sequence of hex digits with embedded wild cards (*,?,[.]). A leading ! char means NOT. Arguments '&' and '|' are interpreted as the respective logical operators. The result is ok, if it matches the pattern and the logical expression.

This command computes a message digest over the provided data. The 'mode' determines the digest algorithm. It defaults to 'SHA-1' if no mode is explicitly requested.

6.2 sign

usage: sign [-m|--mode algorithm][-o|--operation signoperation][[-i|--input input data][-d|--ecprivate ecc private key][-x|--eccpubkeyX ecc public keyX][-y|--eccpubkeyY ecc public keyY][-r|--rSign r component][-s|--sSign s component][[-a|--rsaprivmod rsa private key

modulus][*-e*]*--rsaprivexp* *rsa private key exponent*][*-c*]*--crtparams*][*-p*]*--rsaprimep* *prime factor P*][*-q*]*--rsaprimeq* *prime factor Q*][*-t*]*--primeexpP* *prime exponent dP*][*-u*]*--primeexpQ* *prime exponent dQ*][*-v*]*--crtcoeff* *Crt coefficient qInv*][*-k*]*--rsapubmod* *rsa public key modulus*][*-l*]*--rsapubexp* *rsa public key exponent*][*-g*]*--rsaSign* *rsa signature*][*-w*]*--rsapss-msgrecovery*][*-h*]*--hash* *hash algorithm*][*-b*]*--sm2priv* *sm2 private key*][*-f*]*--sm2pub* *sm2 public key*][*-j*]*--sm2sign* *sm2 signature*][*-n*]*--print*][*-z*]*--eddsa-sk* *EDDSA private key*][*-q*]*--eddsa-pk* *EDDSA public key*][*--eddsa-sign* *EDDSA sign*][*--type* *SM4 Type*][*--SM4Key* *SM4 key*][*--SM4input* *SM4 Input*]]
-m]*--mode* *algorithm*

The mode of operation. Currently RSA, ECDSA, EDDSA and SM2 are supported.

-o]*--operation* *signoperation*

This command supports 2 operations:

1. *genSign* - Generate ECDSA/EDDSA/RSA/SM2 signature. Requires input message and ECC/EDDSA/RSA/SM2 private key
2. *verifySign* - Verification of ECDSA/EDDSA/RSA/SM2 signature. Requires input message, signature and ECC/EDDSA/RSA/SM2 public key
3. *generateMac* - Generation of SM4 mac according to the given type of SM4. Requires input, variant/type and key
4. *verifyMac* - Verification of SM4 mac. Requires input data (message + mac), variant/type and key

-i]*--input* *input data*

Input data for ECDSA

input data Data syntax: *HEX*[*CHARS*[*HEX*...]]

where *HEX* is a possibly empty sequence of pairs of hex digits forming a byte sequence. *CHARS* is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The '*i*' switches modes (HEX/CHARS). Additionally, nested elements of the form '*#*(...)' are parsed. Instead of *()* you can also use the braces *[]*, *{}*, *<>*. The '*#*(' element is replace by the number bytes enclosed in the matching braces.

Examples:

"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64

"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF

-d]*--ecprivate* *ecc private key*

ECC private key

-x]*--eccpubkeyX* *ecc public keyX*

ECC public key X component

-y]*--eccpubkeyY* *ecc public keyY*

ECC public key Y component

-r]*--rSign* *r component*

r part of ECDSA signature

-s]*--sSign* *s component*

s part of ECDSA signature

-a]*--rsaprivmod* *rsa private key modulus*

RSA private key

-e]*--rsaprivexp* *rsa private key exponent*

RSA private key

-c]*--crtparams*

Provide the RSA private key parameters in CRT format

-p]*--rsaprimep* *prime factor P*

RSA prime factor P

-q]*--rsaprimeq* *prime factor Q*

```

        RSA prime factor Q
-t|--primeexpP prime exponent dP
        RSA prime exponent dP
-u|--primeexpQ prime exponent dQ
        RSA prime exponent dQ
-v|--crtcoeff Crt coefficient qInv
        RSA Crt coefficient qInv or PQ
-k|--rsapubmod rsa public key modulus
        RSA public key
-l|--rsapubexp rsa public key exponent
        RSA public key
-g|--rsaSign rsa signature
        RSA signature
-w|--rsapss-msgrecovery
        For RSA-PSS (ISO 9796-2) with message recover
-h|--hash hash algorithm
        Hash algorithm supported for RSA & ECDSA. SHA1, SHA-256 and SHA-512 are
supported
-b|--sm2priv sm2 private key
        SM2 private key
-f|--sm2pub sm2 public key
        SM2 public key
-j|--sm2sign sm2 signature
        SM2 signature
-n|--print Print the signature for the input message or verify signature flag
-z|--eddsa-sk EDDSA private key
        EDDSA private key
-q|--eddsa-pk EDDSA public key
        EDDSA public key
--eddsa-sign EDDSA sign
        EDDSA sign
--type SM4 Mac Type
        ALG_SM4_MAC4_NOPAD
        ALG_SM4_MAC8_NOPAD
        ALG_SM4_MAC16_NOPAD
        ALG_SM4_MAC4_ISO9797_M1
        ALG_SM4_MAC8_ISO9797_M1
        ALG_SM4_MAC16_ISO9797_M1
        ALG_SM4_MAC4_ISO9797_M2
        ALG_SM4_MAC8_ISO9797_M2
        ALG_SM4_MAC16_ISO9797_M2
        ALG_SM4_MAC4_PKCS5
        ALG_SM4_MAC8_PKCS5
        ALG_SM4_MAC16_PKCS5
--SM4Key SM4 key
        Key for SM4 mac generation and verification
--SM4input SM4 Input
        Contains only message for mac generation and message + mac for mac
verification

```

ECDSA:

This command generates the ECDSA signature (r,s) for the message input using ECC private key parameters

And also verifies the signature using ECC public key parameters as input.

For intializing the EC curve parameters for ECDSA, Initialize the following global variables with hex strings using /set-var command, else 'P-256' curve will be used.

ECC_CURVEPARAM_P - The prime p
ECC_CURVEPARAM_A - EC curve coefficient a
ECC_CURVEPARAM_B - EC curve coefficient b
ECC_CURVEPARAM_G - The base point G(Gx,Gy)
ECC_CURVEPARAM_N - The order of G

Generated signature will be stored in SIGNATURE_ECDSA_R
and SIGNATURE_ECDSA_S env variables.

RSA:

This command generates the RSA signature for the message input using RSA private key parameters.

RSA private key parameters can be given in either of the following 2 representations:

1. RSA modulus and RSA private key exponent
2. RSA CRT parameters:
RSA modulus, RSA public exponent, RSA private exponent,
prime factor P, prime factor Q, prime Exponent dP, prime exponent dQ, crt
Coefficient qInv

For RSA-PSS (ISO 9796-2) with message recovery, -w or --rsapss-msgrecovery can be given.

For RSA-PSS (ISO 9796-2) with message recovery, during the verification of the signature:

The input message given will be stored in
INPUT_MESSAGE_FOR_MESSAGE_RECOVERY(ISO9796d2)

The type of message recovery (Total or Partial recovery) will be stored in
MESSAGE_RECOVERY_TYPE

The recovered message will be stored in MESSAGE_RECOVERED

Generated signature will be stored in SIGNATURE_RSA env variable.

Result of the signature verification will be stored in VERIFY_SIGN

SM2:

This command generates the SM2 signature (r,s) for the message input using ECC private key parameters.

And also verifies the signature using ECC public key parameters as input.

For initializing the EC curve parameters for SM2, Initialize the following global variables with hex strings using /set-var command, else default curve parameters will be used.

SM2_CURVEPARAM_P - The prime p
SM2_CURVEPARAM_A - EC curve coefficient a
SM2_CURVEPARAM_B - EC curve coefficient b
SM2_CURVEPARAM_GX - The base point Gx
SM2_CURVEPARAM_GY - The base point Gy
SM2_CURVEPARAM_N - The order of G

Generated signature will be stored in SIGNATURE_SM2, SIGNATURE_SM2_R and
SIGNATURE_SM2_S env variables.

EDDSA:

This command generates the EDDSA signature for the message input using EDDSA private key parameters

And also verifies the signature using EDDSA public key parameters & signature as input.

Generated signature will be stored in SIGNATURE_EDDSA

SM4:

generateMac - This command generates SM4 mac for the given input based on the provided input type

verifyMac -This command verifies the generated mac using key and message (data + mac) as input.

Generated mac will be stored in SM4_MAC

Result of mac verification will be stored in VERIFY_SM4_MAC

Example:

```
sign -m RSA -o genSign -i FDECD34949FBBB42A3BBC23A970580D2 -a
FDECD34949FBBB42A3BBC23A970580DAA3D4A35158968C6DFEF74968157A5261D
A12E4972F3F4D4F23818356922E6CBC770502F6DC473AE8925AC5402168D89191C
6D82CD0705190DE448F8CD7F66B004BDEE18B1553D0D6FDB77CE42079EEE52F9
CE35452C6EB3FF035B34039092A684276BE3FD0FB9ABE09E20D6DFAD9489B -e
00E10E543E32EA809091DA4A4DEFA89B0A2E06BB31D018AF2296C6CA330C463C5
A2A1470B266D7EE47642AD82394B0BD03556205BC9284377C06AB99043C185748F
A3D48F33894F52A3B25D343D4CEA144B9FDF405099316F98E9FF85240D8913AAC9
284B1A0486757EF32E4113469DA01401D38417D4210295D6C2888BD5F3731 -h
SHA1
```

```
sign -m RSA -o verifySign -i FDECD34949FBBB42A3BBC23A970580D2 -g
900DA1E85E87912750E61D12F831FDE7F67B3C4E5D4072B94B36FB1EE1D5DD5E0
7823DCBE3C37ACAB6C992078C88D240ADD0FBD9E3EB9ECA2BF46D02881EEE79
DAC83F455BE0FC59C5625740F0E906BE9CBD5A9102D3AC4CBA6055EDFE92E496
246B01BAA75A1E2055F6F539746277CB42559F8314F3B50EBCBA5F3ACA0CEEA1 -
k
FDECD34949FBBB42A3BBC23A970580DAA3D4A35158968C6DFEF74968157A5261D
A12E4972F3F4D4F23818356922E6CBC770502F6DC473AE8925AC5402168D89191C
6D82CD0705190DE448F8CD7F66B004BDEE18B1553D0D6FDB77CE42079EEE52F9
CE35452C6EB3FF035B34039092A684276BE3FD0FB9ABE09E20D6DFAD9489B -l
010001 -h SHA1
```

Example for SM4 Mac generation and verification:

```
sign -m SM4 -o generateMac --type ALG_SM4_MAC16_NOPAD --SM4Key
2b7e151628aed2a6abf7158809cf4f3c --SM4input
54686520717569636b2062726f776e20666f78206a756d7073206f76657220746865206c
617a7920646f672e
```

```
sign -m SM4 -o verifyMac --type ALG_SM4_MAC16_NOPAD --SM4Key
2b7e151628aed2a6abf7158809cf4f3c --SM4input
54686520717569636b2062726f776e20666f78206a756d7073206f76657220746865206c
617a7920646f672e583829C4127428E0CA06F55080EF0A5F
```

Example for RSA PSS Message Recovery:

```
sign -m RSA -o genSign -i
6162636462636465636465666465666765666768666768696768696A68696A6B696A6B
6C6A6B6C6D6B6C6D6E6C6D6E6F6D6E6F706E6F70716F7071727071727371727374
72737475737475767475767775767778767778797778797A78797A61797A61627A6162
636162636462636465
FAA8ED34EEF1CE38D29814B6EEAA154DC060BB37EB1A51E8AB0398DDADDFD33
4CB9BE20C087B1DDF1F78A39762B5F20A7A73008630913CD2EE60183DE249DD16
9CA4EB3AE0420E5113D730504A73A926BEFBFF32C89858DE5E5B3899FEC5252104
933163625F29635AB8FAA7AA14C4F3C0DD2470DEFCEB392429110A0149A771 -e
0A71B48CDF4A13425E1BAB879F47163892AEB277A9CBC369B1CAD1093C93FE223
3267EC0805A7693F6A506D0F9723F6B1A6F755AECB0B7DE1F44010294186936316A
```

```
AC4BF39B37BF6105DFA0AEA60B82C17306F2179F2ED4704D5A6FBCB141C0C9380
F5A500823CE67E8ED817F8A510059E9541B498C91F41ABE8C106220E72B -w
```

```
sign -m RSA -o verifySign -i
6162636462636465636465666465666765666768666768696768696A6B696A6B
6C6A6B6C6D6B6C6D6E6C6D6E6F6D6E6F706E6F70716F7071727071727371727374
72737475737475767475767775767778767778797778797A78797A61797A61627A6162
636162636462636465 -g ${SIGNATURE_RSA} -k
FAA8ED34EEF1CE38D29814B6EEAA154DC060BB37EB1A51E8AB0398DDADD33
4CB9BE20C087B1DDF1F78A39762B5F20A7A73008630913CD2EE60183DE249DD16
9CA4EB3AE0420E5113D730504A73A926BEFBFF32C89858DE5E5B3899FEC5252104
933163625F29635AB8FAA7AA14C4F3C0DD2470DEFCEB392429110A0149A771 -l 03
-w
```

6.3 keygen

usage: keygen [-m|--mode algorithm][-l key length][-p Public exponent][-o|--eccoperation eccoperation][-d ec private key][-k ec public key][-x ec public key X-component][-y ec public key Y-component][-r|--print]

-m|--mode algorithm

The mode of operation. Currently only RSA, ECC, CURVE25519 and SM2 are supported.

-l key length

Key length for RSA

-p Public exponent

Public exponent for RSA

Public exponent

Data syntax: HEX[|CHARS[|HEX...]]

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The '|' switches modes (HEX/CHARS).

Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replace by the number bytes enclosed in the matching braces.

Examples:

"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64

"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF

-o|--eccoperation eccoperation

The following functions are supported in ECC, CURVE25519 mode:

This command supports 2 functions:

1. keypairgen - EphemeralKeyPair generation, Input for key pair generation will be ignored

2. calcShs - Calculation of shared secret, Input would be ECC private and public keys

-d ec private key

ECDH private key

-k ec public key

ECDH Public key

-x ec public key X-component

ECDH public key X component

-y ec public key Y-component

ECDH public key Y component

-r|--print Print the generate keys

For RSA:

This command calculates the key pair for RSA and sets

env variables for public key and private key.

This also sets CRT RSA key components to environment variables.

Calculated keys will be stored in KEY_RSA_PublicKey

and KEY_RSA_PrivateKey env variables.

Both of these env variables will have modulus and exponent in LV format

[[modulus length][modulus][exponent length][exponent]]

CRT components will be stored in following env variables :

P = KEY_RSA_PrivateKey_P

Q = KEY_RSA_PrivateKey_Q

dP = KEY_RSA_PrivateKey_dP

dQ = KEY_RSA_PrivateKey_dQ

PQ = KEY_RSA_PrivateKey_PQ

For ECC:

For intializing the EC curve parameters, Initialize the following global variables with hex strings using /set-var command, else 'P-256' curve will be used.

ECC_CURVEPARAM_P - The prime p

ECC_CURVEPARAM_A - EC curve coefficient a

ECC_CURVEPARAM_B - EC curve coefficient b

ECC_CURVEPARAM_G - The base point G(Gx,Gy)

ECC_CURVEPARAM_N - The order of G

This command supports 2 functions:

1. keypairgen - ECC KeyPair generation.

Input for key pair generation will be ignored

Key pair components will be stored in following env variables

Private key = KEY_ECC_PrivateKey

Public key x value = KEY_ECC_PublicKeyX

Public key y value = KEY_ECC_PublicKeyY

2. calcShs - Calculation of shared secret, Input would be ECC private and public keys

Shared secret will be stored in following env variable

Shared secret key = ECC_SharedSecret

For SM2:

For initializing the EC curve parameters, Initialize the following global variables with hex strings using /set-var command, else default curve parameters will be used.

SM2_CURVEPARAM_P - The prime p

SM2_CURVEPARAM_A - EC curve coefficient a

SM2_CURVEPARAM_B - EC curve coefficient b

SM2_CURVEPARAM_GX - The base point Gx

SM2_CURVEPARAM_GY - The base point Gy

SM2_CURVEPARAM_N - The order of G

This command calculates the key pair for SM2 and sets env variables for public key and private key.

keypairgen - ECC KeyPair generation.

Input for key pair generation will be ignored

Key pair components will be stored in following env variables

Private key = KEY_SM2_PrivateKey

Public key = KEY_SM2_PublicKey

Public key x value = KEY_SM2_PublicKeyX

Public key y value = KEY_SM2_PublicKeyY

For Curve25519:

This command calculates KEY-PAIR for Curve25519 and sets JCShell env variables for private key & public key.

1. keypairgen - ECDH keypair generation.

Key Pair generated will be stored in following env variables

Private Key = ECDH_Curve25519_Private_Key

Public Key = ECDH_Curve25519_Public_Key

2. calcShs - Calculation of shared secret, input would be ECDH Curve25519 private key and public key

Shared secret will be stored in following env variable

Shared Secret = ECDH_Curve25519_SharedSecret

6.4 mac

usage: mac *[[--m|--mode algorithm][--p|--pad algorithm][--k|--key key][--i|--iv vector][data [pattern..]]**

--m|--mode algorithm

The mode of operation.

The following modes are supported:

M1 - This mode uses single DES in CBC mode. The last block of the cryptographic result is the MAC.

M2 - This mode features 2 DES keys. The first key is used to perform a single DES in CBC mode while the last block of the cryptographic result is again encrypted single DES with the 2nd key.

M3 - This mode features 2 DES keys. The first key is used to perform a single DES in CBC mode while the last block of the chaining mode is triple DES encrypted featuring both keys.

M4 - This mode features 3 DES keys, denoted as Ka, Kb and Kc. The first block is two times encrypted, first with Ka followed by Kc. All subsequent chaining elements are encrypted with Ka. The last block of the cryptographic result is again encrypted using Kb.

M5 - This mode features two instances of mode 'M1', resulting in two intermediate values. These values are bitwise exclusive-ored to obtain the MAC.

M6 - This mode features two instances of mode 'M4', resulting in two intermediate values. These values are bitwise exclusive-ored to obtain the MAC.

M7 - This mode features AES CMAC calculation.

This uses NIST SP 800-38B specification for MAC calculation.

PAD algorithm is ignored for AES CMAC calculation.

--p|--pad algorithm

Defines the padding algorithm to be applied.

If this mode is used in encipherment operation, the provided plaintext data will be padded accordingly. If this mode is used in decipherment operation, the deciphered plaintext data will be checked if the correct padding has been applied.

The following padding algorithms are available:

'NOPAD' - No padding will be performed, i.e. the provided data will be provided unaltered to the cipher engine and thus it must be blocksize aligned.

- 'M1' - Right padded with as few as possible '0' bits necessary, even none, to obtain a data string whose length (in bits) is a positive integer multiple of the blocksize.
 - 'M2' - The data is right padded with a single '1' bit plus as few as possible '0' bits necessary to obtain a data string whose length is a positive integer multiple of the blocksize.
 - 'M3' - The data is right-padded with as few as possible '0' bits, even none, necessary to obtain a data string whose length in bits is a positive integer multiple of the blocksize. The resulting string is then left-padded with a single block. This block consists of the binary representation of the length of the unpadded data string, left-padded with as few bits, even none, as necessary to obtain a block of blocksize length. The right most bit of this block constitutes the least significant bit of the binary representation of the encoded length.
 - 'M4' - Append an 0x80 to the right of the data block, If the resultant data block length is a multiple of BLOCK SIZE, no further padding is required. Append binary zeroes to the right of the data block until the data block length is a multiple of BLOCK SIZE
NOTE : It shall be used only with AES.
 - 'PKCS1' - Padding according to PKCS#1 v1.5 std. It will be used ONLY in case of RSA
 - 'OAEP' - Padding according to PKCS#1v2 OAEP standard. It will be used only in case of RSA
- This parameter is optional, if it is not present, it defaults to M2.

-k|--key key

A sequence of symbols that controls the operation of a cryptographic transformation (e.g., encipherment, decipherment, cryptographic check function computation, signature generation or signature verification)
When mode is RSA, the key will contain the RSA key modulus and exponent.
[len modulus][modulus][len exponent][exponent]

Syntax: HEX[[CHARS[[HEX...]]]

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The 'j' switches modes (HEX/CHARS). Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replaced by the number bytes enclosed in the matching braces.

Examples:

```
"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64
"01#{|ab|#{FE FF}}" ==> 01 05 61 62 02 FE FF
```

-i|--iv vector

The initialization vector to seed the initial transformation function which is applied at the

beginning of the MAC algorithm.

Syntax: `HEX[CHARS[HEX...]]`

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The '[' switches modes (HEX/CHARS). Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces {}, <>. The '#' element is replaced by the number bytes enclosed in the matching braces.

Examples:

```
"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64
"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF
```

data Data syntax: `HEX[CHARS[HEX...]]`

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The '[' switches modes (HEX/CHARS). Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces {}, <>. The '#' element is replaced by the number bytes enclosed in the matching braces.

Examples:

```
"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64
"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF
```

pattern.. Description of expected result. Each argument is pattern describing result as sequence of hex digits with embedded wild cards (*,?,[.]). A leading ! char means NOT. Arguments '&' and '|' are interpreted as the respective logical operators. The result is ok, if it matches the pattern and the logical expression.

This command calculates the MAC (message authentication code) across the provided data. As long as no 'NOPAD' option has been specified for the padding algorithm to be applied, the provided data will be padded before it goes into the cryptographic operation. If no initialization vector is provided, a vector is used with all bits set to '0'. If no mode of operation is provided, the default will apply. If no key is provided, the current session key for integrity (K_CMAC) out of JCShell's repository is used as default. In this case, it is verified that the type of the corresponding key management key indeed belongs to the DES algorithm unless the provision the mode of operation is provided. The provision of data is optional. If no data is provided, only padding pattern will feed the mac algorithm.

6.5 pad

usage: `pad [-p|--pad algorithm][-b|--bs blocksize][data [pattern..]]*`

-p|--pad algorithm

Defines the padding algorithm to be applied.

If this mode is used in encipherment operation, the provided plaintext data will be padded accordingly. If

this mode is used in decipherment operation, the deciphered plaintext data will be checked if the correct padding has been applied.

The following padding algorithms are available:

'NOPAD' - No padding will be performed, i.e. the provided data will be provided unaltered to the cipher engine and thus it must be blocksize aligned.

'M1' - Right padded with as few as possible '0' bits necessary, even none, to obtain a data string whose length (in bits) is a positive integer multiple of the blocksize.

'M2' - The data is right padded with a single '1' bit plus as few as possible '0' bits necessary to obtain a data string whose length is a positive integer multiple of the blocksize.

'M3' - The data is right-padded with as few as possible '0' bits, even none, necessary to obtain a data string whose length in bits is a positive integer multiple of the blocksize. The resulting string is then left-padded with a single block. This block consists of the binary representation of the length of the unpadded data string, left-padded with as few bits, even none, as necessary to obtain a block of blocksize length. The right most bit of this block constitutes the least significant bit of the binary representation of the encoded length.

'M4' - Append an 0x80 to the right of the data block, If the resultant data block length is a multiple of BLOCK SIZE, no further padding is required. Append binary zeroes to the right of the data block until the data block length is a multiple of BLOCK SIZE
NOTE : It shall be used only with AES.

'PKCS1' - Padding according to PKCS#1 v1.5 std. It will be used ONLY in case of RSA

'OAEP' - Padding according to PKCS#1v2 OAEP standard. It will be used only in case of RSA

This parameter is optional, if it is not present, it defaults to M2.

-b|--bs blocksize

Size of a single cipher operation in number of bytes.
If this parameter is not present, it defaults to a blocksize of 8 byte.

data Data syntax: HEX[|CHARS[|HEX...]]

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The '|' switches modes (HEX/CHARS). Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replaced by the number bytes enclosed in the matching braces.

Examples:

```
"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64
"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF
```

pattern.. Description of expected result. Each argument is pattern describing result as sequence of hexdigits with embedded wild cards (*,?,[.]). A leading ! char means NOT. Arguments '&' and '|' are interpreted as the respective logical operators. The result is ok, if it matches the pattern and the logical expression.

This command pads the provided data according to the specified padding algorithm. A pattern may be provided to check the expected result.

6.6 unpad

usage: unpad [-p|--pad algorithm][-b|--bs blocksize] data [pattern..]

-p|--pad algorithm

Defines the padding algorithm to be applied.

If this mode is used in encipherment operation, the provided plaintext data will be padded accordingly. If this mode is used in decipherment operation, the deciphered plaintext data will be checked if the correct padding has been applied.

The following padding algorithms are available:

'NOPAD' - No padding will be performed, i.e. the provided data will be provided unaltered to the cipher engine and thus it must be blocksize aligned.

'M1' - Right padded with as few as possible '0' bits necessary, even none, to obtain a data string whose length (in bits) is a positive integer multiple of the blocksize.

'M2' - The data is right padded with a single '1' bit plus as few as possible '0' bits necessary to obtain a data string whose length is a positive integer multiple of the blocksize.

'M3' - The data is right-padded with as few as possible '0' bits, even none, necessary to obtain a data string whose length in bits is a positive integer multiple of the blocksize. The resulting string is then left-padded with a single block. This block consists of the binary representation of the length of the unpadded data string, left-padded with as few bits, even none, as necessary to obtain a block of blocksize length. The right most bit of this block constitutes the least significant bit of the binary representation of the encoded length.

'M4' - Append an 0x80 to the right of the data block, If the resultant data block length is a multiple of BLOCK SIZE, no further padding is required. Append binary zeroes to the right of the data block until the data block length is a multiple of BLOCK SIZE
NOTE : It shall be used only with AES.

'PKCS1' - Padding according to PKCS#1 v1.5 std. It will be used ONLY in case of RSA

'OAEP' - Padding according to PKCS#1v2 OAEP standard. It will be used only in case of RSA

This parameter is optional, if it is not present,

it defaults to M2.

-b|--bs blocksize

Size of a single cipher operation in number of bytes.
If this parameter is not present, it defaults to a
blocksize of 8 byte.

data Data syntax: HEX[[CHARS[[HEX...]]

where HEX is a possibly empty sequence of pairs of hex
digits forming a byte sequence. CHARS is a possibly empty
sequence of characters. Their ASCII encoding forms a
sequence of bytes. The 'l' switches modes (HEX/CHARS).
Additionally, nested elements of the form '#(...)' are
parsed. Instead of () you can also use the braces [], {},
<>. The '#' element is replaced by the number bytes
enclosed in the matching braces.

Examples:

```
"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64
"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF
```

pattern.. Description of expected result. Each argument is
pattern describing result as sequence of hexdigits
with embedded wild cards (*,?,[.]). A leading ! char
means NOT. Arguments '&' and '|' are interpreted as
the respective logical operators. The result is ok,
if it matches the pattern and the logical expression.

This command un pads the provided data according to the specified
padding algorithm. An error will be raised, if the data is not
correctly padded. A pattern may be provided to check the expected
result.

6.7 encrypt

usage: encrypt [-m|--mode algorithm][-k|--key key][-p|--pad algorithm][-i|--iv vector] [-a|--aad aad] [-h|--hash hash algorithm] [data [pattern..]]*

-m|--mode algorithm

The mode of operation. The size of the provided key
determines if single DES or triple DES with 2 or 3 keys
has to be performed.

The following modes are supported:

DES - Equivalent with 'DES/ECB'

DES/ECB - Perform an encryption/decryption task in
ECB mode (Electronic Code Book), i.e. each
input block is encrypted/decrypted on it's
own.

DES/CBC - Perform an encryption/decryption task in
CBC mode (Cipher Block Chaining), i.e. the
input of any subsequent block depends on the
cryptographic output of the previous block.

AES - Equivalent with 'AES/ECB'

AES/ECB - Perform an encryption/decryption task in
ECB mode (Electronic Code Book), i.e. each
input block is encrypted/decrypted on it's
own.

AES/CBC - Perform an encryption/decryption task in CBC mode (Cipher Block Chaining), i.e. the input of any subsequent block depends on the cryptographic output of the previous block.
Note: Only 'NOPAD' and 'M4' are valid padding modes for AES algorithm.

AES/GCM - Perform an encryption/decryption task in Galois/Counter mode. It is a block cipher mode of operation that uses universal hashing over a binary Galois field to provide authenticated encryption.
Note: 1) Encrypt command Output - Concatenated Cipher text and authentication tag
2) Input data for Decrypt command - Concatenated Cipher text and authentication tag.

RSA - Perform an encryption/decryption task in RSA Mode
Note : Only 'NOPAD' and 'PKCS1' are valid padding modes for RSA algorithm.

SM4 - Equivalent with 'SM4/ECB'

SM4/ECB - A national commercial cryptogram asymmetric cryptographic algorithm.
Perform an encryption/decryption task in ECB mode (Electronic Code Book), i.e. each input block is encrypted/decrypted on it's own.

SM4/CBC - A national commercial cryptogram asymmetric cryptographic algorithm.
Perform an encryption/decryption task in CBC mode (Cipher Block Chaining), i.e. the input of any subsequent block depends on the cryptographic output of the previous block.

SM2 - A national commercial cryptogram asymmetric cryptographic algorithm, with the private key length of 256 bits.

-k|--key key

A sequence of symbols that controls the operation of a cryptographic transformation (e.g., encipherment, decipherment, cryptographic check function computation, signature generation or signature verification)
When mode is RSA, the key will contain the RSA key modulus and exponent.
[len modulus][modulus][len exponent][exponent]

Syntax: HEX[|CHARS[|HEX...]]

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The '|' switches modes (HEX/CHARS). Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replaced by the number bytes enclosed in the matching braces.

Examples:

"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64
"01#{|ab|#{FE FF}}" ==> 01 05 61 62 02 FE FF

-p|--pad algorithm

Defines the padding algorithm to be applied.
If this mode is used in encipherment operation, the

provided plaintext data will be padded accordingly. If this mode is used in decipherment operation, the deciphered plaintext data will be checked if the correct padding has been applied.

The following padding algorithms are available:

- 'NOPAD' - No padding will be performed, i.e. the provided data will be provided unaltered to the cipher engine and thus it must be blocksize aligned.
- 'M1' - Right padded with as few as possible '0' bits necessary, even none, to obtain a data string whose length (in bits) is a positive integer multiple of the blocksize.
- 'M2' - The data is right padded with a single '1' bit plus as few as possible '0' bits necessary to obtain a data string whose length is a positive integer multiple of the blocksize.
- 'M3' - The data is right-padded with as few as possible '0' bits, even none, necessary to obtain a data string whose length in bits is a positive integer multiple of the blocksize. The resulting string is then left-padded with a single block. This block consists of the binary representation of the length of the unpadded data string, left-padded with as few bits, even none, as necessary to obtain a block of blocksize length. The right most bit of this block constitutes the least significant bit of the binary representation of the encoded length.
- 'M4' - Append an 0x80 to the right of the data block, If the resultant data block length is a multiple of BLOCK SIZE, no further padding is required. Append binary zeroes to the right of the data block until the data block length is a multiple of BLOCK SIZE
NOTE : It shall be used only with AES.
- 'PKCS1' - Padding according to PKCS#1 v1.5 std. It will be used ONLY in case of RSA
- 'OAEP' - Padding according to PKCS#1v2 OAEP standard. It will be used only in case of RSA

This parameter is optional, if it is not present, it defaults to M2.

-i|--iv vector

The initialization vector to seed the initial transformation function which is applied at the beginning of the MAC algorithm.

Syntax: HEX[|CHARS[|HEX...]]

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The '|' switches modes (HEX/CHARS). Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replaced by the number bytes enclosed in the matching braces.

Examples:

"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64

"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF

-a|--aad aad

Additional authenticated data.

It is used to protect information that needs to be authenticated, but must be left unencrypted. It indicates how plaintext has to be handled, forwarded or processed. Required in AES/GCM mode.

Syntax: HEX[|CHARS[|HEX...]]

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The '|' switches modes (HEX/CHARS). Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replaced by the number bytes enclosed in the matching braces.

Examples:

"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64

"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF

-h|--hash hash algorithm

Hash algorithms supported for RSA Encryption with OAEP Padding:

SHA1, SHA-224, SHA-256, SHA-384 and SHA-512.

If hash algorithm is not specified, the default hash algorithm selected will be SHA-256.

data Data syntax: HEX[|CHARS[|HEX...]]

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The '|' switches modes (HEX/CHARS). Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replaced by the number bytes enclosed in the matching braces.

Examples:

"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64

"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF

pattern.. Description of expected result. Each argument is pattern describing result as sequence of hex digits with embedded wild cards (*,?,[.]). A leading ! char means NOT. Arguments '&' and '|' are interpreted as the respective logical operators. The result is ok, if it matches the pattern and the logical expression.

This command performs a cryptographic encryption. The 'mode' determines the cipher algorithm to be used. If it is absent, it defaults to 'DES/CBC'.

If 'CBC' is chosen, an initialization vector may be provided. If no initialization vector is given, it defaults to all bits set to '0'. If no 'CBC' mode is chosen, the initialization vector is ignored at all.

The padding algorithm is applied to the input data as long as no 'NOPAD' padding algorithm has been selected.

If no key has been provided, the current session key for confidentiality (K_DEK) out of JCShell's repository is used as default. In this case, the type of the related key management key defines the mode of operation unless the mode option

is provided.

Example :

```
encrypt [-m|--mode algorithm] SM2 [-k|--key]
0345F167FA0449E97FE654A1640A1D30951EE1314E2945693926EEEDC43BB3709F
|Welcome
```

6.8 decrypt

usage: decrypt [-m|--mode algorithm][--p|--pad algorithm][--k|--key key][--i|--iv vector] [--a|--aad aad] [--h|--hash hash algorithm] data [pattern..]

-m|--mode algorithm

The mode of operation. The size of the provided key determines if single DES or triple DES with 2 or 3 keys has to be performed.

The following modes are supported:

DES - Equivalent with 'DES/ECB'

DES/ECB - Perform an encryption/decryption task in ECB mode (Electronic Code Book), i.e. each input block is encrypted/decrypted on it's own.

DES/CBC - Perform an encryption/decryption task in CBC mode (Cipher Block Chaining), i.e. the input of any subsequent block depends on the cryptographic output of the previous block.

AES - Equivalent with 'AES/ECB'

AES/ECB - Perform an encryption/decryption task in ECB mode(Electronic Code Book), i.e. each input block is encrypted/decrypted on it's own.

AES/CBC - Perform an encryption/decryption task in CBC mode (Cipher Block Chaining), i.e. the input of any subsequent block depends on the cryptographic output of the previous block.

Note: Only 'NOPAD' and 'M4' are valid padding modes for AES algorithm.

AES/GCM - Perform an encryption/decryption task in Galois/Counter mode. It is a block cipher mode of operation that uses universal hashing over a binary Galois field to provide authenticated encryption.

Note: 1) Encrypt command Output - Concatenated Cipher text and authentication tag.

2) Input data for Decrypt command - Concatenated Cipher text and authentication tag.

RSA - Perform an encryption/decryption task in RSA Mode

Note : Only 'NOPAD' and 'PKCS1' are valid padding modes for RSA algorithm.

SM4 - Equivalent with 'SM4/ECB'

SM4/ECB - A national commercial cryptogram asymmetric cryptographic algorithm.

Perform an encryption/decryption task in ECB mode (Electronic Code Book), i.e. each input block is encrypted/decrypted on it's own.

SM4/CBC - A national commercial cryptogram asymmetric cryptographic algorithm.

Perform an encryption/decryption task in CBC mode (Cipher Block Chaining), i.e. the

input of any subsequent block depends on the cryptographic output of the previous block.

SM2 - A national commercial cryptogram asymmetric cryptographic algorithm, with the private key length of 256 bits.

-p|--pad algorithm

Defines the padding algorithm to be applied.

If this mode is used in encipherment operation, the provided plaintext data will be padded accordingly. If this mode is used in decipherment operation, the deciphered plaintext data will be checked if the correct padding has been applied.

The following padding algorithms are available:

'NOPAD' - No padding will be performed, i.e. the provided data will be provided unaltered to the cipher engine and thus it must be blocksize aligned.

'M1' - Right padded with as few as possible '0' bits necessary, even none, to obtain a data string whose length (in bits) is a positive integer multiple of the blocksize.

'M2' - The data is right padded with a single '1' bit plus as few as possible '0' bits necessary to obtain a data string whose length is a positive integer multiple of the blocksize.

'M3' - The data is right-padded with as few as possible '0' bits, even none, necessary to obtain a data string whose length in bits is a positive integer multiple of the blocksize. The resulting string is then left-padded with a single block. This block consists of the binary representation of the length of the unpadded data string, left-padded with as few bits, even none, as necessary to obtain a block of blocksize length. The right most bit of this block constitutes the least significant bit of the binary representation of the encoded length.

'M4' - Append an 0x80 to the right of the data block, If the resultant data block length is a multiple of BLOCK SIZE, no further padding is required. Append binary zeroes to the right of the data block until the data block length is a multiple of BLOCK SIZE
NOTE : It shall be used only with AES.

'PKCS1' - Padding according to PKCS#1 v1.5 std. It will be used ONLY in case of RSA

'OAEP' - Padding according to PKCS#1v2 OAEP standard. It will be used only in case of RSA

This parameter is optional, if it is not present, it defaults to M2.

-k|--key key

A sequence of symbols that controls the operation of a cryptographic transformation (e.g., encipherment, decipherment, cryptographic check function computation, signature generation or signature verification)

When mode is RSA, the key will contain the RSA key modulus and exponent.

[len modulus][modulus][len exponent][exponent]

Syntax: HEX[|CHARS[|HEX...]]

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The '|' switches modes (HEX/CHARS). Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replaced by the number bytes enclosed in the matching braces.

Examples:

"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64

"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF

-i|--iv vector

The initialization vector to seed the initial transformation function which is applied at the beginning of the MAC algorithm.

Syntax: HEX[|CHARS[|HEX...]]

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The '|' switches modes (HEX/CHARS). Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replaced by the number bytes enclosed in the matching braces.

Examples:

"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64

"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF

-a|--aad aad

Additional authenticated data.

It is used to protect information that needs to be authenticated, but must be left unencrypted. It indicates how plaintext has to be handled, forwarded or processed. Required in AES/GCM mode.

Syntax: HEX[|CHARS[|HEX...]]

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The '|' switches modes (HEX/CHARS). Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replaced by the number bytes enclosed in the matching braces.

Examples:

"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64

"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF

-h|--hash hash algorithm

Hash algorithms supported for RSA Encryption with OAEP Padding: SHA1, SHA-224, SHA-256, SHA-384 and SHA-512.

data Data syntax: HEX[|CHARS[|HEX...]]

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty

sequence of characters. Their ASCII encoding forms a sequence of bytes. The 'j' switches modes (HEX/CHARS). Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replaced by the number bytes enclosed in the matching braces.

Examples:

```
"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64
"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF
```

pattern.. Description of expected result. Each argument is pattern describing result as sequence of hexdigits with embedded wild cards (*,?,[.]). A leading ! char means NOT. Arguments '&' and '|' are interpreted as the respective logical operators. The result is ok, if it matches the pattern and the logical expression.

This command performs a cryptographic decryption. The 'mode' determines the cipher algorithm to be used. If it is absent, it defaults to 'DES/CBC'. If 'CBC' is chosen, an initialization vector may be provided. If no initialization vector is given, it defaults to all bits set to '0'. If no 'CBC' mode is chosen, the initialization vector is ignored at all. The padding algorithm is applied to the input data as long as no 'NOPAD' padding algorithm has been selected. If no key has been provided, the current session key for confidentiality (K_DEK) out of JCShell's repository is used as default. In this case, the type of the related key management key defines the mode of operation unless the mode option is provided.

Example:

```
decrypt [-m|--mode algorithm] SM2 [-k|--key]
0345F167FA0449E97FE654A1640A1D30951EE1314E2945693926EEEDC43BB3709F
044C7F4B9DF206DAF4DE7B907B455D11978B07B53D7824E62360F3BE60A18798
8D399EC00C4BBB80B0FCDD56174EC398BCCF02DFB56BA3282158A356B6C88C61
AEFD924D453D86C8F90824F137E1CF38968B60DFC8130485FE0F36B96153DB4989
BFB8DD62D6862755
```

where, private key is colored in yellow and ciphered data in red.

6.9 gen-cert

usage: gen-cert [[-c|--cert-type cert-type][-s|--serialno positive integer][-a|--sign-algo algo-identifier][-i|--icountryname issuerCountryName][-o|--iorganizationname issuerOrganizationName][-u|--iorganizationalunitname issuerOrganizationalUnitName][-n|--icommonname issuerCommonName][-d|--start-date start-date][-e|--end-date end-date][-scn subjectCountryName][-son subjectOrganizationName][-soun subjectOrganizationalUnitName][-scomn subjectCommonName][-sserial subjectSerialNo][-isk issuerPrivateKey][-ipk issuerPublicKey][-icurve issuer curve][-scurve subject curve][-eaki][-eski][-eku][-v|--value-eku KeyUsageValue][-ecp][-cpc][-esan-bc ext-san-bCode][-ebc][-pl-ebc num][-eeuku][-ecrldp-bc ext-crldp-bCode][-enc-bc enc-bCode][-f|--fn file-name][-p|--fp file-path]]

-c|--cert-type cert-type
Certificate Type

cert-type CI | EUM | EUICC | DPpb | DPauth | DPTIs | DSauth | DSTIs

```

-s|--serialno positive integer
    serial number assigned to each certificate
-a|--sign-algo algo-identifier
    algorithm identifier for the algorithm used to sign the certificate
algo-identifier
    SHA256WITHECDSA | SHA512WITHECDSA
-i|--icountryname issuerCountryName
    Issuer country name
-o|--iorganizationname issuerOrganizationName
    Issuer organization name
-u|--iorganizationalunitname issuerOrganizationalUnitName
    Issuer organizational unit name
-n|--icommonname issuerCommonName
    Issuer common name
-d|--start-date start-date
    start date of certificate in dd-MM-yyyy format
-e|--end-date end-date
    end date of certificate in dd-MM-yyyy format
--scn subjectCountryName
    Subject country name
--son subjectOrganizationName
    Subject organization name
--soun subjectOrganizationalUnitName
    Subject organizational unit name
--scomn subjectCommonName
    Subject common name
--sserial subjectSerialNo
    Subject common name
--isk issuerPrivateKey
    Issuer private key as byte sequence
--ipk issuerPublicKey
    Issuer public key as byte sequence
--icurve issuer curve
    Issuer EC curve name
issuer curve
    NISTP256 | BPP256r1 | FRP256V1
--scurve subject curve
    Subject EC curve name
subject curve
    NISTP256 | BPP256r1 | FRP256V1
--eaki add Extension Authority Key Identifier
--eski add Extension Subject Key Identifier
--eku add Extension Key Usage
-v|--value-eku KeyUsageValue
    Key Usage values
    OR operator '|' can be used for multiple KeyUsage
    e.g.
        0|5
KeyUsageValue
+-----+-----+
| digitalSignature | 0 |
| nonRepudiation   | 1 |
| keyEncipherment  | 2 |
| dataEncipherment | 3 |
| keyAgreement     | 4 |
| keyCertSign      | 5 |
| cRLSign          | 6 |

```

```
| encipherOnly      | 7 |
| decipherOnly     | 8 |
+-----+-----+
```

```
--ecp add Extension Certificate Policies
--cpc is Extension Certificate Policies Critical
--esan-bc ext-san-bCode
        value of extension SubjectAlternateName in binary Code
--ebc add Extension Basic Constraints
--pl-ebc num
        Path Len value for extension Basic Constraints
--eeeku add Extension Extended Key Usage
--ecrldp-bc ext-crldp-bCode
        value of extension CRLDP in binary Code
--enc-bc enc-bCode
        value of extension Name Constraints in binary Code
-f|--fn file-name
        File Name of the Cert to be generated
        If Certificate file exists, command will overwrite the file.

-p|--fp file-path
        Location of the Cert File to be generated
```

This command generates X.509 Certificate.

6.10 crcgen

usage: `crcgen` `[[-o|--order <order>][--p|--polynom <polynom>][--i|--init <init>][--x|--xor <xor>][--r|--print][data [pattern..]]*`

`-o|--order <order>`

order

`<order>` order. The order defines either 16 bit or 32 bit order, it defaults to 16 bit.

`--p|--polynom <polynom>`

A polynomial value chosen to calculate the CRC remainder.

Syntax: `HEX[CHARS[HEX...]]`

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The 'l' switches modes (HEX/CHARS).

Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replace by the number bytes enclosed in the matching braces.

Examples:

`"0102|ab|03 04|cd"` ==> 01 02 61 62 03 04 63 64

`"01#(|ab|#{FE FF})"` ==> 01 05 61 62 02 FE FF

`--i|--init <init>`

A initial value chosen to calculate the CRC remainder.

Syntax: `HEX[CHARS[HEX...]]`

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The 'l' switches modes (HEX/CHARS).

Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replace by the number bytes

enclosed in the matching braces.

Examples:

```
"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64
"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF
```

-x|--xor <xor>

A xor value chosen to calculate the CRC remainder.

Syntax: HEX[[CHARS[[HEX...]]]

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The '|' switches modes (HEX/CHARS). Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replaced by the number bytes enclosed in the matching braces.

Examples:

```
"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64
"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF
```

-r|--print Print the calculated crc value

data Data syntax: HEX[[CHARS[[HEX...]]]

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The '|' switches modes (HEX/CHARS). Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replaced by the number bytes enclosed in the matching braces.

Examples:

```
"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64
"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF
```

pattern.. Description of expected result. Each argument is pattern describing result as sequence of hexdigits with embedded wild cards (*,?,[.]). A leading ! char means NOT. Arguments '&' and '|' are interpreted as the respective logical operators. The result is ok, if it matches the pattern and the logical expression.

This command performs the CRC calculation for a given input data.

Example:

```
crcgen -o 16 -p 1021 -i FFFF -x FFFF
|2140000f843500000A0000C7697C156E774DE1 F005
/echo ${last.return}

crcgen -o 16 -p 1021 -i FFFF -x FFFF |123456789 906E
/echo ${last.return}
```

7. Provider Security Domain Plugin Commands (Delegate Management)

A Provider Security Domain implies a Delegated Management operation in the case of an applet load, install, delete, registry-update or extradite. Those commands are

overloaded in the Provider Security Domain plugin to execute the delegated management operations.

To activate crypto plugin type below commands in JCShell terminal -

```
/register DMPlugin 0001020304 ProviderSecurityDomain
/app DMPlugin
```

7.1 delete

usage: delete [-v|--verify][-p|--package package-name] CAP-file

-v|--verify

Verify delete receipt. This requires the receipt generation key (key set '71', index 1) to be present

-p|--package package-name

Java package name to look for in the CAP-file for finding appropriate Delete Token information.

package-name

Java package name.

CAP-file Name of the CAP-file which holds Delete Token information.

Delete the applet/package with the given AID via a Security Domain with delegated Management privileges. The CAP-file must include appropriate Delete Token information.

7.2 install

usage: install [-v|--verify][-p|--package package-name][-o|--install-only] CAP-file

-v|--verify

Verify install receipt. This requires the receipt generation key (key set '71', index 1) to be present

-p|--package package-name

Java package name to look for in the CAP-file for finding appropriate Install Token information.

package-name

Java package name.

-o|--install-only

Install the applet only, don't make it selectable.

CAP-file Name of the CAP-file which holds Install Token information.

Install an applet via a Security Domain with delegated Management privileges. The CAP-file must include appropriate Install Token information.

7.3 upload

usage: upload [-p|--progress][-c|--components][-r|--random][-l|--package package-name][-b|--block_length length][-v|--verify][-k|--key key][-j|--mode algorithm][-h|--hashAlgo Hash Algorithm][-f|--clfdb][-e|--pad algorithm][-i|--iv vector] CAP-file

-p|--progress

Display upload progress.

-c|--components

Load CAP-file component wise.

-r|--random

Load with random APDU length.

-l|--package package-name
Package name to look for in the CAP-file
package-name
Java package name.

-b|--block_length length
Max. block (APDU) length sent to the card during package upload.
length Max. block (APDU) length.

-v|--verify
Verify the load receipt. This requires the receipt generation key (key set '71', index 1) to be present

-k|--key key
Required only for ciphered load file data block
A sequence of symbols that controls the operation of a cryptographic transformation (e.g., encipherment, decipherment, cryptographic check function computation, signature generation or signature verification)
When mode is RSA, the key will contain the RSA key modulus and exponent.
[len modulus][modulus][len exponent][exponent]

Syntax: HEX[|CHARS[|HEX...]]

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The '|' switches modes (HEX/CHARS). Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replaced by the number bytes enclosed in the matching braces.

Examples:

"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64
"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF

-j|--mode algorithm
Required only for ciphered load file data block
The mode of operation. The size of the provided key determines if single DES or triple DES with 2 or 3 keys has to be performed.
The following modes are supported:
DES - Equivalent with 'DES/ECB'
DES/ECB - Perform an encryption/decryption task in ECB mode (Electronic Code Book), i.e. each input block is encrypted/decrypted on its own.
DES/CBC - Perform an encryption/decryption task in CBC mode (Cipher Block Chaining), i.e. the input of any subsequent block depends on the cryptographic output of the previous block.
AES - Equivalent with 'AES/ECB'
AES/ECB - Perform an encryption/decryption task in ECB mode (Electronic Code Book), i.e. each input block is encrypted/decrypted on its own.
AES/CBC - Perform an encryption/decryption task in CBC mode (Cipher Block Chaining), i.e. the input of any subsequent block depends on the

cryptographic output of the previous block.

Note: Only 'NOPAD' and 'M4' are valid padding modes for AES algorithm.

RSA - Perform an encryption/decryption task in RSA Mode

Note : Only 'NOPAD' and 'PKCS1' are valid padding modes for RSA algorithm.

-h|--hashAlgo Hash Algorithm

Option is used to decide what SHA digest to be used.

Hash Algorithm

Specifies whether SHA1/ SHA-256 digest to be used.

-f|--clfdb Send encrypted load file data block.

Mandatory Option(s): -k key

Optional Option(s): -j & -e.

-e|--pad algorithm

Required only for ciphered load file data block

Defines the padding algorithm to be applied.

If this mode is used in encipherment operation, the provided plaintext data will be padded accordingly. If this mode is used in decipherment operation, the deciphered plaintext data will be checked if the correct padding has been applied.

The following padding algorithms are available:

'NOPAD' - No padding will be performed, i.e. the provided data will be provided unaltered to the cipher engine and thus it must be blocksize aligned.

'M2' - The data is right padded with a single '1' bit plus as few as possible '0' bits necessary to obtain a data string whose length is a positive integer multiple of the blocksize.

'M4' - Append an 0x80 to the right of the data block, If the resultant data block length is a multiple of BLOCK SIZE, no further padding is required. Append binary zeroes to the right of the data block until the data block length is a multiple of BLOCK SIZE
NOTE : It shall be used only with AES.

'PKCS1' - Padding according to PKCS#1 v1.5 std. It will be used ONLY in case of RSA

This parameter is optional, if it is not present, it defaults to M2.

-i|--iv vector

Required only for ciphered load file data block

The initialization vector to seed the initial transformation function which is applied at the beginning of the MAC algorithm.

Syntax: HEX[[CHARS[[HEX...]]]

where HEX is a possibly empty sequence of pairs of hex digits forming a byte sequence. CHARS is a possibly empty sequence of characters. Their ASCII encoding forms a sequence of bytes. The '|' switches modes (HEX/CHARS).

Additionally, nested elements of the form '#(...)' are parsed. Instead of () you can also use the braces [], {}, <>. The '#' element is replaced by the number bytes enclosed in the matching braces.

Examples:

```
"0102|ab|03 04|cd" ==> 01 02 61 62 03 04 63 64
"01#(|ab|#{FE FF})" ==> 01 05 61 62 02 FE FF
```

CAP-file Name of the CAP-file to be loaded.

Load a package onto the card via a Security Domain with delegated Management privileges. The CAP-file must include appropriate Load Token information.

7.4 registry-update

usage: registry-update [-v|--verify][-p|--package package-name] CAP-file

-v|--verify

Verify Registry update receipt. This requires the receipt generation key (key set '71', index 1) to be present

-p|--package package-name

Java package name to look for in the CAP-file for finding appropriate Registry update Token information.

package-name

Java package name.

CAP-file Name of the CAP-file which holds Registry update Token information.

Registry update process allows GlobalPlatform Registry data associated with an Application, such as Privileges and Implicit Selection parameters, to be modified. This process also allows the restricting of Card Content Management functionality of a specific Security Domain or OPEN itself. The CAP-file must include appropriate Registry Update Token information.

7.5 extradite

usage: extradite [-v|--verify][-p|--package package-name] CAP-file

-v|--verify

Verify extradite receipt. This requires the receipt generation key (key set '71', index 1) to be present

-p|--package package-name

Java package name to look for in the CAP-file for finding appropriate Extradite Token information.

package-name

Java package name.

CAP-file Name of the CAP-file to be extradited.

Extradite an applet to another Security Domain. Corresponds with the Global Platform Install [for extradition] command. The CAP-file must include appropriate Extradite Token information.

8. Simulator Plugin Commands

Simulator plugin supports the commands that are specific to Simulator. To activate Simulator plugin type below commands in JCShell terminal -

```
/app JCOPSimu
```

8.1 simu-reset

usage: `simu-reset [[-c|--counter <JavaCard bytecode counter>]]`
`-c|--counter <JavaCard bytecode counter>`
Bytecode counter value should be non-zero.
`<JavaCard bytecode counter>`
Number of bytecodes to be executed.

Command to reset the Simulator after execution of specified number of bytecodes.

8.2 simu-restart

usage: `simu-restart [[-c|--counter <JavaCard bytecode counter>]]`
`-c|--counter <JavaCard bytecode counter>`
Bytecode counter value should be non-zero.
`<JavaCard bytecode counter>`
Number of bytecodes to be executed.

Command to restart the Simulator after execution of specified number of bytecodes.

8.3 simu-terminate

usage: `simu-terminate [[-c|--counter <JavaCard bytecode counter>]]`
`-c|--counter <JavaCard bytecode counter>`
Bytecode counter value should be non-zero.
`<JavaCard bytecode counter>`
Number of bytecodes to be executed.

Command to terminate the Simulator after execution of specified number of bytecodes.

8.4 simu-change-protocol

usage: `simu-change-protocol [-t|--pt PROTOCOL_TYPE][--m|--pm PROTOCOL_MEDIA]`

`-t|--pt PROTOCOL_TYPE`
Protocol Type provided in hex, in Range `<0x0-0xF>`
`--m|--pm PROTOCOL_MEDIA`
Protocol Media provided in hex, in Range `<0x0-0xF>`

Changes the Simulator behaviour
Changes the return value of `APDU.getProtocol()`
Valid Values are as:

PROTOCOL_MEDIA	PROTOCOL_TYPE	
0x0	0x0	T=0, contact mode
0x0	0x1	T=1, contact mode, block mode
0x8	0x0	T=CL, TYPE_A, T0
0x8	0x1	T=CL, TYPE_A, T1
0x9	0x0	T=CL, TYPE_B, T0
0x9	0x1	T=CL, TYPE_B, T1

Example : `simu-change-protocol -m 0x0 -t 0x0`

8.5 simu-dump-memory

usage: `simu-dump-memory [[-r|--reset][--w|--warm-reset][--s|--restart][--t|--terminate][--c|--counter <JavaCard bytecode counter>]]`
`-r|--reset` Dump Memory with cold reset
`--w|--warm-reset`

Dump Memory with warm reset
-s|--restart
Dump Memory and Restart
-t|--terminate
Dump Memory and Terminate
-c|--counter <JavaCard bytecode counter>
Bytecode counter value should be non-zero.
<JavaCard bytecode counter>
Number of bytecodes to be executed.

Command to create memory dump file with or without reset/restart/termination of the Simulator after execution of specified number of bytecodes.
The memory dump file path is stored in JCShell variable 'last.simu.dump.memory.file'.
Note: warm and cold reset options(-w, -r) are mutually exclusive.

8.6 simu-get-dump-path

usage: *simu-get-dump-path*

Command to get the absolute path including filename of latest memory dump file of the Simulator.
The memory dump file path is stored in JCShell variable 'last.simu.dump.memory.file'.

8.7 start-tearing

usage: *start-tearing* [[-n|--nad <node address>]]
-n|--nad <node address>
Indicates the node address.
NAD is applicable only when Remote Terminal is opened.
<node address>
node address

Start tearing operation.

Example : start-tearing -n 1

8.8 stop-tearing

usage: *stop-tearing* [[-n|--nad <node address>]]
-n|--nad <node address>
Indicates the node address.
NAD is applicable only when Remote Terminal is opened.
<node address>
node address

Stop tearing operation

Example : stop-tearing -n 1

8.9 simu-load-memory-dump

usage: *simu-load-memory-dump* [[-f|--file memory dump file]]
-f|--file memory dump file
Absolute Path including filename of memory dump to be loaded.
memory dump file
memory dump file created as part of simu-dump-memory command.

Command to load memory dump file without termination of the Simulator after execution of simu-dump-memory command.

9. eUICC Plugin commands

eUICC plugin supports the commands that are specific to eUICC features. To activate eUICC plugin type below commands in JCShell terminal –

```
/register euicc 001122334455 Euicc
/app euicc
```

9.1 authenticate

usage: authenticate [-x|--sec-context][--n|--ak-required][--m|--resync-required][--sqnms sqnms][--c|--kc-required][--e|--key-length key-length][--g|--mac-length mac-length][--h|--ck-length ck-length][--d|--ik-length ik-length][--l|--res-len res-length][--i|--keccak-interactions keccak-interactions][--c1 XOR-c1][--c2 XOR-c2][--c3 XOR-c3][--c4 XOR-c4][--c5 XOR-c5][--r1 rotation-constant-r1][--r2 rotation-constant-r2][--r3 rotation-constant-r3][--r4 rotation-constant-r4][--r5 rotation-constant-r5][--o|--op OPorTOP][--j|--opc OPcOrTOPc][--a|--algo algorithm][--k|--key simkey][--r|--random rand][--s|--sqn sqn][--f|--amf amf]

-x|--sec-context

Flag for 2G Security Context. Default 3G Security Context is used.

--n|--ak-required

Anonymity key required

--m|--resync-required

Re-synch mode

--sqnms sqnms

SQNms for Re-Sync mode

sqnms SQNms for Re-Sync mode

--c|--kc-required

Kc Flag

--e|--key-length key-length

Key Length for Tuak in bits

key-length Key Length for Tuak in bits

--g|--mac-length mac-length

MAC Length for Tuak in bits

mac-length MAC Length for Tuak in bits

--h|--ck-length ck-length

CK Length for Tuak in bits

ck-length CK Length for Tuak in bits

--d|--ik-length ik-length

IK Length for Tuak in bits

ik-length IK Length for Tuak in bits

--l|--res-len res-length

RES Length incase of TUAK algorithm

res-length RES Length for TUAK in bits

--i|--keccak-interactions keccak-interactions

keccak function iterations for Tuak

keccak-interactions

keccak function iterations for Tuak

--c1 XOR-c1

XOR Constant 1

XOR-c1 XOR Constant 1

--c2 XOR-c2

XOR Constant 2

XOR-c2 XOR Constant 2

--c3 XOR-c3

XOR Constant 3

XOR-c3 XOR Constant 3

--c4 XOR-c4

```

        XOR Constant 4
    XOR-c4 XOR Constant 4
--c5 XOR-c5
        XOR Constant 5
    XOR-c5 XOR Constant 5
--r1 rotation-constant-r1
        Rotation Constant r1
rotation-constant-r1
        Rotation Constant r1
--r2 rotation-constant-r2
        Rotation Constant r2
rotation-constant-r2
        Rotation Constant r2
--r3 rotation-constant-r3
        Rotation Constant r3
rotation-constant-r3
        Rotation Constant r3
--r4 rotation-constant-r4
        Rotation Constant r4
rotation-constant-r4
        Rotation Constant r4
--r5 rotation-constant-r5
        Rotation Constant r5
rotation-constant-r5
        Rotation Constant r5
-o|--op OPorTOP
        OP for Milenage or TOP for Tuak
    OPorTOP OP for Milenage or TOP for Tuak
-j|--opc OPcOrTOPc
        OPc for Milenage or TOPc for Tuak
    OPcOrTOPc OPc for Milenage or TOPc for Tuak
-a|--algo algorithm
        Algorithm to use, valid options - Tuak or Milenage
    algorithm Algorithm - Tuak/Milenage
-k|--key simkey
        SIM Key
    simkey SIM key
-r|--random rand
        Random Number
    rand Random Number
-s|--sqn sqn
        Sequence number
    sqn Sequence number
-f|--amf amf
        Authentication management field
    amf Authentication management field

```

Command to generate AUTHENTICATE command and expected response for Milenage or Tuak Algorithm.

9.2 get-euicc-info

usage: *get-euicc-info* *[[--p|--payload get-euicc-info2-payload]]* *[[--t|--tag get-euicc-info2-tag]]*
 --p|--payload get-euicc-info2-payload
 Payload to get EUICC info2 (optional)
 get-euicc-info2-payload
 Payload for EUICC info2

```
-t|--tag get-euicc-info2-tag
    This is option for EUICC info2 tag value. (optional)
get-euicc-info2-tag
    EUICC info2 tag value
```

EUICC Profile download get-euicc-info2 command specification.

```
example ::
    get-euicc-info
```

9.3 get-euicc-challenge

```
usage: get-euicc-challenge [--p|--payload get-euicc-challenge-payload][--t|--tag get-euicc-
challenge-tag]]
-p|--payload get-euicc-challenge-payload
    Payload to get EUICC Challenge (optional)
get-euicc-challenge-payload
    Payload for EUICC Challenge
-t|--tag get-euicc-challenge-tag
    This is option for EUICC Challenge tag value. (optional)
get-euicc-challenge-tag
    EUICC Challenge tag value
```

EUICC Profile download get-euicc-challenge command specification.

```
example ::
    get-euicc-challenge
```

9.4 authenticate-server

```
usage: authenticate-server [--s|--serverSignature1 serverSignature1][--euiccInfo2
euiccInfo2][--e|--euiccChallenge euiccChallenge][--f|--eccCurveTypeUsed
eccCurveTypeUsed][--r|--transactionId transactionId][--a|--serverAddress
serverAddress][--c|--serverChallenge serverChallenge][--p|--euiccCiPKIdToBeUsed
euiccCiPKIdToBeUsed][--f|--serverCertificate serverCertificate][--certificatePrivateKey
certificatePrivateKey][--certificatePrivateKeyPemPin certificatePrivateKeyPemPin][--m|--
matchingId matchingId][--t|--tac tac][--g|--gsmSupportedRelease
gsmSupportedRelease][--u|--utranSupportedRelease utranSupportedRelease][--o|--
cdma2000onexSupportedRelease cdma2000onexSupportedRelease][--h|--
cdma2000hrpdSupportedRelease cdma2000hrpdSupportedRelease][--d|--
cdma2000ehrpdsupportedRelease cdma2000ehrpdsupportedRelease][--n|--
eutranSupportedRelease eutranSupportedRelease][--l|--contactlessSupportedRelease
contactlessSupportedRelease][--v|--rspCrISupportedVersion rspCrISupportedVersion][--i|--
imei imei]]*
-s|--serverSignature1 serverSignature1
    ServerSignature1 for Authenticate Server. (optional)
serverSignature1
    ServerSignature1
--euiccInfo2 euiccInfo2
    EUICC Info2 (optional)
euiccInfo2 euiccInfo2
-e|--euiccChallenge euiccChallenge
    EUICC Challenge (optional)
euiccChallenge
    EuiccChallenge
-f|--eccCurveTypeUsed eccCurveTypeUsed
    ECC_curve_type_Used, valid values - NISTP256, BPP256r1, FRP256V1
eccCurveTypeUsed
```

```

        ECC_curve_type_Used
-r|--transactionId transactionId
    Transaction Id (optional)
transactionId
    TransactionId
-a|--serverAddress serverAddress
    ServerAddress to form ServerSigned1.
serverAddress
    ServerAddress
-c|--serverChallenge serverChallenge
    ServerChallenge to form ServerSigned1.
serverChallenge
    ServerChallenge
-p|--euiccCiPKIdToBeUsed euiccCiPKIdToBeUsed
    EuiccCiPKIdToBeUsed for Authenticate Server.
euiccCiPKIdToBeUsed
    EuiccCiPKIdToBeUsed
-f|--serverCertificate serverCertificate
    RSP Server Certificate CERT.XXauth.ECDSA file path.
serverCertificate
    Server Certificate
--certificatePrivateKey certificatePrivateKey
    CertificatePrivateKey of CERT.XXauth.ECDSA used to generate
serverSignature1.
certificatePrivateKey
    CertificatePrivateKey
--certificatePrivateKeyPemPin certificatePrivateKeyPemPin
    CertificatePrivateKey PEM file pin.
certificatePrivateKeyPemPin
    CertificatePrivateKeyPemPin
-m|--matchingId matchingId
    MatchingId for CtxParamsForCommonAuthentication.
matchingId MatchingId
-t|--tac tac
    TAC for DeviceInfo.
tac TAC
-g|--gsmSupportedRelease gsmSupportedRelease
    GsmSupportedRelease for DeviceCapabilities.
gsmSupportedRelease
    GsmSupportedRelease
-u|--utranSupportedRelease utranSupportedRelease
    UtranSupportedRelease for DeviceCapabilities.
utranSupportedRelease
    UtranSupportedRelease
-o|--cdma2000onexSupportedRelease cdma2000onexSupportedRelease
    Cdma2000onexSupportedRelease for DeviceCapabilities.
cdma2000onexSupportedRelease
    Cdma2000onexSupportedRelease
-h|--cdma2000hrpdSupportedRelease cdma2000hrpdSupportedRelease
    Cdma2000hrpdSupportedRelease for DeviceCapabilities.
cdma2000hrpdSupportedRelease
    Cdma2000hrpdSupportedRelease
-d|--cdma2000ehrpdsupportedRelease cdma2000ehrpdsupportedRelease
    Cdma2000ehrpdsupportedRelease for DeviceCapabilities.
cdma2000ehrpdsupportedRelease
    Cdma2000ehrpdsupportedRelease
-n|--eutranSupportedRelease eutranSupportedRelease

```

```

        EutranSupportedRelease for DeviceCapabilities.
eutranSupportedRelease
        EutranSupportedRelease
-l|--contactlessSupportedRelease contactlessSupportedRelease
        ContactlessSupportedRelease for DeviceCapabilities.
contactlessSupportedRelease
        ContactlessSupportedRelease
-v|--rspCrISupportedVersion rspCrISupportedVersion
        RspCrISupportedVersion for DeviceCapabilities.
rspCrISupportedVersion
        RspCrISupportedVersion
-i|--imei imei
        Imei for Device Info.
        imei Imei

```

EUICC Profile download Authentication with Server command specification.

example ::

```

authenticate-server --serverSignature1
197D40C869948689B6A9405BDF0AA039DAB09679FDA93B9DC1EDF8ACA22439A83AB8F59
551921F81A869D43A7424184DEE2458024D1CD7EDEB58853E404B1E57 --euiccInfo2
BF20358203020100A91604142F434E910B9D3A5BD01F2B03B5955864FEC847A0AA16041
42F434E910B9D3A5BD01F2B03B5955864FEC847A0 --euiccChallenge
3F200EF20C574616DE0C37A6F50C6C54 --eccCurveTypeUsed NISTP256 --
transactionId D6529B0DDC8974356F40FD7ADE1742F4 --serverAddress
77777396D63726F736F667433636F6D --serverChallenge
03A7ACDC5F69DD3DE3B4B985658A29BD --euiccCiPKIdToBeUsed
2F434E910B9D3A5BD01F2B03B5955864FEC847A0 --serverCertificate
"F:\jcop_dev_tools_trunk\docs\Architecture\ProfileDownload\SGP.26 1.1
files\Valid test cases\SM-DP+\DPauth\CERT_S_SM_DPauth_ECDSA_NIST.der" -
-certificatePrivateKey
"F:\jcop_dev_tools_trunk\docs\Architecture\ProfileDownload\SGP.26 1.1
files\Valid test cases\SM-DP+\DPauth\SK_S_SM_DPauth_ECDSA_NIST.pem" --
certificatePrivateKeyPemPin "" --matchingId 1122334455667788 --tac
11223344 --gsmSupportedRelease 010000 --utranSupportedRelease 010000 -
-cdma2000onexSupportedRelease 010000 --cdma2000hrpdSupportedRelease
010000 --cdma2000ehrpdsupportedRelease "" --eutranSupportedRelease
010000 --contactlessSupportedRelease 010000 --rspCrISupportedVersion
010000 --imei 1122334455667788

```

9.5 bound-profile-package

usage: bound-profile-package [[-d|--dpOid dpOid]] [--i|--iconType iconType] [--icon icon] [--o|--notificationConfigurationInfo notificationConfigurationInformation] [--O|--profileOwner profileOwner] [--p|--profilePolicyRules profilePolicyRules] [--P|--profileProtectionKeysFlag] [--M|--initialMacChainingValue initialMacChainingValue] [--E|--ppkEnc ppkEnc] [--C|--ppkCmac ppkCmac] [--t|--transactionId transactionId] [--r|--remoteOpId remoteOpId] [--certificatePrivateKey certificatePrivateKey] [--k|--euiccOtpk euiccOtpk] [--s|--smdpSign smdpSign] [--e|--eccCurveTypeUsed eccCurveTypeUsed] [--T|--keyType keyType] [--L|--keyLen keyLen] [--h|--hostId hostId] [--I|--iccid iccid] [--n|--serviceProviderName serviceProviderName] [--N|--profileName profileName] [--c|--profileClass profileClass] [--eld eld] [--f|--profileElementDataDerFilePath profileElementDataDerFilePath]*
-d|--dpOid dpOid
OID in the tree of the SM-DP+ that created the Profile.

```

dpOid DpOid
-i|--iconType iconType
    Icon type for StoreMetadataRequest
iconType IconType
--icon icon
    Data of the icon for StoreMetadataRequest as byte sequence. Size 64 x 64 pixel.
    icon Icon
-o|--notificationConfigurationInfo notificationConfigurationInformation
    TLV structured NotificationConfigurationInformation for StoreMetadataRequest.
    TLV includes tag for NotificationConfigurationInformation.
notificationConfigurationInformation
    NotificationConfigurationInformation
-O|--profileOwner profileOwner
    Profile Owner for StoreMetadataRequest as byte sequence.
profileOwner
    ProfileOwner
-p|--profilePolicyRules profilePolicyRules
    Notification Address for StoreMetadataRequest as byte sequence.
profilePolicyRules
    ProfilePolicyRules
-P|--profileProtectionKeysFlag
    Profile Protection Keys Flag for Bound Profile Package.
-M|--initialMacChainingValue initialMacChainingValue
    The new initial MAC chaining value for ReplaceSessionKeysRequest
initialMacChainingValue
    InitialMacChainingValue
-E|--ppkEnc ppkEnc
    New session key value for encryption/decryption (PPK-ENC) for
ReplaceSessionKeysRequest
    ppkEnc PpkEnc
-C|--ppkCmac ppkCmac
    PpkCmac for StoreMetadataRequest
    ppkCmac PpkCmac
-t|--transactionId transactionId
    The TransactionID generated by the SM-DP+ as byte sequence.(optional)
transactionId
    TransactionId
-r|--remoteOpId remoteOpId
    Remote Operation Type Identifier2 as byte sequence.
remoteOpId RemoteOpId
--certificatePrivateKey certificatePrivateKey
    CertificatePrivateKey of SK.DPpb.ECDSA used to generate smdpSign.
    PrivateKey as byte sequence or pem file containing PrivateKey.
certificatePrivateKey
    CertificatePrivateKey
-k|--euiccOtpk euiccOtpk
    EuiccOtpk to generate SharedSecretKeyAgreement as byte sequence.(optional)
euiccOtpk EuiccOtpk
-s|--smdpSign smdpSign
    SM-DP's signature for InitialiseSecureChannelRequest as byte
sequence.(optional)
    smdpSign SmdpSign
-e|--eccCurveTypeUsed eccCurveTypeUsed
    ECC_curve_type_Used, valid values - NISTP256, BPP256r1, FRP256V1
eccCurveTypeUsed
    ECC_curve_type_Used
-T|--keyType keyType

```

Key type according to GlobalPlatform Card Specification (hex value).
keyType KeyType
-L|--keyLen keyLen
Key length in number of bytes. For current specification key length SHALL by 0x10 bytes.
keyLen KeyLen
-h|--hostId hostId
Host ID value as byte sequence.
hostId HostId
-I|--iccid iccid
Iccid value for StoreMetadataRequest as byte sequence.
iccid Iccid
-n|--serviceProviderName serviceProviderName
Service provider name for StoreMetadataRequest as byte sequence.
serviceProviderName
ServiceProviderName
-N|--profileName profileName
Profile name for StoreMetadataRequest as byte sequence.
profileName
ProfileName
-c|--profileClass profileClass
Profile class for StoreMetadataRequest as byte sequence.
profileClass
ProfileClass
--eld eld eld
eld Eld
-f|--profileElementDataDerFilePath profileElementDataDerFilePath
Profile Element Data DER File Path.
profileElementDataDerFilePath
Profile Element Data DER File Path.

Bound Profile Package bound-profile-package command specification.

example ::

```
bound-profile-package --profileProtectionKeysFlag --
initialMacChainingValue 84E43D2CCD5CC0EC1661FD7665D0FCF0 --ppkEnc
3EF530B078D83A6012A34106B10AF404 --ppkCmac
D3BE93719827BD8821041B8C7CB9BA05 --transactionId
D6529B0DDC8974356F40FD7ADE1742F4 --remoteOpId 01 --
certificatePrivateKey
"F:\jcop_dev_tools_trunk\docs\Architecture\ProfileDownload\SGP.26 1.1
files\Valid test cases\SM-DP+\DPpb\CERT_S_SM_DPpb_ECDSA_NIST.der" --
euiccOtpk
04A4A2FC0E2689D16342B78DF02314C804FD1DF9C65B89972879B88196A08472477D48D
DA543C3C5E585166B4A542697CDD540D1525FF51366C1FF42FE56E0210A --smdpSign
20B6E71D20060FBEAD3F7B68A44DBCACF3AC264589235F7C2ED239F22C9FC2C351EBC23
862204328B86FE5F03731B3788BCCA5B0578CA196D51A39163865CFD9 --
eccCurveTypeUsed NISTP256 --keyType 88 --keyLen 10 --hostId
1122334455667788 --iccid 112233445566778899AA --serviceProviderName
41495254454C --profileName 4F7065726174696F6E616C2D31 --profileClass 02
--eId 8900112233445566778899AABBCCDDEE --profileElementDataDerFilePath
"F:\jcop_dev_tools_trunk\docs\Architecture\ProfileDownload\TelecomProfi
le.der"
```

9.6 prepare-download

usage: prepare-download *[[--x|--pk-euicc-ecdsa-x PK.EUICC.ECDSA.X][--y|--pk-euicc-ecdsa-x PK.EUICC.ECDSA.Y][--pk-euicc-ecdsa-pem-file PK.EUICC.ECDSA PEM file][--pk-euicc-ecdsa-pem-file-pin PK.EUICC.ECDSA PEM file Pin][--transactionId transactionId][--f|--ccRequiredFlag][--b|--bppEuiccOtpk bppEuiccOtpk][--s|--smdpSignature2 smdpSignature2][--h|--hashCc hashCc][--d|--certDPpbPath certDPpbPath][--certificatePrivateKey certificatePrivateKey][--certificatePrivateKeyPemPin certificatePrivateKeyPemPin][--c|--eccCurveTypeUsed eccCurveTypeUsed]]**

--x|--pk-euicc-ecdsa-x PK.EUICC.ECDSA.X
X component of PK.EUICC.ECDSA
PK.EUICC.ECDSA.X
PK.EUICC.ECDSA X component

--y|--pk-euicc-ecdsa-x PK.EUICC.ECDSA.Y
Y component of PK.EUICC.ECDSA
PK.EUICC.ECDSA.Y
PK.EUICC.ECDSA Y component

--pk-euicc-ecdsa-pem-file PK.EUICC.ECDSA PEM file
PEM file of PK.EUICC.ECDSA
PK.EUICC.ECDSA PEM file
PK.EUICC.ECDSA PEM FILE

--pk-euicc-ecdsa-pem-file-pin PK.EUICC.ECDSA PEM file Pin
CertificatePrivateKey PEM file pin.
PK.EUICC.ECDSA PEM file Pin
PK.EUICC.ECDSA PEM FILE Pin

--transactionId transactionId
Transaction Id (optional)
transactionId
TransactionId

--f|--ccRequiredFlag
CC Required Flag for SmdpSigned2.

--b|--bppEuiccOtpk bppEuiccOtpk
Bpp Euicc Otpk for SmdpSigned2.
bppEuiccOtpk
BppEuiccOtpk

--s|--smdpSignature2 smdpSignature2
SmdpSignature2 for Prepare Download Request.
smdpSignature2
SmdpSignature2

--h|--hashCc hashCc
HashCc for Prepare Download Request.
hashCc HashCc

--d|--certDPpbPath certDPpbPath
CertDPpbPath for Prepare Download Request.
certDPpbPath
CertDPpbPath for Prepare Download Request.

--certificatePrivateKey certificatePrivateKey
CertificatePrivateKey of SK.DPpb.ECDSA used to generate smdpSignature2.
certificatePrivateKey
CertificatePrivateKey

--certificatePrivateKeyPemPin certificatePrivateKeyPemPin
CertificatePrivateKey PEM file pin.
certificatePrivateKeyPemPin
CertificatePrivateKeyPemPin

--c|--eccCurveTypeUsed eccCurveTypeUsed
ECC_curve_type_Used, valid values - NISTP256, BPP256r1, FRP256V1
eccCurveTypeUsed

ECC_curve_type_Used

EUICC Profile download prepare-download command specification.

example ::

```
prepare-download --pk-euicc-ecdsa-pem-file
"F:\jcop_dev_tools_trunk\docs\Architecture\ProfileDownload\SGP.26 1.1
files\Valid test cases\euicc\SK_EUICC_ECDSA_NIST.pem" --transactionId
D6529B0DDC8974356F40FD7ADE1742F4 --ccRequiredFlag --bppEuiccOtpk
04A4A2FC0E2689D16342B78DF02314C804FD1DF9C65B89972879B88196A08472477D48D
DA543C3C5E585166B4A542697CDD540D1525FF51366C1FF42FE56E021 --
smdpSignature2 "" --hashCc
1234567890123456789012345678123412345678901234567890123456781234 --
certDPpbPath
"F:\jcop_dev_tools_trunk\docs\Architecture\ProfileDownload\SGP.26 1.1
files\Valid test cases\SM-DP+\DPpb\CERT_S_SM_DPpb_ECDSA_NIST.der" --
certificatePrivateKey
"F:\jcop_dev_tools_trunk\docs\Architecture\ProfileDownload\SGP.26 1.1
files\Valid test cases\SM-DP+\DPauth\SK_S_SM_DPauth_ECDSA_NIST.pem" --
certificatePrivateKeyPemPin "" --eccCurveTypeUsed NISTP256
```

9.7 SCP80-send

usage: SCP80-send [-r|--spi-crypto SPI first byte crypto][-c|--spi-cipher SPI first byte cipher][-n|--spi-counter SPI first byte counter][-o|--spi-por SPI second byte PoR][-y|--spi-por-crypto SPI second byte PoR crypto][-h|--spi-por-cipher SPI second byte cipher][-m|--por-response Type of PoR response][-k|--kic kicValue][-i|--kid kidValue][-s|--keys-kic keys required for Kic][-e|--keys-kid keys required for Kid][-v|--kvnKic Key Version Number][-b|--kvnKid Key Version Number][--implicit-algo-kic Algoirthm value for implicitly known algorithms in KIC][--implicit-algo-kid Algoirthm value for implicitly known algorithms in KID][-t|--tar Toolkit Application Reference][-u|--counter Indicates replay detection and sequence integrity counter][--ms TP-More-Messages-To-Send(TP-MMS)][--lp TP-Loop-Prevention(TP-LP)][--rp TP-Reply-Path(TP-RP)][--udhi TP-User-Data-Header-Indicator(TP-UDHI)][--sri TP-Status-Report-Indication(TP-SRI)][--oa TP-Originating-Address(TP-OA)][-p|--pkt-id Packet Identifier/Reference Number][-f|--format-type Format Type][--scts TP-Service-Centre-Time-Stamp(TP-SCTS)][--dcs TP-Data-Coding-Scheme(TP-DCS)][--pd-id TP-Protocol-Identifier(TP-PID)][--sms Type of SMS][--sca Service Centre Address][-g|--get-response][-d|--data {Secured data}..]

-r|--spi-crypto SPI first byte crypto

Possible values are:

- 0 (no security)
- 1 (Redundancy Checksum)
- 2 (Cryptographic Checksum)

-c|--spi-cipher SPI first byte cipher

Possible values are:

- true (cipher)
- false (no cipher)

-n|--spi-counter SPI first byte counter

Possible values are:

- 0 (no counter available)
- 1 (counter available)
- 2 (Process if and only if counter value is higher than the value in RE)
- 3 (Process if and only if counter value is one higher than the value in RE)

-o|--spi-por SPI second byte PoR

Possible values are:

- 0 (No PoR reply to Sending Entity)
- 1 (PoR required to be sent to SE)
- 2 (PoR required only when an error has occurred)

-y|--spi-por-crypto SPI second byte PoR crypto

Values are:

- 0 (no security)
- 1 (Redundancy Checksum)
- 2 (Cryptographic Checksum)

-h|--spi-por-cipher SPI second byte cipher

Possible values are:

- true (cipher)
- false (no cipher)

-m|--por-response Type of PoR response

The bit 6 of the second octet of spi is used for SMS only.

The possible values:

- 0 (PoR response shall be sent using SMS-DELIVER-REPORT)
- 1 (PoR response shall be sent using SMS-SUBMIT)

This parameter is optional, if not given default value 0 is taken

-k|--kic kicValue

It suggests the algorithm for ciphering.

Possible algorithms supported are:

- 0 (Single DES in CBC mode)
- 1 (Triple DES in outer-CBC mode using two different keys)
- 2 (Triple DES in outer-CBC mode using three different keys)
- 3 (Single DES in ECB mode)
- 4 (AES in CBC mode)
- 5 (Algorithm known implicitly by both entities)
- 6 (Proprietary implementation) - Currently not supported

-i|--kid kidValue

It suggests CC or RC.

Possible options supported are:

0 (Single DES in CBC mode)

1 (Triple DES in outer-CBC mode using two different keys)

2 (Triple DES in outer-CBC mode using three different keys)

3 (Single DES in ECB mode)

4 (AES in CMAC mode)

5 (For redundancy check - CRC16)

6 (For redundancy check - CRC32)

7 (Algorithm known implicitly by both entities)

8 (Proprietary implementation) - Currently not supported

-s|--keys-kic keys required for Kic

Keys which shall be used for ciphering,takes hex values

Key length(AES) : 16,24 or 32

Key length(DES) : 8,16 or 24

If Ciphering option is provided, this option is mandatory.

-e|--keys-kid keys required for Kid

Keys which shall be used for CC,takes hex values

Key length(AES) : 16,24 or 32

Key length(DES) : 8,16 or 24

If Checksum option is provided, this option is mandatory.

-v|--kvnKic Key Version Number

The value is of 1 byte and the range is 01-0F,takes integer value.

Hex value can be provided by prepending 0x

-b|--kvnKid Key Version Number

The value is of 1 byte and the range is 01-0F,takes integer value.

Hex value can be provided by prepending 0x

--implicit-algo-kic Algorithm value for implicitly known algorithms in KIC

it suggests the algorithm to be used for KIC --kic is 5

Possible values are: AES, DES

--implicit-algo-kid Algorithm value for implicitly known algorithms in KID

it suggests the algorithm to be used for KID --kid is 7

Possible values are: AES, DES

-t|--tar Toolkit Application Reference

The TAR value must be of 3 octets/bytes,takes hex values. Example: If TAR value given is 01, then zeroes will be prefixed to make it 000001(3 bytes)

-u|--counter Indicates replay detection and sequence integrity counter

The counter value must be of 5 octets/bytes,takes hex values Example: If counter is given as 02, then zeroes will be prefixed to make it 0000000005(5 bytes)

--ms TP-More-Messages-To-Send(TP-MMS)

Parameter indicating whether or not there are more messages to send(Optional).

Possible values:

0 (more messages are waiting)

1 (no more messages are waiting)

This parameter is optional, if not given default value 1 is taken

--lp TP-Loop-Prevention(TP-LP)

TP-Loop-Prevention(Optional).

The possible values:

0 (message not spawned and forwarded)

1 (message is either spawned or forwarded)

This parameter is optional, if not given default value 0 is taken

--rp TP-Reply-Path(TP-RP)

Parameter indicating that reply path exists.(Optional).

The possible values:

0 (TP-Reply-Path parameter is not set in this SMS-DELIVER)

1 (TP-Reply-Path parameter is set in this SMS-DELIVER)

This parameter is optional, if not given default value 0 is taken

--udhi TP-User-Data-Header-Indicator(TP-UDHI)

TP-User-Data-Header-Indicator(TP-UDHI)(Optional).

The possible values:

0 (contains only short message)

1 (contains header in addition to short message)

This parameter is optional, if not given default value 1 is taken

--sri TP-Status-Report-Indication(TP-SRI)

Parameter indicating if the SME has requested a status report.(Optional)

The possible values:

0 (status report shall not be returned to SME)

1 (status report shall be returned to SME)

This parameter is optional, if not given default value 0 is taken

--oa TP-Originating-Address(TP-OA)

Address of originating SME(Optional).

The value must be of length 2-12 octets. This parameter is optional,default value 038177F9 is taken when not given

- p|--pkt-id Packet Identifier/Reference Number
 - (Optional) Range is 01 - FF,takes hex values
 - Optional parameter and default value is 2F if not given
- f|--format-type Format Type
 - The data can be formatted as compact and expandable(optional).
 - Default is 'compact',The options for expandable format are definite and indefinite.
 - Only Command TLV is supported for indefinite format
 - Possible values:
 - definite
 - indefinite
- scts TP-Service-Centre-Time-Stamp(TP-SCTS)
 - Parameter identifying time when the SC received the message(Optional).
 - This parameter is optional, if not given default value 40707061153500 is taken. It should be of 7 octets, takes hex values.
- dcs TP-Data-Coding-Scheme(TP-DCS)
 - Parameter identifying the coding scheme within the user data.(Optional)
 - This parameter is optional and takes hex values, if not given default value F6 is taken
- pd-id TP-Protocol-Identifier(TP-PID)
 - Parameter identifying the above layer protocol if any(Optional).
 - This parameter is optional and takes hex values, if not given default value 7F is taken
- sms Type of SMS
 - Currently only SMS-PP is supported.(Optional).
 - If this parameter is not given,default SMSPP is taken.
 - Possible value: SMSPP
- sca Service Centre Address
 - Optional paramater
 - Takes hex values, default is 91198901509141
- g|--get-response
 - This option indicates the GET RESPONSE to be issued implicitly.
- d|--data {Secured data}..
 - Application Message, takes hex values(Mandatory)
 - Data can be formatted as compact and expandable, refer format option {Secured data}..
 - List of data separated by space(This option must be always given at last)

Commannd to initialize SCP80 session

Examples:

1. SCP80-send -r 2 -c false -n 1 -o 1 -y 0 -h false -k 1 -i 1 -s
30423042304430443045304530463046 -v 0x01 -b 0x01 -e
0123456789ABCDEF100276FEDCBA0123 -t B00000 -u 02 -d 00A40004023F00
2. SCP80-send -r 0 -c false -n 0 -o 1 -y 2 -h true -m 1 -k 1 -i 1 -s
30423042304430443045304530463046 -v 0x01 -b 0x01 -e
0123456789ABCDEF100276FEDCBA0123 -t B00000 -u 02 -g -d 00A40004023F00
3. SCP80-send -r 1 -c true -n 0 -o 1 -y 0 -h false -k 5 -i 5 -s
30423042304430443045304530463046 -v 0x01 -b 0x01 -e
0123456789ABCDEF100276FEDCBA0123 -t B00000 -u 02 --implicit-algo-kic DES --
implicit-algo-kid DES -d 00A40004023F00

9.8 tr-proactive

usage: tr-proactive [-t|--type type][--cmdDetails Command Details][--devId Device Identities][--r|--result Result][--s|--status channel Status][--b|--bearer Bearer Description][--u|--buffer Command Details][--a|--add Other Address][--l|--channelDataLen Channel Data Length][--h|--channelData Channel Data][--i|--timerId Timer Identifier][--v|--timerVal Timer Value][--u|--duration Duration][--o|--localInfo Local Info][--e|--other Other values to be sent as part of any TR]

-t|--type type

This option supports Terminal Response (TR) for the following UICC proactive commands:

- open (Open Channel)
- send (Send Data)
- event (Setup Event List)
- recv (Receive Data)
- time (Timer Management)
- close (Close Channel)
- poll (Poll Interval)
- refresh (Refresh)
- pli (Provide Local Information)

--cmdDetails Command Details

This option requires command number, type of command and command qualifier as value.

--devId Device Identities

This option requires source device identity and destination device identity as value.

--r|--result Result

This option requires result to be sent as part of TR for the previous proactive command.

--s|--status channel Status

This option requires channel status as value.

-b|--bearer Bearer Description

This option requires bearer type and bearer parameters as value.

-u|--buffer Command Details

This option requires buffer size as value.

The buffer size codes the number of bytes requested by the UICC in an Open Channel command or what the terminal can offer UICC

-a|--add Other Address

This option requires type of address

-l|--channelDataLen Channel Data Length

value indicates the number of bytes available in channel buffer or the number of bytes requested in RECEIVE proactive command.

-h|--channelData Channel Data

Contains data read from or written to specific channel buffer in the terminal.

-i|--timerId Timer Identifier

This option requires timer identifier as value.

-v|--timerVal Timer Value

This option requires timer value, expressed in using the format hour, minute, second.

-u|--duration Duration

Contains polling interval which shall be required for the issue of TR for a POLL INTERVAL command.

Should contain time unit and time interval

-o|--localInfo Local Info

Local Information to be sent as part of TR for PLI. (should be given in TLV format). Refer 102 223.

-e|--other Other values to be sent as part of any TR

Provide any other values as TLV format (for which options are not present)

Terminal Response for Proactive Command. The output of this command is stored in the variable: TR.PROACTIVE.RESPONSE

Example:

```
tr-proactive -t time -c 012700 -d 8281 -r 00
```

9.9 fetch-proactive-data

usage: fetch-proactive-data [-s|--sw2 Bytes to be fetched]

-s|--sw2 Bytes to be fetched

This option requires status word 2 of '91XX' as input.

This command fetches 'XX' bytes of 91XX.

The output of this command will be stored in FETCH.RESPONSE

Client Hello Shall be stored in CLIENT.HELLO

Client Key Exchange Shall be stored in CLIENT.KEY.EXCHANGE

Client Change Cipher Spec Shall be stored in CLIENT.CHANGE.CIPHER.SPEC

Client Finished Shall be stored in CLIENT.FINISHED

Alert from the card will be stored in CLIENT.ALERT (Only alert level and alert description shall be stored)

HTTP POST Request from Card is stored in HTTP.POST.REQUEST

(The HTTP post requests from the card is stored in the above variable(unencrypted)).

Example:

```
fetch-proactive-data -s 3D
```

9.10 scp81-tls

usage: scp81-tls [-m|--minVer Minor Version of TLS][-c|--cipher Type of cipher suite][-p|--psk PreShared Key][-r|--random Server Random][-i|--sessionId Session ID][-t|--tls TLS Handshake type][-h|--host Administartion Host Parameter][-g|--agent Agent ID Parameter][-u|--uri Administration URI Parameter][-y|--retry]

-m|--minVer Minor Version of TLS

Minor Version suggesting the TLS. It is a compulsory option for TLS type SH (Server Hello).

Minor Version supported are: 01, 02 and 03

-c|--cipher Type of cipher suite

Cipher Suites supported are:

008B(3DES_EDE_CBC_SHA)

008C(AES_128_CBC_SHA)

002C(NULL_SHA)

00AE (AES_128_CBC_SHA)

00B0(NULL_SHA_256)

It is a compulsory option for TLS type SH (Server Hello)

-p|--psk PreShared Key

Pre Shared Key for the computation of Master Secret

It is a compulsory option for TLS type SH (Server Hello)

-r|--random Server Random

Server Random used in the generation of keys

It is a compulsory option for TLS type SH (Server Hello)

-i|--sessionId Session ID

Session ID used in the TLS Handshake.It is a compulsory option for TLS type SH (Server Hello)

-t|--tls TLS Handshake type

When this option is set, chnlDataLenOpt and chnlDataOpt is ignored. The following TLS handshakes are supported:

SH (Server Hello)- when this option is set, --minVer|-m, --cipher|-h, --psk|-k, --random|-e, --sessionId|-q is compulsory.

SHD (Server Hello Done)

SCCS (Server Change Cipher Spec)

SF (Server Finished)

-h|--host Administration Host Parameter

Host Value which shall be verified against First HOST field of HTTP Post Request

Compulsory option for TLS type SF (Server Finished)

-g|--agent Agent ID Parameter

Agent Id Parameter which shall be verified against 'X-Admin-From:' field First HTTP Post Request

Compulsory option for TLS type SF (Server Finished)

-u|--uri Administration URI Parameter

Agent Id Parameter which shall be verified against First POST field of HTTP Post Request

Compulsory option for TLS type SF (Server Finished)

-y|--retry This option indicates that first post request of the resumed session use X-Admin-Resume as true

Optional option for TLS type SF (Server Finished)

The command generates Server Hello, Server Hello Done, Server Change Cipher Spec and Server Finished.

Generated Server Hello is stored in the variable SERVER.HELLO

Generated Server Hello Done is stored in the variable SERVER.HELLO.DONE

Generated Server Change Cipher Spec is stored in the variable SERVER.CHANGE.CIPHER.SPEC

Generated Server Finished is stored in the variable SERVER.FINISHED

Examples:

```
scp81-tls -t serverhello -m 03 -c 00B0 -p F0C0FAAC0EF1364A3E5EB4229CF797A3 -r
522093255B228517258D857B084ACBDD8AE5010D041854E0025C559251975150 -i
374858567461664238015315275505860378
```

```
scp81-tls -t serverhellodone
```

```
scp81-tls -t serverchangecepherspec
```

```
scp81-tls -t serverfinished -u 2F6770313F633D -h 6C68 -g 696431
```

9.11 envelope

usage: envelope [-e|--envelopeType Type of Envelope command][[-d|--devId Device Identities][--eventType Type of event][--eventList List of event][--status channel Status][--c|--channelDataLen Channel Data Length][--i|--timerId Timer Identifier][--v|--timerVal Timer Value][--h|--channelDataLenData Channel Data Length][--b|--bearer Bearer Description][--a|--add Other Address]]

-e|--envelopeType Type of Envelope command

Type of Envelope command supported:

EventDownload

TimerExpiration

-d|--devId Device Identities

This option requires source device identity and destination device identity as value.

--eventType Type of event

Type of Event to be sent as part of envelope command - EventDownload.Events supported:

DataAvailabe

channelstatus

--eventList List of event

The event list shall contain any one event

--status channel Status

This option requires channel status as value.

--channelDataLen Channel Data Length

value indicates the status and identifier of the channel on which the event occurred.

-c and -h are mutually exclusive. Note: Can take upto FF value.

--timerId Timer Identifier

Identifier of the timer that has expired.

--timerVal Timer Value

Difference between the time when the command is issued and the time when the timer was initially started.

--channelDataLenData Channel Data Length

Data whose length shall be given as input to channel data length for envelope command.

-c and -h are mutually exclusive.

--bearer Bearer Description

This option requires bearer type and bearer parameters as value.

--add Other Address

This option requires type of address

This option forms the data and sends it as part of envelope command.

Output of this command is stored in the variable ENVELOPE.RESPONSE

Examples:

```
envelope -e eventDownload -d 8281 -t dataavailable -l 09 -s 8100 -h ${SERVER.HELLO}
```

9.12 scp81-http-response

usage: scp81-http-response [-c|--chunk][--rfm][--i--aid AID value][--u--uri Admin URI value][--f--fragSize Fragment size][--s--respStatus Response Status][--a--alert][--l--alertLev Alert Level][--t--alertDes Alert Description][--h--host Administration host][--g--agentID Agent ID][--d--appData {Application Data}..]

-c|--chunk This option indicates that the application data needs to be sent in chunks.

Use option --app-data or -d option to set the data.

--rfm This option indicates that RFM operation so that content type in HTTP Post Response considers Content-Type: application/vnd.etsi.scp.command-data;version=1.0

--i--aid AID value

The targeted security domain AID shall have two components: RID (5 bytes) and PIX (0 - 11 bytes).

--u--uri Admin URI value

URI required for HTTP Post Response

When this option is not set, the uri of first request shall be considered.

--f--fragSize Fragment size

The size based on which fragments shall be divided before sending.

Default size is 512.

--s--respStatus Response Status

HTTP Post Response Status.

The following is supported:

204(No Content)

200(OK): --appData-d is compulsory

This is a compulsory option for --httpResp

--a--alert This option indicates whether the alert should be sent along with HTTP POST RESPONSE for closing the channel.

--l--alertLev Alert Level

This Option indicates the alert level. Refer RFC for details. Default is 01(WARNING)

--t--alertDes Alert Description

This Option indicates the alert Description. Refer RFC for details. Default is 00(Close Notify)

--h--host Administration host

Administration Host

When this option is not set, the uri of first request shall be considered.

`-g|--agentID Agent ID`

Agent ID. When this option is not set, the uri of first request shall be considered.

`-d|--appData {Application Data}.`

Application DataFor Chunk Transfer (j option), list of data can be given separated by space (Data should be given at last)

When `--respStatus|-w` is 200, this option is compulsory.

{Application Data}.

List of data separated by space

This command forms HTTP POST Response

The response of this command is stored in the variable
HTTP.POST.RESPONSE.FRAGMENTS

The fragments are stored in an array. Can be accessed like this:

`${HTTP.POST.RESPONSE.FRAGMENTS[0]}`, `${HTTP.POST.RESPONSE.FRAGMENT S[1]}` etc depending on the number of fragments.

Examples:

```
scp81-http-response -s 200 -u 2F6770313F633D -h 6C68 -g 696431 -d
AE80221880F28000124F10D2760000284101010102010104010000000000
```

```
scp81-http-response -s 204 -f 255 -a
```

9.13 scp81-alert

usage: `scp81-alert [[-l|--alertLev Alert Level] [[-t|--alertDes Alert Description]]]`

`-l|--alertLev Alert Level`

This Option indicates the alert level. Refer RFC for details. Default is 01(WARNING)

`-t|--alertDes Alert Description`

This Option indicates the alert Description. Refer RFC for details. Default is 00(Close Notify)

The output of this command is stored in the variable ALERT

The output is stored in an array. Can be accessed like this `${ALERT[0]}`

Example:

```
scp81-alert -l 01 -t 00
```

10. Generation of .pem file

.pem file for providing EC private key can be generated using openssl command.

Example:

```
ecparam -out "S:\ecprikey.pem" -name prime256v1 -genkey.
```

prime256v1 – P-256 curve. (One of the named curves supported in openssl).
This command takes named curve as a input and generates EC private key and stores in .pem file.

To obtain the associated EC public key, following openssl command can be used.
`ec -in "S:\ecprikey.pem" -text`

11. Generation of PKCS#12 file in openssl

Steps to generate a PKCS12 file in openssl:

1. Generate a pem file having RSA private key
genrsa -out "S:\rsaPrikey.pem" 1024.
2. Generate a new certificate request. It will prompt the user for the relevant field names.
req -new -key "S:\rsaPrikey.pem" -out "S:\my_request.csr" -config C:\OpenSSL-Win32\bin\openssl.cfg".
3. Convert a certificate request into a self-signed certificate using extensions for a CA
x509 -req -days 3650 -in "S:\my_request.csr" -signkey "S:\rsaPrikey.pem" -out "S:\mycert.crt"
4. Generate a PKCS12 file. Options -keypbe and - certpbe allow the algorithm used to encrypt the private key and certificates to be selected.
pkcs12 -keypbe PBE-SHA1-3DES -certpbe PBE-SHA1-3DES -export -in "S:\mycert.crt" -inkey "S:\rsaPrikey.pem" -out "S:\PKCS12.p12"

12. Generation of PKCS#8 file using OpenSSL

OpenSSL supports the PKCS#8 token in DER|PEM output format.

- Steps to generate PKCS#8 token for PEM output file format:

1. Generate a pem file having RSA private key
`OpenSSL> genrsa -out rsaprivatekey.pem 1024`
2. To convert RSA private key from traditional format to pkcs#8.
`OpenSSL> pkcs8 -topk8 -in rsaprivatekey.pem -inform PEM -outform PEM -out pkcs8privatekey.pem`
3. To generate the public key for generated RSA private key.
`OpenSSL> rsa -in rsaprivatekey.pem -pubout -outform PEM -out rsapublickey.pem`

- Steps to generate PKCS#8 token for DER output file format

1. Generate a pem file having RSA private key
`OpenSSL> genrsa -out rsaprivatekey.pem 1024`
2. To convert RSA private key from traditional format to pkcs#8.

```
OpenSSL> pkcs8 -topk8 -in rsaprivatekey.pem -inform PEM -outform
DER -out pkcs8privatekey.der
```

3. To generate the public key for generated RSA private key.

```
OpenSSL> rsa -in rsaprivatekey.pem -pubout -outform DER -out
rsapublickey.der
```

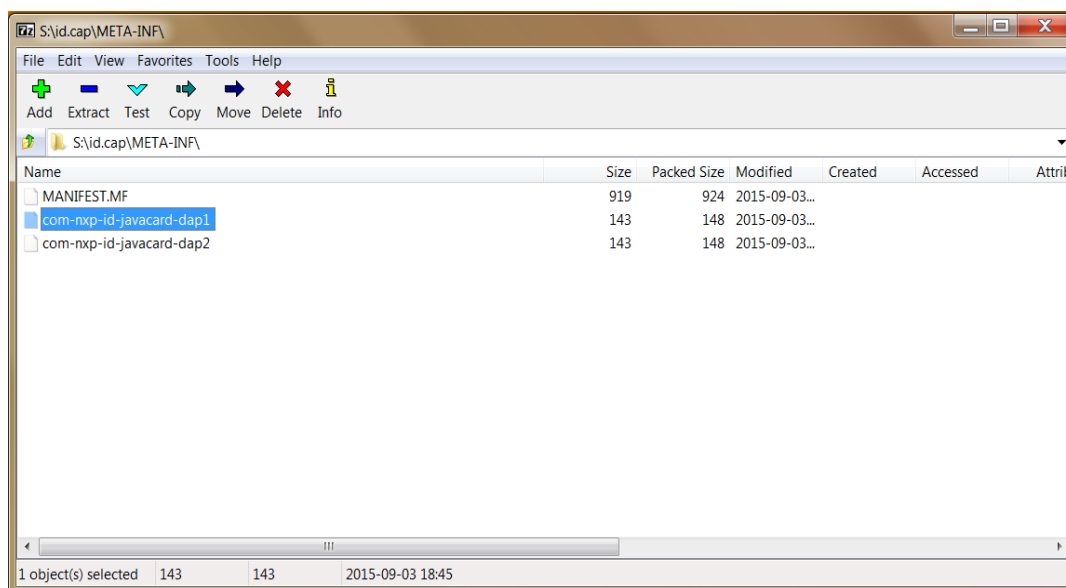
13. Storage of DAP blocks and DM tokens in capfile

In JCShell, DAP blocks and DM tokens are generated using /cap-sign and /cap-xxxx-auth commands. (/cap-load-auth, /cap-delete-auth, /cap-extrdt-auth, /cap-delete-auth, /cap-reg-update)

The resulting DAP blocks/DM tokens information is always stored in the '**meta-inf** or **META_INF**' folder of the capfile.

Examples:

1. /cap-sign -i pwd -a RSAGPSE "S:\id.cap" 112233445567 "S:\DAP5.p12" SHA-256



com-nxp-id-javacard-dap1 => com-nxp-id-javacard is the package name followed by the DAP block number. Subsequent /cap-sign commands generate the DAP blocks with the next sequence number i.e., com-nxp-id-javacard-dap2.

Maximum of 10 DAP blocks are allowed in a capfile.

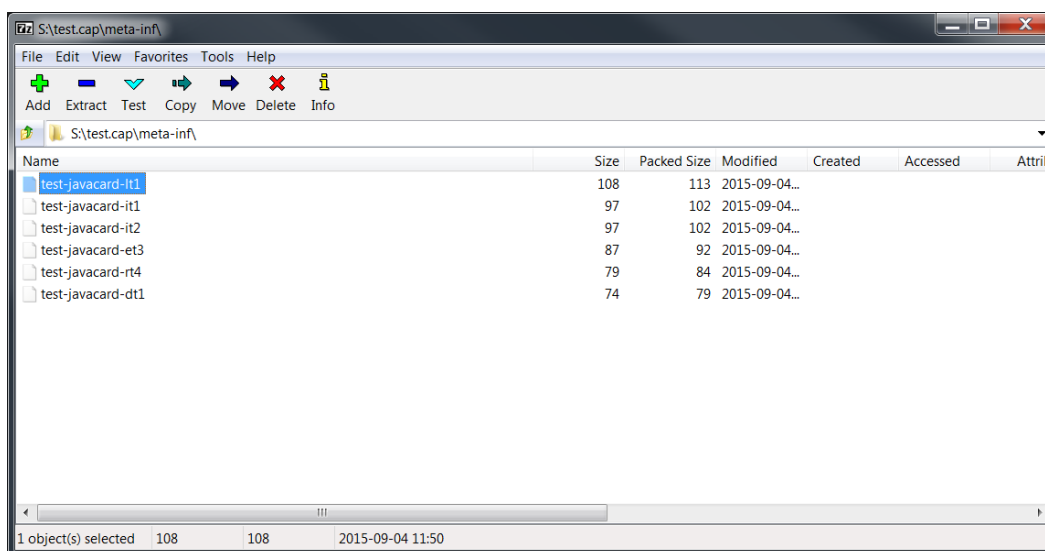
On Uploading the capfile having multiple DAP blocks, JCShell parses the blocks and asks the user to enter the DAP block number which he wishes to send in the upload command APDU. If DAP verification is successful oncard, capfile will be uploaded.

2. Following commands create the DM tokens in the capfile:

```
>/cap-load-auth "S:\test.cap" "S:\WritePrivatekey.pem" SHA-256
```

```
>/cap-inst-auth -i 1122334455667788 |myTestApp "S:\test.cap"
"S:\WritePrivatekey.pem" "SHA-256"
```

- `/cap-del-auth |myTest "S:\test.cap" "S:\WritePrivatekey.pem" SHA-256`
- `/cap-extrdt-auth -s A000000151434169643064 |myTest "S:\test.cap" "S:\WritePrivatekey.pem" SHA-256`
- `/cap-reg-update -q EF#(A00484022000) -i SHA2 "S:\test.cap" "S:\WritePrivatekey.pem"`



`test-javacard-lt1` => test-javacard is the package name followed by the type of token.

- `lt` - load token
- `it` - install token
- `dt` - delete token
- `et` - extradite token
- `rt` - registry update token

14. Ciphred Cap File

In JCShell, [/cap-cipher](#) command can be used to read in a plain cap file in a file format according to JavaCard specification and convert it into a ciphred cap file in an NXP proprietary format. This command reads in the specified cap file, creates the ciphred load file data block and stores it as a ciphred file, excluding debug components. If the command is performed in-place, the plain components except the header component are removed. For more details on input arguments refer [/cap-cipher](#) command.

The `/cap-cipher` command supports AES-128 encryption algorithm with CBC block mode and SHA-256 supported hashing algorithm.

Example:

```
/cap-cipher -i -s |CLSDAppAid -t "T:\output_ciphred_file.cap" "T:\input_plain_file.cap" -k 00112233445566778899001122334455
```

Here “-i” specifies the presence of initialization vector data in META-INF folder of output cipher cap file. Option “-s” specifies to write CLSD AID to META-INF folder, this CLSD AID is read during INSTALL for load. Option “-k” specifies the key for encryption and

"T:\output_ciphered_file.cap" is the output ciphered cap file, and "T:\input_plain_file.cap" the input plain cap file.

[/cap-sign](#) command shall be used to store the DAP blocks into ciphered cap file. The [/cap-sign](#) commands refers the externally provided HASH value and creates corresponding DAP blocks in META-INF folder of ciphered cap file. At most ten DAP blocks are supported concurrently.

Currently only SHA-256 hashing algorithm is supported to sign the ciphered cap file.

NOTE: Storing the Delegated Management tokens are not allowed on ciphered cap file.

Example for CAP Signing:

```
/cap-sign -t "T:\output_signedcipher.cap" -a ECDSA "T:\input_ciphered.cap" |myTestApp  
"T:\dap_privkey.pem" SHA-s256
```

In JCShell, [upload](#) command is used to upload the ciphered cap file to the SE. This requires that the following preconditions are fulfilled:

- Install SSD with Ciphered Load File Data Block Privilege (with TRUSTED PATH privilege).
- Personalize the SSD and select the SSD with CLFDB privilege.
- Store the CL Key to SSD with CLFDB privilege.
- Select ISD and upload the ciphered cap file on SSD with CLFDB privilege.

Example for upload (Assuming the CLSD and CL key are setup properly):

```
upload -s ${SDAID} -h SHA-256 ${CIPHERED_CAP_FILE}
```

Upload command shall be used as follows when ciphered cap file has AID of the target CLSD stored in META-INF folder.

```
upload ${CIPHERED_CAP_FILE}
```

15. Read JCShell Script Contents

JCShell supports the feature to read the contents of the JCShell script. The contents from script is stored to global array variable. The array variable elements can be used within a script as individual commands.

Following is the procedure to read script contents,

1. Set the variable "READ_SCRIPT" to "true"
2. Set the variable "path" to the folder containing script (to read content's)
3. Execute JCShell scripts by specifying filename as shown in Example below.
4. Instead of script execution (behavior of JCShell), the contents of the script (Eg: test) is read to global array variable "script.statements"
5. The variable "READ_SCRIPT" shall be set to "false" to continue with the usual script execution
6. On issuing JCShell command "/list-var", array variable "script.statements" shall be seen containing script contents.
7. The Array elements shall be accessed as "\$script.statements[index]"

Example:

1. Contents of Example Script (test.jcsh):

```
/mode trace on
/term Remote
```

2. Example to read script(test.jcsh) content:

```
- /set-var READ_SCRIPT true
- /set-var path "path to the folder containing script"
- test
- /set-var READ_SCRIPT false
- /list-var
```

Global:_____

```
READ_SCRIPT=true
SESSION.ID=1495D6098E7CE31D946667731ED158AA
script.statements=/mode trace on /term Remote
- /echo ${script.statements[0]}
/mode trace on
- /echo ${script.statements[1]}
/term Remote
```

NOTE: "-" is the Jcshell prompt

16. Additional Resources

To ease the development of JCShell scripts (using .jcsh as file extension) helper tools have been developed and made available to the public.

This includes a syntax highlighter and code completer of JCShell defined commands and script syntax. The tools can be used in combination with programming text editors.

The tools can be found here: <https://github.com/nxp>.

They're made available "as is". Further information can be found at the given link.

17. Known Limitations

- JCShell is not supported on JRE version below 1.6.
- 64 bit jvm is supported only on SComm terminals
- JCShell is not able to connect to a remote server with "/term SCOMM:...".
- In JCShell, not possible to set SSD keys when CM authenticated SCP01
- Extended length APDUs are not supported for PCSC cards on Linux/MacOs
- Cap File Information provided is currently valid only for JCOP242

18. Restrictions

- AES-256 will not work due to restrictions on JCE. To overcome this restriction from JCE, please download “Java Cryptography Extension (JCE) Unlimited Strength Jurisdiction Policy Files 6” from Oracle site and follow installation instructions.

19. Changelog

Table 1. New features and bug fixes for release 5.32.0.X

Synopsis
Support for NFCC Power mode option for /reset command to configure the NFCC in low or full power mode
Support for the Secure Channel Protocol 81 (SCP81) for eUICC
Support SystemOS related communication over SPI - generic logical interface support with configurable NAD byte for Rasp Pi

Table 2. New features and bug fixes for release 5.31.0.X

Synopsis
Support Interface Reset and OS Reset for all versions of T1 protocol implementation
Support for /exec command to display standard error messages
Support for the command for CRC computation for 16-bit and 32-bit data

Table 3. New features and bug fixes for release 5.30.0.X

Synopsis
Support for /set-random-seed command in Mac OSx and Linux

Table 4. New features and bug fixes for release 5.29.0.X

Synopsis
Support for SCP80

Table 5. New features and bug fixes for release 5.28.0.X

Synopsis
Support of ciphered upload (CLFDB) in ProviderSecurityDomain plugin
Support to save the package AID for a cap file in a variable integrated changes

Table 6. New features and bug fixes for release 5.27.0.X

Synopsis
Support for the new option “-r” for “ls” command to get complete data when response is more than 256 bytes and SW returned is 63XX for T0 protocol for UART interface

Synopsis
“scripts” and “tcpserver” folders in JCShell are discontinued
Support to store the response of ‘felica-polling’ command with “last.response.data” variable.
Support to send NCI commands over Mailbox interface
Support in CryptoPlugin support for RSA-PSS with message recover (ISO 9796-2)

Table 7. New features and bug fixes for release 5.26.0.X

Synopsis
Support for new command line option “-m --heapmem” for JCShell to provide JVM heap size
Support for getting JCShell script contents line wise

Table 8. New features and bug fixes for release 5.25.0.X

Synopsis
Support for new command “/simu-get-device-state” for DPD mode for Simulator

Table 9. New features and bug fixes for release 5.24.0.X

Synopsis
Support for Certificate chaining for SCP11 (Amd F)
Support for BRP curve for Prepare-Download command

Table 10. New features and bug fixes for release 5.23.0.X

Synopsis
Support for warm reset of the contactless interface (secondary interface) from wired interface
Support for Set Mode feature
Support for End of APDU feature
Support for SE power tracking events
Support for Deep Power Down mode feature

Table 11. New features and bug fixes for release 5.22.0.X

Synopsis
Support for GPIO pin read(monitor)/write(trigger)
Support for SE Power Tracking events
Support for SM4 MAC generation

Table 12. New features and bug fixes for release 5.21.0.X

Synopsis
Response security in-case of R-APDU chaining
Handling of native commands for Felica
SM2 support for encrypt and decrypt commands

Table 13. New features and bug fixes for release 5.20.0.X

Synopsis
Support for commands 'start-tearing' and 'stop-tearing' in simulator plugin to Tearing sub MTYs for Prepare Tearing command

Table 14. New features and bug fixes for release 5.19.0.X

Synopsis
Support to install a Profile Reading the Metadata parameters and Profile element data from DER File
Support for new extension to JCOP Simulator to load dumps without restart JCOP Simulator executable

Table 15. New features and bug fixes for release 5.18.0.X

Synopsis
Support for Telecom Tuak/Millennium Authentication Procedure [reference TS 102.221]
Rename JCShell command from /set-ifsc-size xxx => to /set-ifsc-ifsd-size xxx

Table 16. New features and bug fixes for release 5.17.0.X

Synopsis
Support for command ' <i>/get-hci-sessionID</i> ' which is available at admin gate and propagated to NFCC during HCI session initialization
Support for command ' <i>notifications</i> ' to list and remove all notifications
Support in ' <i>/mode</i> ' command to throw error when incomplete option string is specified
Support for RSA 2K for ' <i>/cap-sign</i> ' and ' <i>/put-pub-key</i> ' commands

Table 17. New features and bug fixes for release 5.16.0.X

Synopsis
Timer result 'timer.device.time' is populated for all terminals
Support for command ' <i>/manage-upgrade-elf-start</i> ' for the AMD-H (ELF UPgrade)
Support for command ' <i>/manage-upgrade-elf-resume</i> ' for the AMD-H (ELF UPgrade)
Support for command ' <i>/manage-upgrade-elf-abort</i> ' for the AMD-H (ELF UPgrade)
Support for command ' <i>/manage-upgrade-elf-status</i> ' for the AMD-H (ELF UPgrade)
Support for command ' <i>notifications</i> ' to list, remove and List & Remove notifications
Support for Type F commands except polling to be sent as I frame
Support for loading CAP files without Descriptor.cap component

Table 18. New features and bug fixes for release 5.15.0.X

Synopsis
Support for listing all available pending notifications from an eUICC using <i>list-notifications</i> command
Support for removing a notification from pending notification list from an eUICC using <i>delete-notification</i> command

Table 19. New features and bug fixes for release 5.14.0.X

Synopsis
Support of tag 66 (Card Data Tag) for response of /card command.
All Type F Commands Except the Poll Request Command are sent as I Frame over Card emulation type F gate.
Support -g option with Upload command to create a file having the APDU list

Table 20. New features and bug fixes for release 5.13.0.X

Synopsis
Support for extending DAP sign using /cap-sign command
Support of new JCShell variable 'hci.eventresponse' storing all HCI event responses

Table 21. New features and bug fixes for release 5.12.0.X

Synopsis
Support for setting random seed command
Support for extended length usage in combination with load CAP file
Support for extended length usage in combination with Ciphered load CAP file

Table 22. New features and bug fixes for release 5.11.0.X

Synopsis
Support for Edwards ed25519 for Sign generation and verification
Support for ECDH Curve25519 for key exchange
Support for /clear-hci-event to clear HCI events buffer

Table 23. New features and bug fixes for release 5.10.0.X

Synopsis
Support for GP chaining for RSA 2K
Support for storing DES keys in SCP03 session
Support for custom ICV during CLFDB loading

Table 24. New features and bug fixes for release 5.9.0.X

Synopsis

Synopsis
Support for CAP file patching

Table 25. New features and bug fixes for release 5.8.0.X

Synopsis
Bug fixes

Table 26. New features and bug fixes for release 5.7.0.X

Synopsis
Support for querying and setting card RF gates registries in /send command.

Table 27. New features and bug fixes for release 5.6.0.X

Synopsis
Support for /reset with Calypso boards with dummy ATR bytes.

Table 28. New features and bug fixes for release 5.5.0.X

Synopsis
Bug fixes

Table 29. New features and bug fixes for release 5.4.0.X

Synopsis
Support for /reset with Calypso boards
Support for type F command exchange in /send without further arguments
Support for /felica-polling when connected to the MP300/MP500 and the JCOP simulator

Table 30. New features and bug fixes for release 5.3.0.X

Synopsis
Bug fixes

Table 31. New features and bug fixes for release 5.2.0.X

Synopsis
Added support for timing information to /send-jrcp-frame.
%(...) now supports data of length 255 bytes.

Table 32. New features and bug fixes for release 5.1.0.X

Synopsis
Support for sendRaw to Remote terminal
Support for ECDSA Signature generation for 384 bit curve parameter

Table 33. New features and bug fixes for release 5.0.0.X

Synopsis
Gracefully exit JCShell when its process is killed
Support for hash algorithms with ECDSA signature generation

Table 34. New features and bug fixes for release 4.15.0.X

Synopsis
Initial support for the Calypso board

Table 35. New features and bug fixes for release 4.14.0.X

Synopsis
Support for the RC663 terminal

Table 36. New features and bug fixes for release 4.13.0.X

Synopsis
JRCP session ID is now sent to a node with the first communication targeted at it
/get-hci-event now supports setting a timeout when connected to the JCOP simulator
/mode trace=on off now switches on off sending JCShell output to the JCOP simulator for tracing

Table 37. New features and bug fixes for release 4.12.2.X

Synopsis
Prevent JCShell from attempting to disconnect fixed devices from a JRCP Server

Table 38. New features and bug fixes for release 4.12.1.X

Synopsis
Support for the Simulator-specific JRCP JRCP extensions v1.1.

Table 39. New features and bug fixes for release 4.12.0.X

Synopsis
Introduced new commands '/get-transaction-time' and to '/reset-transaction-time' to query and reset the transaction timer for all terminals.
Introduced new command '/switch-contactless-gate' to switch between type A and B when connected to a contactless reader.
Extension of the terminal string of the Remote terminal for setting the contactless gate and the default node address.

Table 40. New features and bug fixes for release 4.11.0.X

Synopsis
Support for sending raw JRCP frames

Table 41. New features and bug fixes for release 4.10.0.X

Synopsis
Unified JRCP node address handling and introduced command to set the default NAD explicitly
Introduced unique session IDs for each terminal connection
Support for JRCP v2.2 (Document version 2.6)

Table 42. New features and bug fixes for release 4.9.0.X

Synopsis
Support for X509 certificate generation
Support for an optional node address parameter in all JRCP related commands
Support for JRCP v2.5

Table 43. New features and bug fixes for release 4.8.0.X

Synopsis
Support for RSA-256
Support to provide a seed to JCShell used in all cryptographic functions
Support for JRCP v2.4

Table 44. New features and bug fixes for release 4.7.0.X

Synopsis
Gracefully handle non-printable characters

Table 45. New features and bug fixes for release 4.6.0.X

Synopsis
Support for Type F (FeliCa) using the MP300 and MP500 readers
Support for optionally specifying the parameter "CLSD AID" for /cap-cipher

Table 46. New features and bug fixes for release 4.5.1.X

Synopsis
Internal code fixes

Table 47. New features and bug fixes for release 4.5.0.X

Synopsis

Synopsis
Support for Type B using the MP300 and MP500

Table 48. New features and bug fixes for release 4.4.0.X

Synopsis
Introduced new command 'create-oce-cert' for the creation of TLV structured OCE certificates for SCP11.
Extended support for operations in the context of context of SCP11.

Table 49. New features and bug fixes for release 4.3.0.X

Synopsis
Integrated the jNUT framework into JCShell.
Leading "/" removed from all commands in the simulator plugin.
Documentation improvements.

Table 50. New features and bug fixes for release 4.2.2.X

Synopsis
Improved communication with the JCOP simulator

Table 51. New features and bug fixes for release 4.2.1.X

Synopsis
Fix signature verification when DAP blocks are present in ciphered CAP files
Close socket connection to the simulator gracefully in case of errors

Table 52. New features and bug fixes for release 4.2.0.X

Synopsis
Use /atr instead of /reset when /card is issued.
Use device-specific JRCP extensions to send the output of /echo to the simulator in addition to printing it to the command line.
The /reset command does no longer print the ATR. Use a subsequent /atr to obtain this information.
In JCShell "SHA-256" deprecates the inexact option "SHA-2" for /cap-sign, upload, and other commands.

Table 53. New features and bug fixes for release 4.1.0.X

Synopsis
\${timer.execution.time} now uses a decimal representation of the execution time.
Added support for loading of custom components

Table 54. New features and bug fixes for release 4.0.0.X

Synopsis
/reset command is supported.
/send-hci-event and /get-hci-event commands are supported for Remote and SCR terminal.
OSCCA crypto algorithms SM2, SM3 and SM4 are supported in crypto plugin. SM2 is a Public Key Cryptographic Algorithm based on Elliptic curves. SM3 is a Cryptographic hash algorithm. SM4 Block cipher algorithm.
Support of AES/GCM encryption and decryption commands in crypto plugin.
Modified the store data command to accommodate the changes for System and Device time.
SCR01 dll version 0.9.6.7 is supported.
The time measured on the device was off by a factor of 1000 when the SCComm driver is chosen. This issue is fixed.
Supports DWP via IO3/4 for SCR Terminal.
/start-timer and /stop-timer commands displays process time.
/switch-swp-gate command is supported for both SCR Terminal and the simulator.
Supports JRCP v2.3 specification
Supports Loading and displaying of JCOP Simulator Plugin commands.
JCOP Plugin provides option to convert projects created with old JCOP plugin (com.ibm.bluez) to support with new JCOP plugin (com.nxp.id).
Converter tool is supported to convert old workspace and JCOP projects to work with new JCOP Plugin.
JCShell supports creation of ciphered CAP files.
JCShell supports loading of ciphered CAP files.
The CAP file conversion makes use of a system environment variable 'COMPRESS_CAP_FILE' which can have values either 'true' or 'false'. If this value is set to true then default compression level will be applied. If this value is set to false or not defined then no compression level will be applied ie. Level 0.

20. Legal Information

20.1 NXP Semiconductors Netherlands B.V.

IMPORTANT: Before installing this software carefully read the following terms and conditions. This Software End-User License Agreement is a binding agreement between you, the end-user and licensee ("Licensee"), and NXP Semiconductors Netherlands B.V. regarding the use of the software accompanying this Agreement, which includes computer software and provided documentation. By installing, copying or otherwise using the computer software and provided documentation, you agree to be bound by the terms of this License Agreement.

20.1.1 END USER LICENSE AGREEMENT

This End User License Agreement (the "Agreement") is made and entered into as of the date of first installation of Licensed Software at Licensee (the "Effective Date") by and between NXP Semiconductors Netherlands B.V. ("NXP") and the legal entity where installation of Licensed Software is made "Licensee". (NXP and Licensee are occasionally referred to herein individually as a "Party" and collectively as the "Parties.")

Whereas, NXP has designed and developed the NXP IC's (as defined below) and Licensee wishes to incorporate the NXP IC's in certain products of Licensee, the "Licensed Products" (as defined below);

Whereas, NXP has developed or acquired rights to certain application software, the "Licensed Software" (as defined below), and Licensee desires to license the Licensed Software for use in certain products of Licensee, the "Licensed Products" (as defined below) solely in conjunction with the NXP IC's; and

Whereas, NXP is willing to license the Licensed Software to Licensee for use in Licensed Products solely in conjunction with the NXP IC's, under the terms and conditions of this Agreement.

The Parties hereby agree to the following terms and conditions:

1.0 DEFINITIONS.

In addition to other terms defined elsewhere in this Agreement, the following terms used herein, when the first letter is capitalized, shall have the meanings set forth in this Section 1. All definitions below shall apply both to their singular or plural forms, as the context may require.

1.1 For NXP, the term "Affiliate" means (i) any Person Controlled by NXP Semiconductors N.V. or (ii) any Person Controlled by any transferee of all or substantially all of the assets of NXP Semiconductors N.V., where "Controlled" means the direct or indirect beneficial ownership of more than fifty percent (50%) of the voting stock, or decision-making authority in the event that there is no voting stock, in another entity; provided, any such Person described in clause (i) or (ii) shall be deemed to be an "Affiliate" only for so long as such Person is Controlled by NXP Semiconductors N.V. or such transferee. "Person" is defined to mean "an individual, corporation, partnership, limited liability company, association, unincorporated association, trust or other entity or organization, including a government or political subdivision or an agency or instrumentality thereof." For Licensee, the term "Affiliate" shall mean any corporation or other legal entity that a party now or hereafter Controls, is Controlled by or is under common Control with; where "Control" means the direct or indirect ownership of more than fifty percent (50%) of the shares or similar interests entitled to vote for the election of directors or other persons performing similar functions.

1.2 "Confidential Information" shall mean the Licensed Software, including where applicable according to Appendix A, all Source Code and Documentation thereof, and any other information provided by NXP to Licensee under this Agreement which has been identified as proprietary and/or confidential information, or that, by its nature and the surrounding circumstances and in good faith, ought to be treated as proprietary and/or confidential.

1.3 "Documentation" shall mean all related comments, annotations, instructions, manuals, and other written materials, whether in printed or

electronic form, including without limitation, customer reference and installation manuals, user's guides, and programmer guides that NXP includes or otherwise provides with the Licensed Software, all as set forth on Appendix A.

1.4 "End User" shall mean a person or business enterprise which acquires the Licensed Product for its ordinary personal purposes or use in its normal business operations and not for resale or transfer to others in its ordinary course of business.

1.5 "Executable Code" shall mean the machine-executable version of the Licensed Software as set forth on Appendix A, and further such Executable Code, in whole or in part, which results from linking, loading or assembling (or other similar process), as required, the Object Code into machine language, executable form.

1.6 "Licensed Executable Code" shall mean the Executable Code as set forth on Appendix A.

1.7 "Licensed Object Code" shall mean the Object Code as set forth on Appendix A,

1.8 "Licensed Source Code" shall mean the Source Code as set forth on Appendix A.

1.9 "Licensed Product" shall mean any product based on NXP IC's produced by or for Licensee which incorporates any Licensed Software, in whole or in part.

1.10 "Licensed Software" shall mean all the software, as set forth on Appendix A.

1.11 "NXP IC's" shall mean NXP's integrated circuits that include NXP's JCOP operating system, manufactured and sold by or for NXP.

1.12 "Object Code" shall mean all or any portion of the machine-readable or machine language version of the Licensed Software, and all corrections, updates, and revisions and improvements thereto, which NXP provides to Licensee hereunder.

1.13 "Source Code" means the human-readable version of software, including without limitation, associated flow charts, algorithms, comments and other written instructions and technical documentation.

1.14 "Term" shall mean the effective period of the Agreement as further defined in Section 7.1.

1.15 "Update" shall mean any modification, enhancement, error correction, maintenance release and/or revisions to the Licensed Software which may be provided by NXP to Licensee, at NXP sole discretion, hereunder.

2.0 LIMITED LICENSE.

2.1 License to Executable Code. Subject to the terms and conditions of this Agreement, NXP hereby grants to Licensee during the Term of this Agreement a personal, non-exclusive and non-transferable royalty-free copyright license, without the right to sublicense, to use the provided Licensed Executable Code (a) to install all components of the Licensed Executable Code on a hard disk or other storage medium or on a network fileserver, (b) to create backup copies of the Licensed Executable Code, and (c) to install and use the Licensed Executable Code solely in-house. The Licensed Executable Code may only be used in conjunction with or for devices that are developed, produced, sold and / or authorized by NXP.

2.2 License to Object Code. Subject to the terms and conditions of this Agreement, NXP hereby grants to Licensee during the Term of this Agreement a non-exclusive, personal, non-transferable royalty-free copyright license, without the right to sublicense, to use the Licensed Object Code from only of the Licensed Software - to the extent the Licensed Object Code version is licensed according to Appendix A- for use in conjunction with NXP IC's in the Licensed Product, and distribute, directly or indirectly, said Licensed Product incorporating such Licensed Software, solely as embedded in the Licensed Product in Executable Code form only, to End Users.

2.3 License to Source Code. Subject to the terms and conditions of this Agreement, NXP hereby grants to Licensee during the Term of this Agreement a non-exclusive, personal, non-transferable royalty-free copyright license, without the right to sublicense, to use and reproduce the Licensed Source Code from only of the Licensed Software for the sole purpose of: (i) creating modifications of the Licensed Source Code, (ii) compiling the Licensed Source Code and/or the modifications of the Licensed Source Code into Object Code, and (iii) using such generated Object Code in accordance with the license for Licensed Object Code as granted under the previous Section 2.2 of this Agreement.

2.4 License to Documentation. Subject to the terms and conditions of this Agreement, NXP hereby grants to Licensee during the Term of this Agreement a non-exclusive, personal, non-transferable royalty-free copyright license, without the right to sublicense, to use only within Licensee's organization the Documentation solely to support and exercise the license rights granted under Sections 2.1 and 2.2.

3.0 LICENSE RESTRICTIONS.

3.1 Licensee shall not copy, reproduce or distribute Licensed Software and/or Documentation except as expressly provided under Article 2.0.

3.2 Except as expressly provided under Article 2.0, Licensee shall not, and shall not permit any third party to, modify, adapt, alter, translate, merge with any other software, or create derivative works from any portion of the Licensed Software or Documentation, nor reverse assemble, decompile, disassemble or otherwise attempt to derive Source Code of the Licensed Executable Code or the Licensed Object Code, except and solely to the extent such action is expressly permitted by mandatory applicable law in spite of this provision.

3.3 Licensee shall not, and shall not permit any third party to, assign, sublicense, lease, rent, loan, transfer, disclose, or otherwise make available the Licensed Software or Documentation.

3.5 All licenses granted to Licensee under this Agreement shall immediately end upon the expiration or termination of this Agreement.

3.6 The Licensed Software is licensed (under copyright only) and not sold, and any and all references to "sale" or "sold" of any Licensed Software shall be deemed to mean a copyright license, and no ownership or assignment of any intellectual property rights is intended nor shall be implied thereby.

3.7 Other than the limited copyright license granted to Licensee under Article 2.0 herein, no other rights or licenses are granted, or implied by estoppel or otherwise, under any intellectual property rights of NXP and/or its Affiliates or any intellectual property residing in the Licensed Software, Documentation or any other Confidential Information furnished by NXP.

3.8 Notwithstanding anything to the contrary herein, this Agreement shall not be construed as conferring any right, license or immunity, either directly or by implication, estoppel or otherwise to Licensee or any third party acquiring any Licensed Software, Documentation or other Confidential Information furnished by NXP:

- a) under any NXP IPR or intellectual property rights of any third party other than explicitly granted under this Agreement;
- b) for the combination of such Licensed Software, Documentation or other Confidential Information with one or more other items (including items acquired from NXP and/or its Affiliates) even if such items have no substantial use other than as part of such combination, unless infringement of NXP IPR necessarily results from applying one or more of such items furnished by NXP to Licensee.
- c) with respect to any trademark, trade or brand name, a corporate name of NXP and/or its Affiliate(s), or any other name or mark, or contraction abbreviation or simulation thereof.
- d) under any intellectual property rights of NXP and/or its Affiliates covering a standard set by a standard setting body and/or agreed to between at least two companies.
- e) under any intellectual property rights of NXP and/or its Affiliates with respect to which NXP and/or its Affiliates has informed Licensee or has published (in a datasheet or other specifications concerning the Product or elsewhere) a statement that a separate license has to be obtained and/or that no implied license is granted; or
- f) under any intellectual property rights of NXP and/or its Affiliates relating to manufacturing or technical information except as expressly required under this Agreement.

3.9 Licensee's rights under the Agreement are conditional upon Licensee not performing, and Licensee shall not perform any actions in a manner that would require any Licensed Software or any derivative work thereof, to be licensed under Open License Terms. These actions include but are not limited to:

- (i) combining the Licensed Software or a derivative work thereof with Open Source Software, by means of incorporation or linking or otherwise; or
- (ii) distributing the Licensed Software or a derivative work thereof with Open Source Software; or
- (iii) using Open Source Software to create a derivative work of the Licensed Software, insofar as these actions would require the Licensed Software or a derivative work thereof to be licensed under Open License Terms.

As used herein, "Open Source Software" means any software that is licensed under Open License Terms. "Open License Terms" means terms in any license agreement or grant that require as a condition of use, modification and/or distribution of a work that:

- 1. source code will be made available, or
- 2. permission will be granted for creating derivative works, or
- 3. a royalty-free license is granted to any party under any intellectual property rights regarding that work and/or any other work that contains, is combined with, requires or is based on that work.

Notwithstanding any limitation of liability under this Agreement Licensee shall without any limitations indemnify NXP and its Affiliates against and hold NXP and its Affiliates harmless from any damage or costs arising from or in connection with any violation or breach of the provisions of this Section 3 and Licensee shall reimburse all costs and expenses incurred by NXP and/or its Affiliates in defending any claim, demand, suit or proceeding arising from or in connection with such violation or breach.

3.10 Any use of any other software that might be delivered or installed together with the Licensed Software and that is not included in the Licensed Software (i.e. any library for cryptographic algorithms) is not part of this Agreement and mandatorily requests a separate license agreement with NXP.

4.0 OWNERSHIP RIGHTS AND OTHER RIGHTS.

4.1 All intellectual property rights covering the Licensed Software, Documentation and other Confidential Information provided hereunder, and all ownership rights in and to such intellectual property rights, Licensed Software, Documentation and other Confidential Information, shall remain solely and exclusively with NXP, its Affiliates or its third party suppliers, whether or not it was developed specifically for the Licensee.

4.2 Licensee shall not remove or alter any NXP, its Affiliates, or third party supplier copyright notices, proprietary information notices or restricted rights notices contained in any of the Licensed Software, Documentation, and Confidential Information, and shall affix to any media containing a copy or all or any portion thereof, all copyright notices, proprietary information notices and restricted rights notices as were affixed to the original media.

4.3 NXP and its Affiliates shall have the right to use any feedback provided by Licensee to NXP relating to the Licensed Software and its related Documentation without any limitation for any purpose. In the event that Licensee at any time creates any Modification or any intellectual property rights related thereto, Licensee hereby grants NXP and NXP's Affiliates a fully paid-up, royalty free, perpetual, irrevocable, world-wide, non-exclusive license to use, modify, sell, copy, license, distribute, and make derivative work from the Modifications and to grant sub-licenses on the same terms to any third party.

4.4 Licensee agrees and acknowledges that the Licensed Software shall be owned by NXP and/or its Affiliates.

5.0 AUDIT.

During the Term of this Agreement and for three years thereafter, NXP, or its designated representative, shall have the right, upon reasonable notice, to audit all relevant books and records of Licensee to ensure Licensee's compliance with the terms and conditions of this Agreement. Licensee shall willingly co-operate and provide all such assistance in connection with such audit as NXP and/or its auditors may require. The audit will be conducted at NXP' expense, unless the audit reveals that Licensee has breached the license terms under this Agreement.

6.0. THIRD PARTY RIGHTS.

6.1 The Software is supplied with, and uses, third party software or materials which are governed by their own separate license agreements set forth in Appendix B (together "DELIVERED THIRD PARTY SOFTWARE"). This License does not apply to such Delivered Third Party Software and they are not included in the term "LICENSED SOFTWARE" under this License. You agree to comply with all terms and conditions imposed on you in respect of such Delivered Third Party Software ("Third Party Terms") set forth in Appendix C. Any provisions in this License which differ from Third Party Terms are offered by NXP alone and shall not supersede or modify any Third Party Terms. In addition to Third Party Terms, the disclaimer of warranty and limitation of liability provisions in this License shall apply to all Licensed Software and Delivered Third Party Software.

6.2 NXP HEREBY DISCLAIMS ANY AND ALL WARRANTIES EXPRESS OR IMPLIED FROM ANY THIRD PARTIES REGARDING ANY DELIVERED THIRD PARTY SOFTWARE, ANY THIRD PARTY MATERIALS INCLUDED IN THE SOFTWARE, ANY THIRD PARTY MATERIALS FROM WHICH THE SOFTWARE IS DERIVED (COLLECTIVELY "OTHER CODE"), AND THE USE OF ANY OR ALL THE OTHER CODE IN CONNECTION WITH THE SOFTWARE, INCLUDING (WITHOUT LIMITATION) ANY WARRANTIES OF SATISFACTORY QUALITY OR FITNESS FOR A PARTICULAR PURPOSE.

6.3 NO THIRD PARTY LICENSORS OF OTHER CODE SHALL HAVE ANY LIABILITY FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING WITHOUT LIMITATION LOST PROFITS), HOWEVER CAUSED AND WHETHER MADE UNDER CONTRACT, TORT OR OTHER LEGAL THEORY, ARISING IN ANY WAY OUT OF THE USE OR DISTRIBUTION OF THE OTHER CODE OR THE EXERCISE OF ANY RIGHTS GRANTED UNDER EITHER OR BOTH THIS LICENSE AND THE LEGAL TERMS APPLICABLE TO ANY SEPARATE FILES, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGES.

7.0 TERM AND TERMINATION.

7.1 This Agreement shall commence on the Effective Date and shall have a term of two (2) years, unless terminated sooner as set forth herein (the "Term").

7.2 Without prejudice to any rights or remedies of NXP, NXP may, by written notice to Licensee, terminate with immediate effect this Agreement, without any liability, in NXP's sole discretion at any time.

7.3 The license shall terminate automatically, if:

- (a) Licensee violates or breaches any of the provisions of this Agreement;
- (b) a voluntary or involuntary petition in bankruptcy or winding up is filed against Licensee, any proceedings in insolvency or bankruptcy (including reorganization) are instituted against Licensee, a trustee or receiver is appointed over Licensee, or any assignment is made for the benefit of creditors of Licensee; or
- (c) and to the extent as required by applicable third party rights according to Section 6 of this Agreement.

7.4 Upon termination or expiration of this Agreement, Licensee shall immediately return all Licensed Software, Documentation and other Confidential Information to NXP.

8.0 CONFIDENTIALITY.

8.1 Licensee shall retain in confidence and not disclose to any third person or employees except to those Licensee permanent employees who have a legitimate need to know any Confidential Information that it receives from NXP hereunder, and Licensee shall not use the Confidential Information for any purpose whatsoever except for the purpose of this Agreement.

8.2 Licensee represents and warrants that Licensee shall have as of the Effective Date, and shall maintain thereafter, with each of its employees who will have access to any Confidential Information, prior to their access to any Confidential Information, a written, signed agreement legally sufficient and effective to enable Licensee and each such employee to comply with all the provisions of this Agreement.

8.3 Except as expressly permitted in Article 2.0, Licensee shall not, and shall not permit anyone else to use, reproduce, duplicate, or otherwise disclose, distribute or disseminate any Confidential Information. Licensee agrees to take all reasonable steps to prevent disclosure and unauthorized use of Confidential Information and to use no less than same degree of care that it uses with respect to its own confidential information of similar kind or nature. Licensee shall promptly notify NXP of any unauthorized access to or any unauthorized use or disclosure of Confidential Information.

8.4 Licensee's obligations of confidentiality under this Agreement shall not apply to information that Licensee can establish by legally sufficient evidence: (i) is already in the public domain at the time it was disclosed or subsequently enters the public domain through no fault of the Licensee; (ii) is known to Licensee or in its possession, as shown by dated documentation prior to receipt; (iii) is independently developed by Licensee, as shown by dated documentation; or (iv) is rightfully received by Licensee on a non-confidential basis from a third party, as shown by dated documentation, without breach of this Agreement by Licensee.

9.0 WARRANTY.

9.1 The Licensed Software and related Documentation is provided to Licensee AS IS WITHOUT WARRANTY OF ANY KIND, which is accepted by Licensee. NXP makes no warranties to Licensee and shall not indemnify Licensee or hold it harmless for any reason related to the Licensed Software or Documentation, or otherwise be liable to Licensee. Licensee acknowledges and agrees that the Licensed Software and Documentation is provided AS IS and accepts all risks of utilizing the Licensed Software and Documentation under the conditions set forth in this Agreement. The foregoing states the entire liability of NXP in connection with any defective, error, omission or non-conforming Licensed Software or Documentation supplied hereunder.

9.2 NXP EXPRESSLY DISCLAIMS ALSO ON BEHALF AND FOR THE BENEFIT OF ITS AFFILIATES ALL WARRANTIES, EXPRESS OR IMPLIED, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE, AND NON-INFRINGEMENT OF INTELLECTUAL PROPERTY RIGHTS. NXP SHALL HAVE NO LIABILITY TO LICENSEE, OR ITS SUBSIDIARIES, AFFILIATES, OR ANY OTHER THIRD PARTY FOR ANY DAMAGES, INCLUDING WITHOUT LIMITATION, DAMAGES RESULTING OR ALLEGED TO HAVE RESULTED FROM ANY DEFECT, ERROR OR OMISSION IN THE LICENSED SOFTWARE, THIRD PARTY APPLICATION SOFTWARE AND/OR DOCUMENTATION, OR AS A RESULT OF ANY INFRINGEMENT OF ANY INTELLECTUAL PROPERTY RIGHT OF ANY THIRD PARTY.

10.0 LIMITATION OF LIABILITY

10.1 EXCEPT FOR LICENSEE'S LIABILITY FOR BREACH OF ARTICLES 2, 3, 4, 8, OR 9 HEREOF, NO PARTY SHALL BE LIABLE TO THE OTHER FOR ANY INDIRECT, INCIDENTAL, PUNITIVE, SPECIAL OR CONSEQUENTIAL DAMAGES (INCLUDING LOST PROFITS OR LOST SAVINGS) WHETHER OR NOT SUCH DAMAGES ARE BASED ON TORT, WARRANTY, CONTRACT OR ANY OTHER LEGAL THEORY - EVEN IF SUCH PARTY HAS BEEN ADVISED, OR IS AWARE, OF THE POSSIBILITY OF SUCH DAMAGES. IN NO EVENT SHALL NXP BE LIABLE FOR ANY DAMAGE, COSTS OR EXPENSES ASSOCIATED WITH WARRANTY OR INTELLECTUAL PROPERTY INFRINGEMENT CLAIMS WHETHER FOR THE REPLACEMENT OR REPAIR OF PRODUCTS, INCLUDING LABOR, INSTALLATION OR OTHER COSTS INCURRED BY LICENSEE AND, IN PARTICULAR, ANY COSTS RELATED TO THE REMOVAL OR REPLACEMENT OF ANY PRODUCTS SOLDERED OR OTHERWISE PERMANENTLY AFFIXED TO ANY PRINTED CIRCUIT BOARD, EXCESS PROCUREMENT COSTS, OR REWORK CHARGES.

10.2 THE LICENSED SOFTWARE IS NOT DESIGNED, AUTHORIZED OR WARRANTED TO BE SUITABLE FOR USE IN MEDICAL, MILITARY, AIR CRAFT, SPACE OR LIFE SUPPORT EQUIPMENT NOR IN APPLICATION WHERE FAILURE OR MALFUNCTION OF THE LICENSED SOFTWARE CAN REASONABLY BE EXPECTED TO RESULT IN A PERSONAL INJURY, DEATH OR SEVERE PROPERTY OR ENVIRONMENTAL DAMAGE. INCLUSION AND/OR USE OF LICENSED SOFTWARE IN SUCH EQUIPMENT OR APPLICATIONS, WITHOUT PRIOR AUTHORIZATION IN WRITING OF NXP, IS NOT PERMITTED AND FOR LICENSEE'S OWN RISK. LICENSEE AGREES TO FULLY INDEMNIFY NXP FOR ANY DAMAGES RESULTING FROM SUCH INCLUSION OR USE.

11.0 GENERAL.

11.1 The provisions of Articles 1, 3, 4, 5, 6, 7, 8, 9, 10 and 11 shall survive the expiration or termination of this Agreement.

11.2 Licensee shall not assign its rights or obligations under this Agreement without the prior written consent of NXP, and any purported assignment without such consent shall be void and have no force or effect. Subject to the foregoing, this Agreement shall bind and inure to the benefit of the respective parties hereto and their successors and assigns. Without limiting the generality of the foregoing, a change in the majority ownership or control of Licensee, by the transfer, directly or indirectly, of fifty percent (50%) or more of the shares of Licensee entitled to vote for the election of directors or other persons performing similar functions, or by equivalent change in ownership or control of Licensee if a partnership or other non-corporate form, shall be deemed an assignment of the Agreement by Licensee for the purpose of this Section 10.2. In the event of any assignment or attempted assignment by Licensee, this Agreement shall terminate and all Licensed Software, Documentation, Confidential Information and related materials shall be returned to NXP within ten (10) days. NXP may assign this Agreement to any of its Affiliates or any third party without any consent of Licensee.

11.3 The Parties hereto intend to establish a relationship of licensee and licensor and as such are independent contractors with neither party having authority to act as an agent or legal representative of the other to create any obligation, express or implied, on behalf of the other. No principal/agent, joint venture or partnership relationship is created between them by this Agreement.

11.4 Each Party shall comply with all applicable federal, state and local laws, including, but not limited to those pertaining to U.S. and member states of the European Union's Export Administration or the export or import controls or restrictions of other applicable jurisdictions in performing under this Agreement. Licensed Software, Documentation and other Confidential Information may be subject to United States and/or member states of the European Union's export control laws and may be subject to other foreign export and import control laws and regulations. Licensee agrees to strictly comply with all applicable export and import control laws and regulations, and, in particular, Licensee will not export or re-export the Licensed Software without all required, if any, United States and member states of the European Union's and foreign governmental licenses, approvals, or waivers.

11.5 Licensee acknowledges that a breach of this Agreement may cause irreparable damage for which recovery of money damages would be inadequate, and that, in addition to any and all remedies available at law and equity, NXP shall be entitled to seek injunctive relief to protect its rights under this Agreement.

11.6 Each Party acknowledges and agrees that this Agreement does not confer any rights to use any of the other Party's names, trademarks, or logos for any reason, including but not limited to, in connection with it advertising, publicity or other marketing activities.

11.7 The failure on the part of either Party to exercise, or any delay in exercising, any right or remedy arising from this Agreement shall not operate as a waiver thereof; nor shall any single or partial exercise of any right or remedy arising therefrom preclude any other or future exercise thereof or the exercise of any other right or remedy arising from this Agreement or by law.

11.8 The headings and captions to Articles and Sections of this Agreement are for reference and only and shall not affect the construction or interpretation of this Agreement.

11.9 This Agreement shall be governed by and construed in accordance with the laws of Germany, without regard to provisions concerning conflicts of law. Licensee submits and consents to, for the exclusive benefit of NXP, the jurisdiction of the state and federal courts of Germany. Parties however also agree that NXP, at NXP's sole discretion, may also elect to bring legal action regarding any dispute arising from or in connection with this Agreement before any court in any country (or political subdivision thereof) which may have jurisdiction and Licensee herewith irrevocably and unconditionally submits to the exclusive jurisdiction of the courts chosen by NXP. The United Nations Convention on Contracts for the International Sale of Goods shall not apply to this Agreement.

11.10 All notices or communications to be given under this Agreement shall be in writing and shall be deemed to be validly given by delivery thereof to its recipient if (i) delivered upon hand delivery, or (ii) by

deposit in the mail of the home country of the Party, postage prepaid, by certified, registered or first class mail or equivalent, or (iii) by nationally recognized overnight courier service (e.g., FedEx, UPS) prepaid, all of which are addressed to the parties at their addresses set forth above or such other address that a Party may notify the other party from time to time, in accordance with this Section 11.10. Any written notice is deemed to have been received, if sent by hand delivery, certified or registered, first class (or equivalent) mail or prepaid overnight courier, at the time of its delivery.

11.11 Neither Party shall publicize or disclose the existence or terms of this Agreement to any third Party, other than on a confidential basis to its legal and financial advisors, without the prior written consent of the other, except as otherwise may be required by law.

11.12 In the event that any provision(s) of this Agreement shall be held invalid or unenforceable by a court of competent jurisdiction or by any future legislative or administrative action, such holding or action shall not negate the validity or enforceability of any other provisions hereof.

11.13 This Agreement may be executed in one or more counterparts, each of which will be deemed an original, but all of which together will constitute one and the same instrument.

11.14 This Agreement, including its Appendices, sets forth the entire agreement between the parties relating to the subject matter herein, and supersedes and replaces all prior or contemporaneous agreements, discussions and understandings, written or oral, with respect to such subject matter. Each Party represents that it is duly authorized to enter into the Agreement. The terms and conditions of this Agreement may not be modified, or amended.

20.1.2 APPENDIX A

1. LICENSED SOFTWARE

A. EXECUTABLE CODE

NXP JavaCard Library and Utilities (JCShell)
Windows reader libraries
TCP JRCP server for Linux, Windows, and MAC OSX

B. OBJECT CODE

NXP JavaCard Library and Utilities (JCShell)

C. SOURCE CODE

None

2. DOCUMENTATION

A. DOCUMENTATION

User Manual

20.1.3 APPENDIX B

1. DELIVERED THIRD PARTY SOFTWARE (Covered By Third Party Terms, See Appendix C)

A. EXECUTABLE CODE

None

B. OBJECT CODE

Apache Commons IO 2.5 – Licensed under Apache License 2.0
bouncycastle 1.52 - Licensed under MIT License
commons-cli 1.2 - Licensed under Apache License 2.0
eddsa 0.2.0 – Licensed under CC0 1.0 Universal
FTDI D2XX Drivers 2.12.24 (Windows), 1.3.6 (Linux), 1.2.2 (macOS)
jline 2.12.1 - Licensed under BSD License
jna 4.1.0 - Licensed under Apache License 2.0
milena 1.0.0 – Licensed under MIT License
org.fusesource.jansi 1.11 - Licensed under Apache License 2.0
org.fusesource.hawtjni - Licensed under Apache License 2.0

C. SOURCE CODE

None

2. DOCUMENTATION

A. DOCUMENTATION

User Manual

20.1.4 APPENDIX C

THIRD PARTY LICENSES

=====

=====

Apache Commons IO 2.5

Project homepage: <http://commons.apache.org/io>

Sources:

<http://apache.mirror1.spango.com/commons/io/source/commons-io-2.5-src.tar.gz>

Notice.txt :

Apache Commons IO
Copyright 2002-2016 The Apache Software Foundation

This product includes software developed at
The Apache Software Foundation (<http://www.apache.org/>).

License:

Apache License 2.0 (See below)

=====

=====

bouncycastle 1.52

Project homepage: <http://www.bouncycastle.org/java.html>

Sources: https://www.bouncycastle.org/latest_releases.html

License:

Copyright (c) 2000 - 2015 The Legion of the Bouncy Castle Inc.
(<http://www.bouncycastle.org>)

Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files (the "Software"), to deal in the Software without restriction, including without limitation the rights to use, copy, modify, merge, publish, distribute, sublicense, and/or sell copies of the Software, and to permit persons to whom the Software is furnished to do so, subject to the following conditions:

The above copyright notice and this permission notice shall be included in all copies or substantial portions of the Software.

THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE SOFTWARE.

=====

=====

commons-cli 1.2

Project homepage: <https://commons.apache.org/proper/commons-cli/>

Sources: https://commons.apache.org/proper/commons-cli/download_cli.cgi

License:

Licensed under Apache License 2.0

APPENDIX: How to apply the Apache License to your work.

To apply the Apache License to your work, attach the following boilerplate notice, with the fields enclosed by brackets "[]" replaced with your own identifying information. (Don't include the brackets!) The text should be enclosed in the appropriate comment syntax for the file format. We also recommend that a file or class name and description of purpose be included on the same "printed page" as the copyright notice for easier identification within third-party archives.
Copyright [yyyy] [name of copyright owner]

Licensed under the Apache License, Version 2.0 (the "License"); you may not use this file except in compliance with the License. You may obtain a copy of the License at

<http://www.apache.org/licenses/LICENSE-2.0>

Unless required by applicable law or agreed to in writing, software distributed under the License is distributed on an "AS IS" BASIS, WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied. See the License for the specific language governing permissions and limitations under the License.

NOTICE.txt:

Apache Commons CLI
Copyright 2001-2016 The Apache Software Foundation

This product includes software developed at
The Apache Software Foundation (<http://www.apache.org/>).

=====

eddsa 0.2.0

Project homepage: <https://github.com/str4d/ed25519-java>

Sources : <https://github.com/str4d/ed25519-java>

License:
Creative Commons Legal Code

CC0 1.0 Universal

CREATIVE COMMONS CORPORATION IS NOT A LAW FIRM AND DOES NOT PROVIDE LEGAL SERVICES. DISTRIBUTION OF THIS DOCUMENT DOES NOT CREATE AN ATTORNEY-CLIENT RELATIONSHIP. CREATIVE COMMONS PROVIDES THIS INFORMATION ON AN "AS-IS" BASIS. CREATIVE COMMONS MAKES NO WARRANTIES REGARDING THE USE OF THIS DOCUMENT OR THE INFORMATION OR WORKS PROVIDED HEREUNDER, AND DISCLAIMS LIABILITY FOR DAMAGES RESULTING FROM THE USE OF THIS DOCUMENT OR THE INFORMATION OR WORKS PROVIDED HEREUNDER.

Statement of Purpose

The laws of most jurisdictions throughout the world automatically confer exclusive Copyright and Related Rights (defined below) upon the creator and subsequent owner(s) (each and all, an "owner") of an original work of authorship and/or a database (each, a "Work").

Certain owners wish to permanently relinquish those rights to a Work for the purpose of contributing to a commons of creative, cultural and scientific works ("Commons") that the public can reliably and without fear of later claims of infringement build upon, modify, incorporate in other works, reuse and redistribute as freely as possible in any form whatsoever and for any purposes, including without limitation commercial purposes. These owners may contribute to the Commons to promote the ideal of a free culture and the further production of creative, cultural and scientific works, or to gain reputation or greater distribution for their Work in part through the use and efforts of others.

For these and/or other purposes and motivations, and without any expectation of additional consideration or compensation, the person associating CC0 with a Work (the "Affirmer"), to the extent that he or she is an owner of Copyright and Related Rights in the Work, voluntarily elects to apply CC0 to the Work and publicly distribute the Work under its terms, with knowledge of his or her Copyright and Related Rights in the Work and the meaning and intended legal effect of CC0 on those rights.

1. Copyright and Related Rights. A Work made available under CC0 may be protected by copyright and related or neighboring rights ("Copyright and Related Rights"). Copyright and Related Rights include, but are not limited to, the following:

- i. the right to reproduce, adapt, distribute, perform, display, communicate, and translate a Work;
- ii. moral rights retained by the original author(s) and/or performer(s);
- iii. publicity and privacy rights pertaining to a person's image or likeness depicted in a Work;
- iv. rights protecting against unfair competition in regards to a Work, subject to the limitations in paragraph 4(a), below;
- v. rights protecting the extraction, dissemination, use and reuse of data in a Work;
- vi. database rights (such as those arising under Directive 96/9/EC of the European Parliament and of the Council of 11 March 1996 on the legal protection of databases, and under any national implementation thereof, including any amended or successor version of such directive); and
- vii. other similar, equivalent or corresponding rights throughout the world based on applicable law or treaty, and any national

implementations thereof.

2. Waiver. To the greatest extent permitted by, but not in contravention of, applicable law, Affirmer hereby overtly, fully, permanently, irrevocably and unconditionally waives, abandons, and surrenders all of Affirmer's Copyright and Related Rights and associated claims and causes of action, whether now known or unknown (including existing as well as future claims and causes of action), in the Work (i) in all territories worldwide, (ii) for the maximum duration provided by applicable law or treaty (including future time extensions), (iii) in any current or future medium and for any number of copies, and (iv) for any purpose whatsoever, including without limitation commercial, advertising or promotional purposes (the "Waiver"). Affirmer makes the Waiver for the benefit of each member of the public at large and to the detriment of Affirmer's heirs and successors, fully intending that such Waiver shall not be subject to revocation, rescission, cancellation, termination, or any other legal or equitable action to disrupt the quiet enjoyment of the Work by the public as contemplated by Affirmer's express Statement of Purpose.

3. Public License Fallback. Should any part of the Waiver for any reason be judged legally invalid or ineffective under applicable law, then the Waiver shall be preserved to the maximum extent permitted taking into account Affirmer's express Statement of Purpose. In addition, to the extent the Waiver is so judged Affirmer hereby grants to each affected person a royalty-free, non transferable, non sublicensable, non exclusive, irrevocable and unconditional license to exercise Affirmer's Copyright and Related Rights in the Work (i) in all territories worldwide, (ii) for the maximum duration provided by applicable law or treaty (including future time extensions), (iii) in any current or future medium and for any number of copies, and (iv) for any purpose whatsoever, including without limitation commercial, advertising or promotional purposes (the "License"). The License shall be deemed effective as of the date CC0 was applied by Affirmer to the Work. Should any part of the License for any reason be judged legally invalid or ineffective under applicable law, such partial invalidity or ineffectiveness shall not invalidate the remainder of the License, and in such case Affirmer hereby affirms that he or she will not (i) exercise any of his or her remaining Copyright and Related Rights in the Work or (ii) assert any associated claims and causes of action with respect to the Work, in either case contrary to Affirmer's express Statement of Purpose.

4. Limitations and Disclaimers.

- a. No trademark or patent rights held by Affirmer are waived, abandoned, surrendered, licensed or otherwise affected by this document.
- b. Affirmer offers the Work as-is and makes no representations or warranties of any kind concerning the Work, express, implied, statutory or otherwise, including without limitation warranties of title, merchantability, fitness for a particular purpose, non infringement, or the absence of latent or other defects, accuracy, or the present or absence of errors, whether or not discoverable, all to the greatest extent permissible under applicable law.
- c. Affirmer disclaims responsibility for clearing rights of other persons that may apply to the Work or any use thereof, including without limitation any person's Copyright and Related Rights in the Work. Further, Affirmer disclaims responsibility for obtaining any necessary consents, permissions or other rights required for any use of the Work.
- d. Affirmer understands and acknowledges that Creative Commons is not a party to this document and has no duty or obligation with respect to this CC0 or use of the Work.

For more information, please see <https://creativecommons.org/publicdomain/zero/1.0/>

=====

FTDI D2XX Drivers 2.12.24 (Windows), 1.3.6 (Linux), 1.2.2 (macOS)

Project homepage: <http://www.ftdichip.com/Drivers/D2XX.htm>

Sources : <http://www.ftdichip.com/Drivers/D2XX.htm>

Driver Licence Terms:

IMPORTANT NOTICE: PLEASE READ CAREFULLY BEFORE INSTALLING THE RELEVANT SOFTWARE: This licence agreement ("Licence") is a legal agreement between you ("Licencee" or "you") and Future Technology Devices International Limited of 2 Seaward Place, Centurion Business Park, Glasgow, Scotland, G41 1HH (UK Company Number SC136640) ("Licensor" or "we") for use of driver software provided by the Licensor ("Software").

BY INSTALLING OR USING THIS SOFTWARE YOU AGREE TO THE TERMS OF THIS LICENCE WHICH WILL BIND YOU. IF YOU DO NOT AGREE TO THE TERMS OF THIS LICENCE, WE ARE UNWILLING TO LICENSE THE SOFTWARE TO YOU AND YOU MUST DISCONTINUE INSTALLATION OF THE SOFTWARE NOW.

1. Grant and scope of licence

1.1 In consideration of you agreeing to abide by the terms of this Licence, the Licensor hereby grants to you a non-exclusive, non-transferable, royalty free licence to use the Software on the terms of this Licence.

1.2 In this Licence a "Genuine FTDI Component" means an item of hardware that was manufactured for, and sold by, the Licensor or a member of the Licensor's group of companies. It does not include any counterfeit or fake products.

1.3 If you are a manufacturer of a device that includes a Genuine FTDI Component (each a "Device") then you may install the Software onto that device. If you are a seller or distributor of a Device then You may distribute the Software with the Device. If you are a user of a Device then you may install the Software on the Device, or onto a computer system in order to use the Device.

1.4 In each of those cases you may:

1.4.1 install and use the Software for your purposes only; and

1.4.2 only use the Software in conjunction with products based on and/or incorporating a Genuine FTDI Component.

1.5 The Software will not function properly on or with a component that is not a Genuine FTDI Component. Use of the Software as a driver for, or installation of the Software onto, a component that is not a Genuine FTDI Component, including without limitation counterfeit components, MAY IRRETRIEVABLY DAMAGE THAT COMPONENT. It is the Licensee's responsibility to make sure that all chips it installs the Software on, or uses the Software as a driver for, are Genuine FTDI Components. If in doubt then contact the Licensor.

2. If a custom vendor ID and/or product ID or description string are used, it is the responsibility of the product manufacturer to maintain any changes and subsequent WHQL re-certification as a result of making these changes.

3. Licensee's undertakings

3.1 Except as expressly set out in this Licence or as permitted by any local law, you undertake:

3.1.1 not to copy the Software, except where such copying is incidental to normal use of the Software or where it is necessary for the purpose of back-up or operational security;

3.1.2 not to rent, lease, sub-license, loan, translate, merge, adapt, vary or modify the Software or any part of it;

3.1.3 not to make alterations to, or modifications of, the whole or any part of the Software nor permit the Software or any part of it to be combined with, or become incorporated in, any other programs;

3.1.4 not to disassemble, de-compile, reverse engineer or create derivative works based on the whole or any part of the Software;

3.1.5 to keep all copies of the Software secure;

3.1.6 to include the copyright notice of the Licensor on all entire and partial copies of the Software in any form; and

3.1.7 not to provide, or otherwise make available, the Software in any form, in whole or in part (including, but not limited to, program listings, object and source program listings, object code and source code) to any person.

4. Intellectual property rights

4.1 You acknowledge that all intellectual property rights in the Software throughout the world belong to the Licensor, that rights in the Software are licensed (not sold) to you, and that you have no rights in, or to, the Software other than the right to use them in accordance with the terms of this Licence.

5. Warranty

5.1 To the maximum extent permitted by applicable law, the software is provided "as is".

5.2 All implied warranties, implied conditions and/or implied licences are excluded from this Licence, including but not limited to implied warranties of quality and/or fitness for purpose (in all cases) to the fullest extent permitted by law.

5.3 You acknowledge that the Software has not been developed to meet your individual requirements and that the Software may not be uninterrupted or free of bugs or errors.

6. Licensor's liability

6.1 To the maximum extent permitted by applicable law, in no event shall the Licensor be liable for any:

6.1.1 special loss or damage;

6.1.2 incidental loss or damage;

6.1.3 indirect or consequential loss or damage;

6.1.4 loss of income;

6.1.5 loss of business;

6.1.6 loss of profits;

6.1.7 loss of revenue;

6.1.8 loss of contracts;

6.1.9 business interruption;

6.1.10 loss of the use of money or anticipated savings;

6.1.11 loss of information;

6.1.12 loss of opportunity;

6.1.13 loss of goodwill or reputation; and/or

6.1.14 loss of, damage to or corruption of data;

(in each case) of any kind howsoever arising and whether caused by delict (including negligence), breach of contract or otherwise.

6.2 FTDI's total liability to you in relation to the Software shall not exceed 500 US Dollars.

6.3 Nothing in this Licence limits or excludes liability for death or personal injury or for fraud.

7. Termination

7.1 The Licensor may terminate this Licence immediately if:

7.1.1 you fail to comply with any of the terms and conditions of the Licence; or

7.1.2 you commence or participate in any legal proceedings against the Licensor.

7.2 Upon termination:

7.2.1 all rights granted to you under this Licence shall cease;

7.2.2 you must cease all activities authorised by this Licence; and

7.2.3 you must immediately delete or remove the Software from all computer equipment in your possession and immediately destroy all copies of the Software then in your possession, custody or control.

8. Transfer of rights and obligations

8.1 You may not transfer, assign, charge or otherwise dispose of this Licence, or any of your rights or obligations arising under it.

8.2 The Licensor may transfer, assign, charge, sub-contract or otherwise dispose of this Licence, or any of his rights or obligations arising under it, at any time during the term of the Licence.

9. Waiver

9.1 If the Licensor fails, at any time during the term of this Licence, to insist on strict performance of any of your obligations under this Licence, or if the Licensor fails to exercise any of the rights or remedies to which he is entitled under this Licence, this shall not constitute a waiver of such rights or remedies and shall not relieve you from compliance with such obligations.

9.2 A waiver by the Licensor of any default shall not constitute a waiver of any subsequent default.

9.3 No waiver by the Licensor of any of these terms and conditions shall be effective unless it is expressly stated to be a waiver and is communicated to you in writing.

10. Severability

If any of the terms of this Licence are determined by any competent authority to be invalid, unlawful or unenforceable to any extent, such term, condition or provision will to that extent be severed from the remaining terms, conditions and provisions which will continue to be valid to the fullest extent permitted by law.

11. Entire agreement

11.1 This Licence constitutes the whole agreement between us and supersedes any previous arrangement, understanding or agreement between us, relating to the licensing of the Software.

11.2 Each party acknowledges that in entering into this Licence it does not rely on any statement, representation, warranty or understanding other than those expressly set out in this Licence. Each party agrees that it will have no remedy in respect of any statement, representation, warranty or understanding that is not expressly set out in this Licence. Each party agrees that its only remedy in respect of those representations, statements, assurances and warranties that are set out in this Licence will be for breach of contract in accordance with the terms of this Licence.

11.3 The parties agree that nothing in this Licence will limit or exclude any liability they may have for fraud.

12. Miscellaneous

12.1 This Licence does not create a partnership or joint venture between the parties to it, nor authorise a party to act as agent for the other.

12.2 This Licence does not create any legal rights enforceable by any third party.

12.3 This Licence may only be varied by express written legal agreement between the parties.

13. Law and jurisdiction

This Licence, its subject matter or its formation (including non-contractual disputes or claims) shall be governed by and construed in accordance with Scots law and submitted to the non-exclusive jurisdiction of the Scottish courts.

=====

=====

groovy-all 2.3.8

Project homepage: <http://groovy-lang.org>

Sources: <https://github.com/java-native-access/jna>

License:
Licensed under Apache License 2.0

=====

=====

jline 2.12.1

Project homepage: <http://jline.sourceforge.net/>

Sources: <http://github.com/jline/jline2>

License:

Copyright (c) 2002-2015, the original author or authors.
All rights reserved.

<http://www.opensource.org/licenses/bsd-license.php>

Redistribution and use in source and binary forms, with or without modification, are permitted provided that the following conditions are met:

Redistributions of source code must retain the above copyright notice, this list of conditions and the following disclaimer.

Redistributions in binary form must reproduce the above copyright notice, this list of conditions and the following disclaimer in the documentation and/or other materials provided with the distribution.

Neither the name of JLine nor the names of its contributors may be used to endorse or promote products derived from this software without specific prior written permission.

THIS SOFTWARE IS PROVIDED BY THE COPYRIGHT HOLDERS AND CONTRIBUTORS "AS IS" AND ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE ARE DISCLAIMED. IN NO EVENT SHALL THE COPYRIGHT OWNER OR CONTRIBUTORS BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

=====

=====

jna 4.1.0

Project homepage: <https://github.com/java-native-access/jna>

Sources: <https://github.com/java-native-access/jna>

License:

This copy of JNA is licensed under the Apache (Software) License, version 2.0 ("the License"). See the License for details about distribution rights, and the specific rights regarding derivative works.

You may obtain a copy of the License at:

<http://www.apache.org/licenses/>

A copy is also included in the downloadable source code package containing JNA, in file "ASL2.0", under the same directory as this file.

ASL2.0:

Apache License
Version 2.0, January 2004
<http://www.apache.org/licenses/>

TERMS AND CONDITIONS FOR USE, REPRODUCTION, AND DISTRIBUTION

1. Definitions.

"License" shall mean the terms and conditions for use, reproduction, and distribution as defined by Sections 1 through 9 of this document.

"Licensor" shall mean the copyright owner or entity authorized by the copyright owner that is granting the License.

"Legal Entity" shall mean the union of the acting entity and all other entities that control, are controlled by, or are under common control with that entity. For the purposes of this definition, "control" means (i) the power, direct or indirect, to cause the direction or management of such entity, whether by contract or otherwise, or (ii) ownership of fifty percent (50%) or more of the outstanding shares, or (iii) beneficial ownership of such entity.

"You" (or "Your") shall mean an individual or Legal Entity exercising permissions granted by this License.

"Source" form shall mean the preferred form for making modifications, including but not limited to software source code, documentation source, and configuration files.

"Object" form shall mean any form resulting from mechanical transformation or translation of a Source form, including but not limited to compiled object code, generated documentation, and conversions to other media types.

"Work" shall mean the work of authorship, whether in Source or Object form, made available under the License, as indicated by a copyright notice that is included in or attached to the work (an example is provided in the Appendix below).

"Derivative Works" shall mean any work, whether in Source or Object form, that is based on (or derived from) the Work and for which the editorial revisions, annotations, elaborations, or other modifications represent, as a whole, an original work of authorship. For the purposes of this License, Derivative Works shall not include works that remain separable from, or merely link (or bind by name) to the interfaces of, the Work and Derivative Works thereof.

"Contribution" shall mean any work of authorship, including the original version of the Work and any modifications or additions to that Work or Derivative Works thereof, that is intentionally submitted to Licensor for inclusion in the Work by the copyright owner or by an individual or Legal Entity authorized to submit on behalf of the copyright owner. For the purposes of this definition, "submitted" means any form of electronic, verbal, or written communication sent to the Licensor or its representatives, including but not limited to communication on electronic mailing lists, source code control systems, and issue tracking systems that are managed by, or on behalf of, the Licensor for the purpose of discussing and improving the Work, but excluding communication that is conspicuously marked or otherwise designated in writing by the copyright owner as "Not a Contribution."

"Contributor" shall mean Licensor and any individual or Legal Entity on behalf of whom a Contribution has been received by Licensor and subsequently incorporated within the Work.

2. Grant of Copyright License. Subject to the terms and conditions of this License, each Contributor hereby grants to You a perpetual, worldwide, non-exclusive, no-charge, royalty-free, irrevocable copyright license to reproduce, prepare Derivative Works of, publicly display, publicly perform, sublicense, and distribute the Work and such Derivative Works in Source or Object form.

3. Grant of Patent License. Subject to the terms and conditions of this License, each Contributor hereby grants to You a perpetual, worldwide, non-exclusive, no-charge, royalty-free, irrevocable (except as stated in this section) patent license to make, have made, use, offer to sell, sell, import, and otherwise transfer the Work, where such license applies only to those patent claims licensable by such Contributor that are necessarily infringed by their Contribution(s) alone or by combination of their Contribution(s) with the Work to which such Contribution(s) was submitted. If You institute patent litigation against any entity (including a cross-claim or counterclaim in a lawsuit) alleging that the Work or a Contribution incorporated within the Work constitutes direct or contributory patent infringement, then any patent licenses granted to You under this License for that Work shall terminate as of the date such litigation is filed.

4. Redistribution. You may reproduce and distribute copies of the Work or Derivative Works thereof in any medium, with or without modifications, and in Source or Object form, provided that You meet the following conditions:

(a) You must give any other recipients of the Work or Derivative Works a copy of this License; and

(b) You must cause any modified files to carry prominent notices stating that You changed the files; and

(c) You must retain, in the Source form of any Derivative Works that You distribute, all copyright, patent, trademark, and attribution notices from the Source form of the Work, excluding those notices that do not pertain to any part of the Derivative Works; and

(d) If the Work includes a "NOTICE" text file as part of its distribution, then any Derivative Works that You distribute must include a readable copy of the attribution notices contained within such NOTICE file, excluding those notices that do not pertain to any part of the Derivative Works, in at least one of the following places: within a NOTICE text file distributed as part of the Derivative Works; within the Source form or documentation, if provided along with the Derivative Works; or, within a display generated by the Derivative Works, if and wherever such third-party notices normally appear. The contents of the NOTICE file are for informational purposes only and do not modify the License. You may add Your own attribution notices within Derivative Works that You distribute, alongside or as an addendum to the NOTICE text from the Work, provided that such additional attribution notices cannot be construed as modifying the License.

You may add Your own copyright statement to Your modifications and may provide additional or different license terms and conditions for use, reproduction, or distribution of Your modifications, or for any such Derivative Works as a whole, provided Your use, reproduction, and distribution of the Work otherwise complies with the conditions stated in this License.

5. Submission of Contributions. Unless You explicitly state otherwise, any Contribution intentionally submitted for inclusion in the Work by You to the Licensor shall be under the terms and conditions of this License, without any additional terms or conditions. Notwithstanding the above, nothing herein shall supersede or modify the terms of any separate license agreement you may have executed with Licensor regarding such Contributions.

6. Trademarks. This License does not grant permission to use the trade names, trademarks, service marks, or product names of the Licensor, except as required for reasonable and customary use in describing the origin of the Work and reproducing the content of the NOTICE file.

7. Disclaimer of Warranty. Unless required by applicable law or agreed to in writing, Licensor provides the Work (and each Contributor provides its Contributions) on an "AS IS" BASIS, WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied, including, without limitation, any warranties or conditions of TITLE, NON-INFRINGEMENT, MERCHANTABILITY, or FITNESS FOR A PARTICULAR PURPOSE. You are solely responsible for determining the appropriateness of using or redistributing the Work and assume any risks associated with Your exercise of permissions under this License.

8. Limitation of Liability. In no event and under no legal theory, whether in tort (including negligence), contract, or otherwise, unless required by applicable law (such as deliberate and grossly negligent acts) or agreed to in writing, shall any Contributor be liable to You for damages, including any direct, indirect, special, incidental, or consequential damages of any character arising as a result of this License or out of the use or inability to use the Work (including but not limited to damages for loss of goodwill, work stoppage, computer failure or malfunction, or any and all other commercial damages or losses), even if such Contributor has been advised of the possibility of such damages.

9. Accepting Warranty or Additional Liability. While redistributing the Work or Derivative Works thereof, You may choose to offer, and charge a fee for, acceptance of support, warranty, indemnity, or other liability obligations and/or rights consistent with this License. However, in accepting such obligations, You may act only on Your own behalf and on Your sole responsibility, not on behalf of any other Contributor, and only if You agree to indemnify, defend, and hold each Contributor harmless for any liability incurred by, or claims asserted against, such Contributor by reason of your accepting any such warranty or additional liability.

END OF TERMS AND CONDITIONS

=====
=====

mileneage 1.0.0

Project homepage: <https://github.com/brake/milenage>

Sources: <https://github.com/brake/milenage>

License: MIT License

Copyright © 2015-2016 Constantin Roganov

Permission is hereby granted, free of charge, to any person obtaining a copy of this software and associated documentation files (the "Software"), to deal in the Software without restriction, including without limitation the rights to use, copy, modify, merge, publish, distribute, sublicense, and/or sell copies of the Software, and to permit persons to whom the Software is furnished to do so, subject to the following conditions:

The above copyright notice and this permission notice shall be included in all copies or substantial portions of the Software.

THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY, FITNESS FOR A

PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE SOFTWARE.

=====

org.fusesource.jansi 1.11
Project homepage: <https://github.com/fusesource/jansi>
Sources: <https://github.com/fusesource/jansi>
License: Apache License 2.0

=====

org.fusesource.hawtjni
Project homepage: <https://github.com/fusesource/hawtjni>
Sources: <https://github.com/fusesource/hawtjni>
License: Apache License 2.0

=====

Please be aware that important notices concerning this document and the product(s) described herein, have been included in the section 'Legal information'.

© NXP Semiconductors Netherlands B.V.. 2020. All rights reserved.

For more information, please visit: <http://www.nxp.com>
For sales office addresses, email to: salesaddresses@nxp.com

Date of release: 13/01/2020
Document identifier: <DOC_ID>